



MASTER THESIS

Procedural Generation of Solvable Cooperative Levels for the Laser Learning Environment

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Abstract

This thesis develops a SAT-based framework for the procedural generation of solvable, cooperative levels for the Laser Learning Environment (LLE), a multi-agent reinforcement learning benchmark. We formalise bounded-horizon LLE solvability as a propositional satisfiability decision problem and prove the reduction correct; we introduce a strict-beam-semantics counterfactual that turns the **cooperation-required** property into a second decidable problem on the same level; and we refine the resulting binary criterion into a profile taxonomy that classifies the dependency structure of a cooperative level into one of five patterns (asymmetric, mutual, chain, distributed, fully coupled). These decision procedures are embedded inside a family of procedural generators (Random, Constrained Random, Constructive, Level-6-Style) that emit only solver-certified levels with the requested properties. The empirical evaluation compares two SAT encodings, characterises per-generator rejection rates and cooperation-profile distributions, and shows that off-the-shelf value-decomposition MARL algorithms (IQL, VDN, QMIX) can learn certified cooperative levels at 5×5 and that enlarging the generated training pool restores generalisation. A final set of curriculum experiments finds that curriculum **ordering** offers no advantage over direct training on a reachable target, and that no curriculum — including a staged budget of two million steps — lets these algorithms solve the mutually-cooperative LLE Level 6, on which direct training also scores zero; we trace this negative result to the base task being unlearnable by value-decomposition methods rather than to a failure of curriculum design.

Keywords: multi-agent reinforcement learning, procedural content generation, SAT solving, cooperative levels, Laser Learning Environment, curriculum learning.

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Chapter 1

Introduction

1.1 Context

Cooperative Multi-Agent Reinforcement Learning (MARL) studies settings in which several agents must learn behaviours whose value appears only at the team level. In such settings, the environment is not a neutral container: it determines which coordination patterns are possible, which ones are necessary, and how difficult they are to discover.

This dependence on environment structure is especially strong in sparse-reward cooperative tasks. When reward is issued only after the team objective has been achieved, a training level is useful only if it exposes a meaningful coordination challenge while remaining actually solvable. Unsolvable levels provide no valid signal, and levels that admit independent solutions fail to test the cooperative mechanism they are meant to study.

We instantiate this work on the Laser Learning Environment (LLE), a cooperative MARL benchmark whose mechanics are formalised in Section 3.1; its hand-crafted Level 6 serves as the canonical hard target throughout the thesis. We expect the methodology developed here — coupling procedural generation with a formal verification oracle — to generalise to any MARL setting whose target structural properties are expressible as decision problems, but we evaluate it on LLE only and leave broader transfer to future work.

1.2 Motivation

In cooperative environments whose mechanics create explicit inter-agent dependencies, level design is particularly challenging. A useful training level should not merely appear cooperative — it should provably contain the intended coordination structure.

Procedural Content Generation (PCG) offers a way to scale level creation beyond what manual design allows, but only if the generated instances satisfy the properties that matter for training. Two properties are central. First, a level must be **solvable**. Second, success should **require cooperation** in the specific sense induced by the environment’s mechanics. Without those guarantees, generation risks producing levels that are invalid, trivial, or misaligned with the training objective.

Generated levels are also useful beyond standalone training: the same configuration knobs that control individual levels — grid size, number of agents, number of lasers, wall budget, cooperation profile,

horizon — also define a natural difficulty axis. A formally-verified generator is therefore a candidate building block for **curriculum learning**: rather than train directly on a hard hand-crafted target, an agent can be trained on a sequence of progressively harder generated levels.

The thesis delivers three contributions and an empirical study. The contributions are a bounded-horizon SAT encoding of LLE solvability with a correctness proof and complexity placement; a strict-semantics counterfactual that turns the informal “cooperation required” intuition into a decidable property, refined into a five-label profile taxonomy; and a solver-in-the-loop generator family that accepts only certified levels and exposes the structural axes above as user-facing parameters. The empirical study covers SAT-encoding cost, generator acceptance rates and profile distributions, learnability of generated cooperative levels for off-the-shelf MARL, and a curriculum-transfer study toward LLE Level 6.

1.3 Research Questions and Scope

This thesis addresses the following question: how can we automatically generate levels for a cooperative MARL environment that are provably solvable and provably require the target cooperative interaction, and how can such levels be used to support MARL training?

More precisely, the work is organised around six research questions:

- **RQ1:** How can we formally verify that a level is solvable by the agents?
- **RQ2:** How can we formally verify that solving a level genuinely requires cooperation between agents, rather than allowing independent solutions?
- **RQ3:** How can formal verification be embedded inside a procedural generator so that every accepted level comes with certified properties?
- **RQ4:** Can we control the cooperation structure of generated levels by targeting specific profiles such as asymmetric, mutual, chain, distributed, or fully coupled dependencies?
- **RQ5:** Can MARL agents trained exclusively on procedurally generated cooperative levels reach a non-trivial greedy success rate on a held-out pool of generated levels at a fixed training budget, and how does that performance vary across IQL, VDN, and QMIX?
- **RQ6:** Does a staged curriculum of generated levels of growing geometric and cooperative complexity yield a higher greedy success rate on hand-crafted LLE Level 6 than a baseline trained directly on Level 6 (or on the union of all stage pools) at the same total training budget?

The thesis instantiates this framework on LLE, focusing on a restricted but explicit subset of its mechanics. The formal model covers agent movement, inter-agent blocking interactions, and a bounded-horizon solvability criterion. LLE additionally supports collectible gems and an incentive-scoring layer on top of plain exit-reaching, as well as void tiles distinct from walls; gems and the incentive layer are outside the scope of this thesis, and void tiles are treated as walls (impassable to agents) rather than as a separate tile type. These restrictions are deferred to future work.

1.4 Thesis Structure

The remainder of this thesis is organised as follows. Chapter 2 positions the work relative to cooperative MARL benchmarks, procedural content generation, and compilation-based planning literature. Chapter 3 fixes the technical setting: the LLE mechanics, the SAT background, and the formal model of bounded-horizon solvability and the cooperation requirement.

The three contribution chapters then develop the original technical content of the thesis in logical order. Chapter 4 introduces the SAT-based solver: a propositional encoding of bounded-horizon LLE solvability, with correctness proofs and a complexity-theoretic positioning. Chapter 5 builds on that encoding to define the cooperation detector based on a strict counterfactual semantics, and extends it with a profile analyzer that classifies the kind of cooperation a level exhibits. Chapter 6 then uses both decision procedures as acceptance oracles inside a family of procedural generators that produce levels certified to satisfy the advertised properties.

Chapter 7 reports the empirical evaluation: the SAT encoding comparison, generator acceptance rates and cooperation-profile distributions, the learnability of generated cooperative levels for off-the-shelf MARL algorithms, and the curriculum-transfer study toward the hand-crafted LLE Level 6. Chapter 8 concludes with a summary and directions for future work.

Chapter 2

Related Work

2.1 Cooperative MARL and Coordination-Critical Benchmarks

This thesis is motivated by a benchmark-design question within cooperative Multi-Agent Reinforcement Learning (MARL): how can one generate training instances whose coordination structure is both non-trivial and formally controlled?

In the fully cooperative setting, all agents optimise a shared return, so the quality of the environment matters as much as the quality of the learning algorithm. If the environment admits only trivial solutions, it does not meaningfully test coordination. If it contains unsolvable instances, it provides no useful training signal at all. For the present thesis, the central issue is therefore not MARL in general, but the design of instances in which inter-agent dependence is both structurally present and formally decidable.

The general framework for sequential multi-agent decision-making originates in stochastic games [1], generalised to the Markov game model that has become the standard formalism for MARL [2]. The single-agent reinforcement-learning machinery on which cooperative MARL builds is covered comprehensively in [3].

Two algorithm families dominate the cooperative MARL literature. **Value-decomposition** methods factor a shared team value into per-agent components to enable decentralised execution: VDN [4] uses an additive decomposition, and QMIX [5] generalises this to a state-dependent monotonic mixing network. **Centralised-critic actor-critic** methods take a different route, training a joint critic that conditions on full state at training time while each agent retains a local actor at execution time: MADDPG [6] is the canonical deterministic-policy instance, COMA [7] refines the credit-assignment signal with a counterfactual baseline, and MAVEN [8] extends QMIX with latent-variable exploration to escape the monotonicity bottleneck on coordination-critical tasks.

Both families perform well when individual contributions are roughly additive, and both struggle on the coordination-critical, low-reward bottlenecks that LLE is designed to expose [9]. The empirical chapter of this thesis (Chapter 7) accordingly uses three points along the value-decomposition spectrum – independent Q -learning (IQL, no credit assignment), VDN (additive), and QMIX (monotonic mixing) – to train on generated cooperative levels and on the curriculum-transfer target.

2.2 The Laser Learning Environment

The Laser Learning Environment (LLE), whose mechanics we formalise in Section 3.1, was introduced precisely to study coordination-critical multi-agent tasks [9]. It sits alongside other cooperative-MARL benchmarks such as SMAC [10] (StarCraft micro-management with partially observable team play) and Overcooked [11] (a constrained cooperative kitchen with temporal synchronisation), but differs in that its hardness comes from explicit **inter-agent blocking** rather than from partial observability or large action spaces. The LLE paper identifies three properties that make the benchmark difficult for value-based MARL methods: **perfect coordination**, **interdependence**, and **zero-incentive dynamics**. Together, these properties create bottlenecks in which one agent must perform a locally unrewarded action that enables another agent to progress.

This benchmark framing is directly relevant to the present work. The thesis does not attempt to improve MARL training algorithms on LLE. Instead, it addresses an upstream question left open by the benchmark paper: how can we generate LLE levels that are guaranteed to be solvable and that contain the beam-blocking dependency on which the benchmark relies?

The LLE paper therefore plays two roles in this thesis. First, it justifies why LLE is an interesting target domain. Second, it provides the conceptual vocabulary used here to discuss cooperation: the key object is not generic teamwork, but a concrete interdependence mechanism created by coloured lasers and same-colour blocking.

2.3 Dependency Structures in Cooperative MARL

The same-colour beam-truncation mechanism on which LLE relies admits a “cooperation required / not required” verdict per level, but cooperative behaviour at finer granularity can vary considerably: in a level with several agents and several lasers, helping relations can form a one-way edge, a mutual pair, a directed chain, a fan-in (one beneficiary, multiple helpers), or a fully connected graph. The cooperation profile analyzer of Chapter 5 extracts this dependency graph from a SAT model of a solution and labels it as, respectively, **asymmetric**, **mutual**, **chain**, **distributed**, or **fully coupled**. The structural categories themselves are standard graph-theory vocabulary; to our knowledge, this is the first taxonomy that **recovers** such structures from a SAT certificate of solvability in LLE. The prior MARL literature considers related but distinct structural notions.

The closest precedent is the **coordination graph** introduced by C. Guestrin, M. G. Lagoudakis, and R. Parr [12], which decomposes a joint Q -function over agent subsets connected by hyper-edges. Coordination graphs are a representational tool: they assume the structure is known a priori and use it to make value computation tractable. Our taxonomy operates in the opposite direction. The structure is not given; it is **recovered** from a SAT certificate of solvability, and the label is a property of the level produced rather than a modelling assumption made about the agents. The two notions are therefore complementary: a coordination graph tells the **learner** which agent subsets interact, while our profile tells the **level designer** what kind of cooperation a generated instance contains.

2.4 Procedural Generation Under Structural Constraints

Procedural Content Generation (PCG) is useful in reinforcement-learning settings because it can replace a small fixed benchmark set with a larger and more diverse stream of instances [13]. Within PCG, search-based methods — surveyed by J. Togelius, G. N. Yannakakis, K. O. Stanley, and C. Browne

[14] — phrase content creation as an optimisation problem over a content space, which is conceptually close to the solver-driven acceptance loop adopted in this thesis. The closest precedent for the present declarative-constraint approach is the Answer Set Programming generator of A. M. Smith and M. Mateas [15], which uses an ASP solver as the acceptance oracle for game content. A complementary line of work surveyed under the PCGML banner [16] uses machine-learned generators trained on existing levels, but typically provides no formal guarantee on the produced output. The difficulty for the present problem is not merely to produce varied levels, but to produce levels that satisfy logically defined properties.

PCG has been increasingly developed **for** reinforcement learning, where the role of the generated content is not to entertain a human player but to provide training material for a learning agent. S. Risi and J. Togelius [17] survey this line and argue that PCG is a natural lever for moving beyond fixed-benchmark RL toward generalisation across infinite environment families. Within that line, PCGRL [18] inverts the relationship between PCG and RL: rather than using PCG to feed RL agents, it trains an RL agent to **act as** the level designer. The thesis here runs in the opposite direction — a fixed solver-in-the-loop generator produces certified levels that feed RL training — and PCGRL is therefore a useful contrast rather than a comparable system.

What distinguishes this thesis from the lines above is the **verification** step. A constructive or search-based generator may bias generation toward interesting layouts, but without a verifier it cannot certify that a sampled level is solvable or that success genuinely depends on cooperation. The present thesis therefore adopts a constraint-aware view of PCG: generation is coupled to a formal decision procedure, and the solver acts as an acceptance oracle rather than as a post-hoc descriptive tool.

2.5 Curriculum Learning and Generated Environments

The general idea of training on a sequence of progressively harder tasks predates the modern deep-learning era; Y. Bengio, J. Louradour, R. Collobert, and J. Weston [19] formalised **curriculum learning** as a meta-learning strategy in which the order of training examples is itself a design choice. In reinforcement learning specifically, the survey by S. Narvekar, B. Peng, M. Leonetti, J. Sinapov, M. E. Taylor, and P. Stone [20] catalogues the design space along three axes — the **task generator**, the **sequencing policy**, and the **transfer mechanism** — and shows that curriculum learning consistently helps on long-horizon and sparse-reward problems, the regime in which LLE Level 6 sits.

A more recent line of work tightens the coupling between curriculum and environment generation. **POET** [21] co-evolves a population of environments and a population of agents in an open-ended loop: environments that are too easy or too hard for the current agent population are eliminated, and surviving environments act as a self-organising curriculum. **PAIRED** [22] extends this idea to **unsupervised environment design**: an adversary parameterises environments to maximise the regret of a protagonist agent against an antagonist baseline, which provably keeps the generated environments solvable while remaining at the frontier of the protagonist’s ability.

The present thesis sits adjacent to this line of work. Like POET and PAIRED, we generate environments rather than reuse a fixed test set, and we use those environments as a curriculum toward a hard target. Unlike POET and PAIRED, the generator here is not adversarial and not adaptive to the learner’s current policy: it is a **static** solver-in-the-loop generator that emits levels certified to satisfy fixed structural properties (solvability, cooperation, profile) at a chosen difficulty configuration. The curriculum used in Chapter 7 is hand- staged — four manually ordered configurations of growing grid size and cooperation requirement.

2.6 SAT-based Planning

A second compilation lineage directly relevant to this thesis is **SAT-based planning** (we recall the propositional satisfiability background in Section 3.2). H. Kautz and B. Selman [23] introduced SATPLAN, encoding bounded-horizon STRIPS planning instances as propositional formulas; subsequent work refined both the encodings and the search strategies [24]. The treatment by J. Rintanen, K. Heljanko, and I. Niemelä [25] covers parallel-plan encodings and modern algorithmic refinements that drove SAT-based planners to competitive performance with dedicated planners. The bounded-horizon LLE encoding developed in this thesis sits in the same conceptual family: a state-transition decision problem reduced to SAT, with the encoding choices materially affecting solver performance.

2.7 Compilation-Based Multi-Agent Path Finding

The closest methodological precedent is not PCG for MARL, but compilation-based Multi-Agent Path Finding (MAPF). In standard MAPF, agents move on a discrete graph from start vertices to goal vertices while avoiding collisions. The computational difficulty comes from the interaction between multiple agents and the optimality criterion imposed on the solution.

The survey by P. Surynek [26] shows that MAPF has become a major testbed for compilation-based solving: instead of searching directly in the original state space, one reduces a MAPF instance to a target formalism (CSP, SAT, or MILP) and relies on the target solver to handle the combinatorial burden. MAPF research is not exclusively compilation-based; the dominant search-based alternative, Conflict-Based Search [27], achieves optimal solutions through a two-level constraint-satisfaction tree without reducing to a target formalism. We adopt the compilation route rather than CBS-style search because the property of interest in this thesis is the **existence** of a valid joint plan within a fixed horizon, not its makespan optimality: a SAT decision procedure matches the question we ask, whereas CBS would need substantial extension to encode the laser-propagation and strict-counterfactual semantics introduced in Chapter 5.

Within the compilation route, encoding design materially affects solver performance. The work of M. Frommknecht and P. Surynek [28] studies SAT-based MAPF under the makespan objective using an MDD-SAT formulation and compares solver-facing choices such as eager versus lazy encodings and informative initial assignments. The point of citing this paper is not that it solves the same problem, but that it confirms a recurring lesson: performance is not determined only by the underlying decision problem; it also depends on how the problem is encoded and on how the resulting CNF interacts with the chosen SAT solver. The empirical chapter of this thesis (Chapter 7) compares two alternative uniqueness encodings in the same spirit.

At the same time, the distance from standard MAPF must be stated explicitly. Standard MAPF encodings reason about graph motion and collisions. The present thesis must additionally encode time-dependent laser propagation, same-colour immunity, and a strict-counterfactual semantics used to define cooperation. The MAPF literature therefore supplies a methodological template, not a drop-in solution.

2.8 Formal Methods Coupled to Reinforcement Learning

A separate line of work uses formal methods as an explicit component of the reinforcement learning loop rather than as a post-hoc analysis. The canonical example is **shielded RL** [29], where a formally synthesised reactive shield filters the agent’s candidate actions at runtime so that no policy update can ever violate a stated temporal-logic specification. The shield is computed once from the specification and the environment model, then inserted between the agent and the environment as a hard constraint.

The connection to this thesis is methodological rather than direct. Shielded RL inserts a formal acceptance oracle on the **action** side of the loop, at runtime; we insert one on the **training- material** side, at generation time. Both follow the same recipe — encode a target property in a decidable formalism, run a solver, and reject anything that fails — but they apply it to opposite ends of the agent-environment interface. Shielded RL therefore does not provide a drop-in technique, but it does demonstrate that “formal-methods filter on top of an RL system” is a working pattern.

2.9 Positioning of the Thesis

Taken together, the MARL benchmarking literature, the PCG and curriculum-learning literature, and the SAT-planning / compilation-based MAPF literature mark the conceptual neighbourhood of this thesis. None of them, however, fully covers the problem we address: each line studies one side of the problem in isolation. The remaining gap can be summarised under four headings.

- **Cooperative MARL benchmarks and inter-agent structure.** The LLE paper [9] establishes the benchmark and explains why its coordination bottlenecks are difficult for MARL algorithms, but it does not provide a formal generator for certified cooperative levels. The coordination-graph line [12] gives a representation for known inter-agent dependencies but does not classify the **types** of dependency that emerge from a specific cooperation mechanism.
- **Compilation-based multi-agent planning.** The MAPF compilation literature [26], [28] shows that SAT is an effective backend for multi-agent planning and that encoding design matters in practice, but it solves a different problem (path optimality on a collision graph) and does not address either the laser-propagation semantics or cooperation as a semantic property of the instance.
- **Procedural generation, curriculum learning, and environment design.** The PCG-for-RL line [17], [18] frames generation as a tool for RL generalisation, and the environment-design line [21], [22] treats curriculum generation as adaptive co-evolution. Neither embeds a decision procedure that **proves** each generated instance satisfies a target structural property.
- **Formal methods on top of RL.** The shielded-RL line [29] shows that formal methods integrate cleanly with RL when used as a runtime **action** filter, but does not address the upstream question of whether the **training material** itself satisfies the specification.

This thesis sits at the intersection of those four lines. It transfers the compilation-based SAT mindset from MAPF into the LLE setting, formalises bounded-horizon solvability for an LLE-specific model, introduces a strict-semantics counterfactual that turns a benchmark-level intuition about cooperation into a decidable property inside procedural generation, and classifies the resulting dependency structures into a taxonomy that downstream generators can target as a parameter.

Chapter 3

Background and Formalization

3.1 The Laser Learning Environment

The Laser Learning Environment (LLE) [9] is a grid-based cooperative multi-agent benchmark designed to study coordination-critical tasks. A level consists of a rectangular grid containing walls, coloured laser sources, agent start positions, and exit tiles. Each agent is assigned a colour and must reach an exit tile.

The central mechanic is the laser beam. Each source emits a directional beam that propagates cell by cell until it reaches a wall or the grid boundary. An agent whose colour matches the beam is *immune* to it: it may occupy a cell the beam traverses without being harmed, but its physical presence truncates the beam at that cell. Agents of other colours cannot occupy a cell crossed by an active beam.

This truncation is the source of the cooperative structure studied in this thesis. By occupying a position along its own beam, an agent shortens the beam and opens cells beyond it for teammates who are not immune to that colour. Crucially, this action is locally unrewarded: the helper agent receives no direct benefit from blocking its own beam. The reward is issued only when all agents simultaneously occupy their exits, which requires the full team to coordinate successfully.

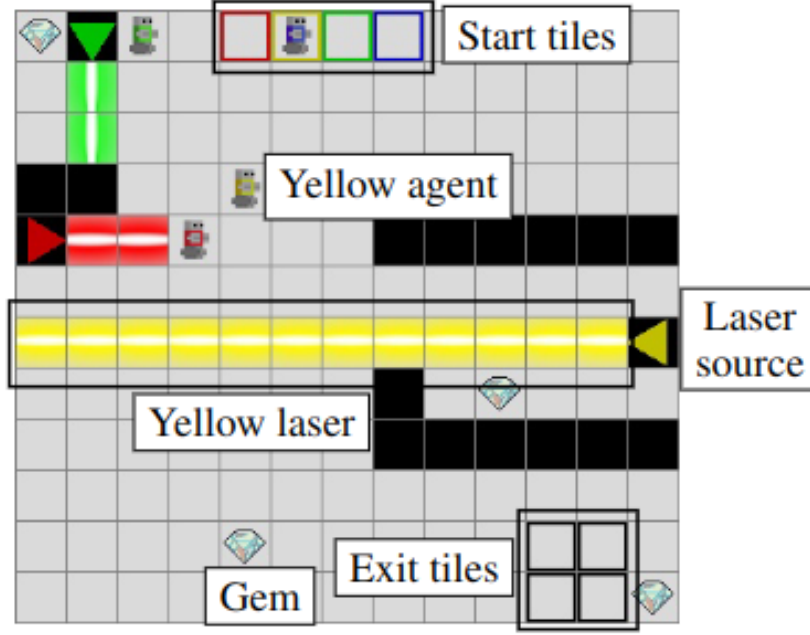


Figure 1: LLE Level 6, the canonical hard target used throughout this thesis. Agents must reach the exit tiles. Laser beams block movement for non-immune agents, but an immune agent can truncate its own beam by occupying a cell along its path, opening a route for a teammate.

LLE additionally supports collectible gems and void tiles. Neither is used in this thesis: gems and the incentive-scoring layer they enable are out of scope, and the formal model developed below does not introduce a void-tile type. Levels that contain void tiles can still be handled by treating each void tile as a wall, since both are impassable to agents.

We positioned LLE within the cooperative-MARL benchmark landscape in Chapter 2; the present chapter makes the mechanics precise through a formal model suitable for reasoning about solvability and cooperation within a bounded time horizon.

3.2 Boolean Satisfiability

Boolean Satisfiability (SAT) is the problem of deciding whether a propositional formula has a satisfying assignment — a mapping from Boolean variables to true or false that makes the formula evaluate to true. It is the canonical NP-complete decision problem [30], where NP is the class of decision problems whose yes-instances admit a polynomial-time-checkable certificate.

In practice, SAT solvers operate on formulas in *Conjunctive Normal Form* (CNF). A CNF formula is a conjunction of *clauses*, where each clause is a disjunction of *literals* and a literal is either a variable x or its negation $\neg x$. For example:

$$(x_1 \vee \neg x_2) \wedge (\neg x_1 \vee x_3) \wedge (x_2 \vee x_3)$$

is a CNF formula over three variables. It is satisfied by setting $x_1 = \text{true}$, $x_2 = \text{false}$, $x_3 = \text{true}$, among other assignments.

Modern SAT solvers based on the DPLL procedure [31] and Conflict-Driven Clause Learning (CDCL) [32] can handle industrial instances with millions of variables and clauses. Given a CNF formula, a solver either:

- returns a *satisfying assignment* that witnesses the formula is satisfiable (SAT), or

- certifies that *no* satisfying assignment exists (UNSAT).

The UNSAT certificate is as informative as the SAT witness: it proves the non-existence of a solution. This property is central to the cooperation detection mechanism developed in Chapter 5, where UNSAT under a modified encoding is used to certify that no solution exists without the cooperative interaction.

In this thesis, propositional variables are introduced to represent agent positions, laser beam states, and laser activity at each time step. The constraints of the bounded-horizon solvability problem are then encoded as CNF clauses, and a SAT solver is used as a decision oracle. The solver used throughout is Minisat22, a descendant of the MiniSat solver of Eén and Sörensson [32], accessed through the PySAT interface [33]. The full encoding — variables, clauses, and the correctness argument — is developed in Chapter 4.

3.3 Problem Formalization

We now define the semantic objects and decision problems used throughout the thesis. Two properties are central: **solvability** — whether there exists a valid joint trajectory bringing all agents to exits within a bounded horizon — and the **cooperation requirement** — whether every such trajectory must rely on the same-colour beam-truncation mechanism. Both are formally decidable and certified by the solver for every accepted level; the SAT encoding itself is developed in Chapter 4. Throughout, we assume $1 \leq n_a \leq 4$ in line with the LLE engine; this bound is an engine artefact, not a limitation of the encoding, which remains correct for any finite n_a .

Definition 3.1 (LLE Level)

An LLE level is a tuple $L = (H, W, C, s, \mathcal{W}, \mathcal{S}, \mathcal{E})$ where:

- $H, W \in \mathbb{N}^+$ are the height and width of the grid.
- $P = \{(x, y) \mid 0 \leq x < W, 0 \leq y < H\}$ is the set of all grid positions, with x the column index and y the row index (image-coordinates convention).
- $C = \{0, 1, \dots, n_a - 1\}$ is the set of agent colours, with $1 \leq n_a \leq 4$.
- $s : C \rightarrow P$ assigns an initial position to each agent, and s is injective.
- $D = \{N, S, E, W\}$ is the set of laser directions.
- $\mathcal{W} \subseteq P$ is the set of wall positions.
- $\mathcal{S} \subseteq C \times D \times P$ is the set of laser sources. A source $(c, d, p) \in \mathcal{S}$ emits a laser of colour c in direction d from position p .
- $\mathcal{E} \subseteq P$ is the set of exit positions; we assume $|\mathcal{E}| = |C|$ as an input invariant of LLE levels.
- In the instances studied in this thesis, each colour appears in at most one laser source.
- The sets $s(C)$, $\mathcal{W}$, $\mathcal{E}$, and

$$P_{\text{src}} = \{p \in P \mid \exists c \in C, d \in D : (c, d, p) \in \mathcal{S}\}$$

are pairwise disjoint.

We also write

$$C_{\text{src}} = \{c \in C \mid \exists d \in D, p \in P : (c, d, p) \in \mathcal{S}\}$$

for the set of colours that actually have a source.

Definition 3.2 (Active Beams)

Fix a time step t and a joint position map $p_t : C \rightarrow P$.

For a source $(c, d, p_s) \in \mathcal{S}$, consider the grid ray starting at p_s and extending in direction d .

- Under the **standard beam semantics**, the beam of source (c, d, p_s) is active on every cell of that ray up to, but not including, the first wall or the cell occupied by the unique agent of colour c (if any such cell lies on the ray).
- Under the **strict beam semantics**, the beam of source (c, d, p_s) is active on every cell of that ray up to, but not including, the first wall.

For $c \in C_{\text{src}}$, a laser of colour c is active at a position when the beam of the unique source of colour c is active there.

Definition 3.3 (Valid Joint Trajectory)

A **horizon** $T_{\text{max}} \in \mathbb{N}$ is the maximum number of joint moves allowed in a candidate solution.

A **joint trajectory** of length T_{max} is a sequence of joint positions $\sigma = (p_0, p_1, \dots, p_{T_{\text{max}}})$ where each $p_t : C \rightarrow P$ assigns a position to every agent at time step t .

A joint trajectory is **valid** if it satisfies the following conditions:

- **Initialization:** $p_0(c) = s(c)$ for all $c \in C$.
- **Movement:** For all $t < T_{\text{max}}$ and all $c \in C$, $p_{t+1}(c)$ is reachable from $p_t(c)$ in one step: the agent may stay in place or move to a 4-neighbouring cell. Walls and laser-source cells are the only blocked destinations; exit tiles and cells crossed by the agent's own-colour laser are admissible.
- **No collision:** $p_{t(c_1)} \neq p_{t(c_2)}$ for all t and all distinct $c_1, c_2 \in C$.
- **Conservative hand-off rule:** For all $t < T_{\text{max}}$ and all distinct $c_1, c_2 \in C$, agents may not enter a cell occupied by another agent at the previous time step; that is, $p_{t+1}(c_1) \neq p_{t(c_2)}$ and $p_{t(c_1)} \neq p_{t+1}(c_2)$. This follows the LLE engine convention: agents act simultaneously without coordinating their moves, so an agent cannot enter a cell that another agent only vacates in the same step — it must wait one step. The same rule rules out direct swaps.
- **Laser safety:** No agent c_1 occupies a cell at time t where a laser of colour $c_2 \in C_{\text{src}}$, with $c_2 \neq c_1$, is active under the standard beam semantics of Definition 3.2. The restriction $c_2 \neq c_1$ encodes **immunity**: an agent is unaffected by — and may stand inside — a laser of its own colour.
- **Stay on exit:** If $p_{t(c)} \in \mathcal{E}$ for some $t < T_{\text{max}}$, then $p_{t+1}(c) = p_{t(c)}$. This follows the LLE engine convention: once an agent enters an exit tile, it remains there for the rest of the episode.

Definition 3.4 (Solvability)

A level L is **solvable** with horizon T_{max} if there exists a valid joint trajectory σ of length T_{max} such that the set of occupied positions at time T_{max} is exactly the set of exits:

$$\{p_{T_{\text{max}}}(c) \mid c \in C\} = \mathcal{E}$$

A level is **solvable** without qualification if it is solvable for some $T_{\text{max}} \in \mathbb{N}$.

The restriction to a bounded horizon is natural for the SAT encoding. In the model studied here, beam activity at time t is a deterministic function of the joint position map p_t , so the full game state is determined by the joint agent positions. Any valid trajectory that visits the same joint position map twice can have the intervening segment excised without affecting reachability of later states; the stay-on-exit rule does not break this argument, since any agent on an exit at the start of the loop must remain on that same exit throughout the loop. Hence, if a level is solvable at all, it admits a valid trajectory whose length is at most the number of collision-free joint position maps, which is bounded by $|P|^{n_a}$. A sufficient $T_{\max}$ therefore always exists. In practice $T_{\max}$ is chosen by the generation process, and all formal guarantees in this thesis are stated relative to that choice.

Definition 3.5 (Bounded-Horizon LLE Solvability Problem)

The **bounded-horizon LLE solvability problem** is the following decision problem.

- **Input:** an LLE level L and a horizon $T_{\max} \in \mathbb{N}$.
- **Question:** does L admit a valid joint trajectory of length $T_{\max}$ whose final occupied positions are exactly the exits?

Definition 3.6 (Strict Beam Semantics)

A **strict trajectory** of length $T_{\max}$ is defined exactly as in Definition 3.3, except that beam activity is computed using the strict beam semantics of Definition 3.2: same-colour occupancy no longer truncates the corresponding beam.

In the model studied here, the relevant cooperative action is precisely this same-colour beam-truncation mechanism: an agent occupies a position that would otherwise allow its own beam to continue, thereby opening a path for another agent.

Definition 3.7 (Cooperation Requirement with Horizon $T_{\max}$)

A level L is said to **require cooperation** with horizon $T_{\max}$ if:

- L is solvable under the standard semantics.
- L admits no strict trajectory of length $T_{\max}$ whose final positions occupy all exit tiles.

When the horizon is clear from context, we simply say that L requires cooperation. This bounded form is the one used throughout the rest of the methods chapter, since both the SAT solver and the cooperation detector operate at a fixed finite horizon.

Chapter 4

SAT-based Solver

4.1 Notation

We reuse the level objects introduced in Section 3.3: the grid dimensions H and W , the position set P , the colour set C , the direction set D , the wall set $\mathcal{W}$, the source set $\mathcal{S}$, the exit set $\mathcal{E}$, and the start map s .

The SAT reduction introduces a finite horizon $T_{\max} \in \mathbb{N}$, which is the maximum number of joint moves allowed in the bounded decision problem, together with the discrete time-step set $T = \{0, 1, \dots, T_{\max}\}$.

The sets P_{src} and C_{src} are as defined in Section 3.3. We recall that each colour appears in at most one laser source; under this assumption, when a source of colour c exists, its position and direction are uniquely determined by c . We write $s(c)$ for the initial position of agent $c \in C$, as defined in Definition 3.1.

4.2 Propositional Variables

We introduce three families of propositional variables.

- $a_{c,x,y,t}$: true iff agent $c \in C$ occupies position $(x, y) \in P$ at time step $t \in T$.
- $b_{c,d,x,y,t}$: true iff the beam emitted by the source $(c, d, p_s) \in \mathcal{S}$ is active at position $(x, y) \in P$ at time $t \in T$.
- $l_{c,x,y,t}$: true iff a laser of colour $c \in C_{\text{src}}$ is active at position $(x, y) \in P$ at time $t \in T$.

4.3 Constraints and Logical Encoding

We now formalise the constraint families used in the reduction. Each group of clauses corresponds to one logical component of the bounded-horizon decision problem.

1. Initialization

- **Agents:**

Each agent $c \in C$ is placed at its designated starting position $s(c)$ at $t = 0$; all other positions are unoccupied by that agent:

$$\bigwedge_{c \in C} \bigwedge_{(x,y) \in P} \begin{cases} a_{c,x,y,0} & \text{if } (x,y) = s(c) \\ \neg a_{c,x,y,0} & \text{otherwise} \end{cases}$$

- **Laser sources:**

For each laser source $(c, d, (x_s, y_s)) \in \mathcal{S}$, the beam is always active at its origin, at every time step:

$$\bigwedge_{(c,d,(x_s,y_s)) \in \mathcal{S}} \bigwedge_{t \in T} b_{c,d,x_s,y_s,t}$$

2. Agent Movements

We define the set of positions reachable from (x, y) in one step as (x, y) itself together with its four grid neighbours, excluding walls and laser-source positions:

$$\text{next}(x, y) = \{(x', y') \in \{(x, y), (x, y-1), (x+1, y), (x, y+1), (x-1, y)\} \mid (x', y') \in P, (x', y') \notin \mathcal{W}, (x', y') \notin P_{\text{src}}\}$$

- **Forward consistency:**

If agent c is at position (x, y) at time t , it must be at some position in $\text{next}(x, y)$ at time $t+1$:

$$\begin{aligned} \bigwedge_{c \in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y) \in P} a_{c,x,y,t} &\rightarrow \bigvee_{(x',y') \in \text{next}(x,y)} a_{c,x',y',t+1} \\ &\Leftrightarrow \\ \bigwedge_{c \in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y) \in P} \neg a_{c,x,y,t} &\vee \bigvee_{(x',y') \in \text{next}(x,y)} a_{c,x',y',t+1} \end{aligned}$$

Two formulations are proposed to enforce that each agent occupies at most one position at each time step. The **global** formulation uses a single grid-wide uniqueness rule; the **local** formulation uses a sparser uniqueness rule restricted to $\text{next}(x, y)$ and pairs it with a backward-consistency clause that pins occupancy at $t+1$ to a predecessor in $\text{next}(x, y)$.

- **Global uniqueness:**

No two distinct positions can both be occupied by agent c at time t . The clauses at $t=0$ are already implied by initialization, so the family ranges over $t=1, \dots, T_{\max}$:

$$\bigwedge_{c \in C} \bigwedge_{t=1}^{T_{\max}} \bigwedge_{(x_1,y_1) \in P} \bigwedge_{\substack{(x_2,y_2) \in P, \\ (x_2,y_2) \neq (x_1,y_1)}} \neg a_{c,x_1,y_1,t} \vee \neg a_{c,x_2,y_2,t}$$

- **Local uniqueness:**

No two distinct positions in $\text{next}(x, y)$ can simultaneously be occupied by agent c at time $t+1$:

$$\bigwedge_{c \in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y) \in P} \bigwedge_{(x',y') \in \text{next}(x,y)} \bigwedge_{\substack{(x'',y'') \in \text{next}(x,y), \\ (x'',y'') \neq (x',y')}} \neg a_{c,x',y',t+1} \vee \neg a_{c,x'',y'',t+1}$$

- **Backward consistency (local formulation only):**

If agent c is at position (x, y) at time $t+1$, it must have been at some position in $\text{next}(x, y)$ at time t :

$$\begin{aligned}
& \bigwedge_{c \in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y) \in P} a_{c,x,y,t+1} \rightarrow \bigvee_{(x',y') \in \text{next}(x,y)} a_{c,x',y',t} \\
& \quad \Leftrightarrow \\
& \bigwedge_{c \in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y) \in P} \neg a_{c,x,y,t+1} \vee \bigvee_{(x',y') \in \text{next}(x,y)} a_{c,x',y',t}
\end{aligned}$$

- **No simultaneous occupation:**

Two distinct agents cannot share the same position at the same time, nor can they move to a position already occupied by the other agent:

$$\begin{aligned}
& \bigwedge_{c_1 \in C} \bigwedge_{c_2 \in C, c_2 \neq c_1} \bigwedge_{(x,y) \in P} \bigwedge_{t \in T} \neg a_{c_1,x,y,t} \vee \neg a_{c_2,x,y,t} \\
& \bigwedge_{c_1 \in C} \bigwedge_{c_2 \in C, c_2 \neq c_1} \bigwedge_{(x,y) \in P} \bigwedge_{t=0}^{T_{\max}-1} \neg a_{c_1,x,y,t+1} \vee \neg a_{c_2,x,y,t} \\
& \bigwedge_{c_1 \in C} \bigwedge_{c_2 \in C, c_2 \neq c_1} \bigwedge_{(x,y) \in P} \bigwedge_{t=0}^{T_{\max}-1} \neg a_{c_1,x,y,t} \vee \neg a_{c_2,x,y,t+1}
\end{aligned}$$

- **Victory condition:**

Each exit must be occupied by at least one agent at time $T_{\max}$:

$$\bigwedge_{(x,y) \in \mathcal{E}} \bigvee_{c \in C} a_{c,x,y,T_{\max}}$$

Combined with the no-collision clauses below and the equality $|\mathcal{E}| = n_a$, this clause family yields a bijection between agents and exits at time $T_{\max}$: the disjunctions force a surjection from C onto $\mathcal{E}$, and a surjection between two finite sets of equal size is automatically a bijection.

- **Stay on exit:**

Once an agent reaches an exit, it remains there for all subsequent time steps:

$$\bigwedge_{c \in C} \bigwedge_{(x,y) \in \mathcal{E}} \bigwedge_{t=0}^{T_{\max}-1} \neg a_{c,x,y,t} \vee a_{c,x,y,t+1}$$

3. Laser Activity

We define $\text{next}_{d(x,y)}$ as the position immediately adjacent to (x, y) in direction d :

$$\text{next}_{d(x,y)} = \begin{cases} (x, y - 1) & \text{if } d = N \\ (x + 1, y) & \text{if } d = E \\ (x, y + 1) & \text{if } d = S \\ (x - 1, y) & \text{if } d = W \end{cases}$$

- **Walls block beams:**

A beam cannot be active at a wall position:

$$\bigwedge_{(c,d,p_s) \in \mathcal{S}} \bigwedge_{(x,y) \in \mathcal{W}} \bigwedge_{t \in T} \neg b_{c,d,x,y,t}$$

- **Beam propagation:**

Beam propagation clauses are instantiated only when the successor cell $(x', y') = \text{next}_{d(x,y)}$ lies inside the grid, is not a wall, and is not itself a source cell. Source cells are handled separately by the initialization clauses above. Under these conditions, the beam is active at (x', y') iff it is active at (x, y) and no agent of colour c occupies (x', y') :

$$\begin{aligned}
& \bigwedge_{(c,d,p_s) \in \mathcal{S}} \bigwedge_{t \in T} \bigwedge_{\substack{(x,y) \in P \setminus \mathcal{W}, \\ \text{next}_{d(x,y)} \in P \setminus \mathcal{W}, \\ \text{next}_{d(x,y)} \notin P_{\text{src}}}} b_{c,d,x',y',t} \leftrightarrow (b_{c,d,x,y,t} \wedge \neg a_{c,x',y',t}) \\
& \quad \Updownarrow \\
& \bigwedge_{(c,d,p_s) \in \mathcal{S}} \bigwedge_{t \in T} \bigwedge_{\substack{(x,y) \in P \setminus \mathcal{W}, \\ \text{next}_{d(x,y)} \in P \setminus \mathcal{W}, \\ \text{next}_{d(x,y)} \notin P_{\text{src}}}} \neg b_{c,d,x,y,t} \vee a_{c,x',y',t} \vee b_{c,d,x',y',t} \\
& \quad \bigwedge_{(c,d,p_s) \in \mathcal{S}} \bigwedge_{t \in T} \bigwedge_{\substack{(x,y) \in P \setminus \mathcal{W}, \\ \text{next}_{d(x,y)} \in P \setminus \mathcal{W}, \\ \text{next}_{d(x,y)} \notin P_{\text{src}}}} b_{c,d,x,y,t} \vee \neg b_{c,d,x',y',t} \\
& \quad \bigwedge_{(c,d,p_s) \in \mathcal{S}} \bigwedge_{t \in T} \bigwedge_{\substack{(x,y) \in P \setminus \mathcal{W}, \\ \text{next}_{d(x,y)} \in P \setminus \mathcal{W}, \\ \text{next}_{d(x,y)} \notin P_{\text{src}}}} \neg a_{c,x',y',t} \vee \neg b_{c,d,x',y',t}
\end{aligned}$$

- **Link between beam and laser variables:**

Since each colour has at most one source in the instances considered here, the laser variable $l_{c,x,y,t}$ is true at a position iff the beam of the unique source of colour c is active there:

$$\begin{aligned}
& \bigwedge_{(c,d,p_s) \in \mathcal{S}} \bigwedge_{(x,y) \in P} \bigwedge_{t \in T} b_{c,d,x,y,t} \leftrightarrow l_{c,x,y,t} \\
& \quad \Updownarrow \\
& \bigwedge_{(c,d,p_s) \in \mathcal{S}} \bigwedge_{(x,y) \in P} \bigwedge_{t \in T} (\neg b_{c,d,x,y,t} \vee l_{c,x,y,t}) \wedge (\neg l_{c,x,y,t} \vee b_{c,d,x,y,t})
\end{aligned}$$

- **Agents cannot step on active lasers:**

An agent of colour c_1 cannot occupy a position where an active laser of some source colour $c_2 \in C_{\text{src}}$, with $c_2 \neq c_1$, is present. Agents are immune only to lasers of their own colour:

$$\begin{aligned}
& \bigwedge_{c_1 \in C} \bigwedge_{\substack{c_2 \in C_{\text{src}}, \\ c_2 \neq c_1}} \bigwedge_{(x,y) \in P} \bigwedge_{t \in T} l_{c_2,x,y,t} \rightarrow \neg a_{c_1,x,y,t} \\
& \quad \Updownarrow \\
& \bigwedge_{c_1 \in C} \bigwedge_{\substack{c_2 \in C_{\text{src}}, \\ c_2 \neq c_1}} \bigwedge_{(x,y) \in P} \bigwedge_{t \in T} \neg l_{c_2,x,y,t} \vee \neg a_{c_1,x,y,t}
\end{aligned}$$

4.4 Clause Complexity and Polynomial Size

To show that the reduction has polynomial size, it is enough to bound the number of generated clauses as a function of the input parameters. Let

$$\begin{aligned} n &= |C|, \\ p &= |P| = HW, \\ s &= |\mathcal{S}|, \\ e &= |\mathcal{E}|, \\ \tau &= T_{\max} + 1 \end{aligned}$$

and write

$$V = P \setminus (\mathcal{W} \cup P_{\text{src}})$$

for the walkable cells.

We count **clauses** rather than **literals**. A CNF formula is a conjunction of clauses, each clause being a disjunction of literals. The total literal count is at most a constant factor larger than the clause count in our encoding, since every clause family generated below contains at most $\max(|\text{next}(u)| + 1, n_a, 4) \leq 6$ literals per clause. Counting clauses is therefore sufficient to establish a polynomial bound on the formula size, and the same bound transfers to literals up to a constant factor. The main families admit the following clause bounds.

- **Initialization.** The agent-initialisation clauses contribute exactly np unit clauses, and the laser-source initialisation contributes exactly $s\tau$ unit clauses.
- **Movement constraints.** Forward consistency contributes at most $n(\tau - 1)p$ clauses. The two uniqueness formulations differ here:

$$\text{global uniqueness} = n(\tau - 1) \binom{p}{2}$$

$$\text{local uniqueness} = n(\tau - 1) \sum_{u \in V} \binom{|\text{next}(u)|}{2}$$

Since $|\text{next}(u)| \leq 5$, the local uniqueness term is bounded by

$$n(\tau - 1)10 |V| \leq 10n(\tau - 1)p$$

and backward consistency contributes at most $n(\tau - 1)p$ further clauses. The collision and conservative hand-off clauses contribute at most

$$\binom{n}{2}(\tau + 2(\tau - 1))p = \binom{n}{2}(3\tau - 2)p$$

The exit condition contributes exactly e clauses, and the stay-on-exit condition contributes exactly $ne(\tau - 1)$ clauses.

- **Laser constraints.** The wall-blocking clauses contribute exactly $s |\mathcal{W}| \tau$ unit clauses (one per source-wall-time triple). Beam propagation contributes at most $3s\tau p$ clauses, since each admissible propagation edge yields the three CNF clauses of the iff above; wall successors are handled separately by the wall-blocking family. The beam/laser linking clauses contribute exactly $2s\tau p$ clauses. The laser-safety clauses contribute at most $ns\tau p$ clauses.

Therefore the total number of clauses is polynomial in n , p , s , and τ . More precisely, with the global formulation the dominant term is

$$O(n\tau p^2 + n^2\tau p + ns\tau p)$$

while with the local formulation it is

$$O(n\tau p + n^2\tau p + ns\tau p)$$

The same loops introduce $O((n + s)p\tau)$ propositional variables (agent, beam, and laser indices), also polynomial in the input parameters. The prose above quantifies agent-position predicates over the full position set P ; the implementation may safely restrict these to the walkable subset V , since clauses generated at walls and laser-source cells are vacuous under the initialization and the next filtering. Beam variables at wall cells are forced false by the wall-blocking family and play no role in any satisfying assignment; the implementation may skip them without affecting correctness.

Since each clause is generated by a simple bounded computation inside these loops, the reduction itself is computable in polynomial time as well. This justifies the claim that bounded-horizon LLE solvability is polynomial-time reducible to SAT.

4.5 Correctness of the Reduction

Proposition 4.1 (Correctness of the SAT Reduction)

Let L be an LLE level and let $T_{\max}$ be a horizon. For either movement formulation described above, the CNF formula $\Phi(L, T_{\max})$ is satisfiable if and only if there exists a valid joint trajectory of length $T_{\max}$ for L .

Proof.

For soundness, assume $\Phi(L, T_{\max})$ is satisfiable. Initialization fixes exactly one start position for each agent at time 0. For the global formulation, forward consistency together with pairwise exclusion ensures by induction on time that each agent occupies exactly one legal position at every later step. For the local formulation, the same conclusion follows from forward consistency, local exclusivity, and backward consistency. We may therefore define a joint trajectory $p_{t(c)}$ by reading off the unique true agent variable for each colour c and time t . The movement clauses enforce legal motion between consecutive steps; the collision clauses enforce both simultaneous separation and the conservative hand-off rule; the laser clauses enforce safety with respect to active beams; and the exit clauses enforce the terminal condition. Hence the extracted trajectory is valid.

For completeness, assume a valid joint trajectory of length $T_{\max}$ is given. Set each $a_{c,x,y,t}$ according to whether agent c occupies (x, y) at time t in the trajectory. Set beam variables $b_{c,d,x,y,t}$ and laser variables $l_{c,x,y,t}$ according to the deterministic beam dynamics induced by the same agent positions. Every clause family is then satisfied by construction: initialization matches the start state; the movement, uniqueness, and collision clauses match the trajectory semantics; the propagation iff and beam-laser link clauses hold because b and l are set exactly to those deterministic values; the agent-laser blocking clauses hold because trajectory validity already

forbids any agent from standing on an active laser of a different colour; and the final positions occupy all exits. Therefore $\Phi(L, T_{\max})$ is satisfiable. $\square$

4.6 Complexity-Theoretic Consequences

We can now state the consequence for the decision problem introduced in Definition 3.5.

The bounded-horizon LLE solvability problem lies in **NP**. A candidate trajectory can be verified in polynomial time by simulating the joint execution and checking that all agents occupy the exit tiles at the end without violating the movement, collision, and laser constraints defined in Section 3.3.

Combined with the polynomial-time construction above and Proposition 4.1, this shows that bounded-horizon LLE solvability is polynomial-time many-one reducible to SAT:

$$\text{LLE-Solvability} \leq_p \text{SAT}$$

Thus bounded-horizon LLE solvability is **at most as hard as SAT**: any SAT algorithm yields an algorithm for this decision problem with only polynomial overhead from the reduction.

Whether bounded-horizon LLE solvability is also **NP-hard** remains open in the present work. Establishing NP-hardness would require a polynomial-time reduction in the opposite direction, from a known NP-hard problem to LLE solvability. This thesis does not claim such a result.

It is also important to distinguish proved statements from standard complexity-theoretic beliefs. The question whether $P = NP$ remains open. Accordingly, statements here about worst-case difficulty should be read only through the formal claims we have established: SAT is NP-complete, bounded-horizon LLE solvability belongs to NP, and the reduction above places bounded-horizon LLE solvability within the polynomial-time many-one image of SAT.

In practice, this positioning explains why a SAT-based approach is attractive: the solver inherits the strong empirical performance of modern CDCL SAT solvers on many structured instances, even though the worst-case complexity remains exponential.

Chapter 5

Cooperation Detection

5.1 Strict SAT Encoding

Recall from Definition 3.6 that strict beam semantics changes only one aspect of the dynamics: same-colour occupancy no longer truncates the corresponding beam. Same-colour immunity is unchanged.

Accordingly, the strict SAT encoding keeps the standard laser-safety clauses and replaces only the same-colour beam-propagation rule. For every source $(c, d, p_s) \in \mathcal{S}$, every admissible propagation edge from (x, y) to (x', y') — in the sense of Chapter 4: successor inside the grid, not a wall, and not a source cell — and every time step $t \in T$, the strict encoding uses

$$b_{c,d,x',y',t} \leftrightarrow b_{c,d,x,y,t}$$

instead of the standard equivalence

$$b_{c,d,x',y',t} \leftrightarrow (b_{c,d,x,y,t} \wedge \neg a_{c,x',y',t}).$$

Thus the beam continues through agents of the matching colour instead of stopping at them. We denote the resulting CNF formula by $\Phi_{\text{strict}}(L, T_{\max})$ and the corresponding solver by StrictSolver.

5.2 Why This Captures Cooperation

Under the LLE mechanics studied here, the cooperative action of interest is for an agent to occupy a cell that would otherwise allow its own beam to continue, thereby making another agent's path safe. Standard solvability allows this beam-truncation mechanism; strict solvability removes it.

Therefore, if a level is solvable under the standard semantics but unsatisfiable under the strict semantics, every successful standard solution must rely on at least one same-colour beam-truncation step.

5.3 Formal Theorem and Proof

Theorem 5.1 (Cooperation Detection Criterion)

Let L be an LLE level and $T_{\max}$ a time horizon. Then L requires cooperation with horizon $T_{\max}$ if and only if $\Phi(L, T_{\max})$ is satisfiable and $\Phi_{\text{strict}}(L, T_{\max})$ is unsatisfiable.

Proof.

($\rightarrow$) Assume that L requires cooperation with horizon $T_{\max}$. By Definition 3.7, L is solvable under the standard semantics, so $\Phi(L, T_{\max})$ is satisfiable. Suppose for contradiction that $\Phi_{\text{strict}}(L, T_{\max})$ is also satisfiable. Then there exists a strict trajectory whose final positions occupy all exit tiles. Since strict beam semantics differs from the standard one only by removing same-colour beam truncation, such a trajectory is also a valid standard trajectory that succeeds without using that mechanism. This contradicts Definition 3.7. Therefore $\Phi_{\text{strict}}(L, T_{\max})$ is unsatisfiable.

($\leftarrow$) Assume that $\Phi(L, T_{\max})$ is satisfiable and $\Phi_{\text{strict}}(L, T_{\max})$ is unsatisfiable. The first condition implies that L is solvable under the standard semantics. Suppose, for contradiction, that some successful standard trajectory uses no same-colour beam-truncation step — that is, no agent c ever stands on a cell of the unblocked beam path of the source of colour c . Under that hypothesis the two semantics produce identical beam states for this trajectory: the standard rule truncates the beam of colour c only at a cell occupied by agent c , and by assumption no such occupancy occurs, so each beam reaches the same wall or grid boundary as it would under the strict rule. The same variable assignment therefore satisfies $\Phi_{\text{strict}}(L, T_{\max})$, contradicting unsatisfiability. Hence every successful standard trajectory must use at least one same-colour beam-truncation step, so L requires cooperation with horizon $T_{\max}$. $\square$

5.4 Horizon-Dependence of the Cooperation Criterion

Theorem 5.1 binds cooperation to a fixed horizon $T_{\max}$. This is not an artefact of the SAT encoding: it is the only sense in which cooperation is decidable in this framework. Solvability itself is a horizon-indexed property (Definition 3.4), and so is the companion cooperation requirement (Definition 3.7). The horizon is therefore a true parameter of the cooperation label, and the same level can switch classification when $T_{\max}$ changes.

The mechanism behind this sensitivity is straightforward. The strict SAT encoding refuses same-colour beam truncation but leaves every other movement and timing constraint identical to the standard one. Hence the strict solver can find a satisfying trajectory simply because there exists a path long enough to walk **around** the laser geometry rather than **through** it, even though the short, natural solution would step into the beam and shield another agent. As soon as the horizon $T_{\max}$ admits such a detour, the strict formula becomes satisfiable and the level stops being labelled cooperative — regardless of what the natural solution does.

Example: detour bypass. Figure 2 shows a 4×4 grid that makes this concrete. Two agents start in the top row; their only exits sit in the bottom row, and a single red laser covers two cells of row 1. Running the cooperation analyzer on this level at three different horizons reproduces all three failure modes at once (Table 1).

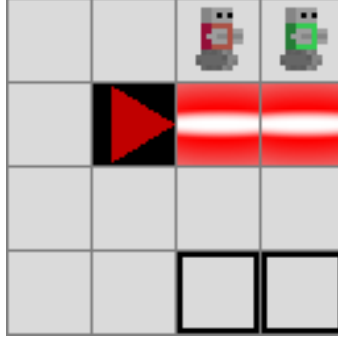


Figure 2: Horizon-dependence demo level. Two agents in the top row; a red laser spans two cells of row 1; two exits in the bottom row.

$T_{\max}$	Solver result (standard / strict)	Profile label
2	UNSAT	— (rejected as unsolvable)
3	SAT, strict UNSAT	asymmetric, edges $\{(0, 1)\}$
9	SAT, strict SAT	independent

Table 1: Cooperation classification of the level in Figure 2 at three horizons. With $T_{\max} = 2$ no agent has time to reach an exit, so the standard solver returns UNSAT and cooperation is never tested. With $T_{\max} = 3$ — the tightest horizon at which the level is solvable — cooperation is required: the only feasible plan uses red same-colour truncation. With $T_{\max} = 9$ — the smallest horizon admitting a detour around the beam — the strict solver finds a trajectory in which the blue agent walks around the beam, so cooperation is no longer flagged.

The same level therefore receives three qualitatively different labels depending on the chosen horizon — **unsolvable**, **cooperative**, or **non-cooperative** — without the level itself changing. The phenomenon generalises: any level whose natural short solution uses beam truncation can be made to look non-cooperative by raising $T_{\max}$ until a geometric detour fits, and can be made to look unsolvable by lowering $T_{\max}$ until even the natural solution does not.

Practical consequence. The criterion is **sound** but **horizon-sensitive**. Choosing $T_{\max}$ too generously under-detects cooperation by allowing geometric detours; choosing it too tightly makes the standard solver itself UNSAT, so the level is rejected as unsolvable before cooperation is even tested. Both failure modes shift the operating point in opposite directions, and neither contradicts Theorem 5.1, which states cooperation strictly **relative to** a chosen horizon.

The recipe used throughout this thesis is to pick $T_{\max}$ as the smallest horizon at which a representative short solution of the geometry is expected to fit, with a small additive slack. For the parameter configurations of Chapter 7 this means horizons proportional to the grid diameter (concrete values are listed alongside each experimental configuration). Under that choice, the criterion is tight enough that accepted levels reliably exhibit the intended beam-truncation behaviour at their stated horizon, and loose enough that the underlying standard solver does not spuriously reject solvable instances.

5.5 Practical Algorithm

The cooperation detector runs two SAT calls on the same level:

1. Run $\text{Solver}(L, T_{\max})$. If the result is UNSAT, the level is unsolvable for that horizon, so it is rejected before cooperation is considered.
2. Otherwise, run $\text{StrictSolver}(L, T_{\max})$. If this call is UNSAT, the level requires cooperation for the same horizon; if it is SAT, a strict trajectory exists, so the level is solvable without any same-colour beam-truncation step and is therefore non-cooperative.

Both calls share the same bounded horizon and differ only in the beam-propagation clauses. For benchmark levels, the horizon can be chosen from known solution lengths; for generated levels, it is the user-supplied generation parameter $T_{\max}$.

5.6 Cooperation Profiles

Theorem 5.1 yields a **binary** answer: a level either requires cooperation or it does not. Once the binary criterion holds, however, several qualitatively different cooperation structures fall under that single label, and the generators of Chapter 6 rely on this finer distinction to target specific patterns. The cooperation profile analyzer operates on top of the binary detector and produces a richer classification by combining three pieces of derived data:

1. a **helper-event set** extracted from one satisfying assignment of $\Phi(L, T_{\max})$;
2. a **necessary-helper set** identified by colour-wise counterfactual SAT calls; and
3. a **dependency graph** between agents, built from the helper events.

The decision procedure produces one of eight labels:

- **unsolvable**: $\Phi(L, T_{\max})$ is UNSAT.
- **independent**: cooperation is not required.
- **cooperative**: cooperation is required, but the extracted plan exhibits no observable helper event (a fallback used when the dependency graph happens to be empty).
- **asymmetric**: at least one one-way helping relation exists, with none of the richer structures below.
- **mutual**: two agents help each other.
- **chain**: helper events form a directed path with no branching.
- **distributed**: at least one agent benefits from two or more distinct helpers.
- **fully coupled**: every agent belongs to a single strongly connected component of the dependency graph.

Every label is constructible within LLE’s $1 \leq n_a \leq 4$ regime: `mutual` and `fully_coupled` need at least two agents; `chain` and `distributed` need at least three; the remaining labels are reachable from two agents upward. The taxonomy therefore does not require any structural configuration that LLE cannot host.

The remainder of this section makes each underlying object precise and gives one geometric example per family.

5.6.1 Selective-Strict Semantics

Some profile decisions require asking whether a single agent is **individually** indispensable as a helper. To support this, the SAT encoding offers a **selective-strict** laser mode, parameterised by a set of colours $K \subseteq C_{\text{src}}$ — chosen distinct from the source-set symbol $\mathcal{S}$ to avoid confusion:

- for every source $(c, d, p_s) \in \mathcal{S}$ with $c \in K$, the beam-propagation clauses use the strict equivalence of Chapter 4;

- for every source $(c, d, p_s) \in \mathcal{S}$ with $c \notin K$, the standard equivalence is used.

When $K = \emptyset$, the encoding coincides with the standard semantics; when $K = C_{\text{src}}$, it coincides with the strict semantics of Definition 3.6. Intermediate choices forbid same-colour truncation for the colours in K while leaving the remaining beams untouched. Selective-strict is the SAT lever that lets us single out one helper at a time without affecting the other beams.

The same argument used in the proof of Theorem 5.1 generalises to this intermediate encoding — the selective-strict clauses for the colours in K are exactly the strict clauses of Chapter 4 restricted to those colours, while every other colour keeps its standard clauses, so the model-transfer construction of the theorem applies colour by colour. Concretely, Φ with the selective-strict encoding parameterised by K is satisfiable if and only if L admits a successful standard trajectory in which no agent of colour $c \in K$ ever stands on the unblocked beam path of the source of colour c . We therefore use the selective-strict encoding as a sound decision oracle for “ L remains solvable when each colour in K is individually barred from truncating its own beam”, which is exactly what the necessary-helper analysis below requires.

5.6.2 Helper Events from a SAT Model

Given a satisfying assignment of $\Phi(L, T_{\text{max}})$, the analyzer recovers the joint trajectory $\sigma = (p_0, \dots, p_{T_{\text{max}}})$. For each time step t and each ordered pair of distinct agents (c, c') , a **helper event** with helper c , beneficiary c' , and time t is recorded if two geometric conditions hold:

1. agent c stands at time t on a cell that lies on the unblocked beam path of the source of colour c , so the beam of colour c is truncated by c at position $p_{t(c)}$; and
2. agent c' stands at time t on a cell strictly downstream of $p_{t(c)}$ on the same beam path, so c' would lie inside the un-truncated beam and is therefore protected by c .

Helper events are an **observable of the chosen joint plan**, not an invariant of the level. A cooperative level typically admits many joint plans, and different plans may yield different helper-event sets.

5.6.3 Necessary Helpers

The **necessary-helper set** is, by contrast, a property of the level itself. For every colour $c \in C_{\text{src}}$ the analyzer runs one selective-strict SAT call with $K = \{c\}$. If the call returns UNSAT, the colour c is added to the set. Operationally, c is necessary when the level cannot be solved as soon as c alone is barred from helping with its own beam, even though every other agent retains that ability.

The necessary-helper set does not depend on which standard model the solver returns: each counterfactual is its own one-shot satisfiability check, independent of the joint plan chosen elsewhere.

5.6.4 Dependency Graph

Helper events induce a directed graph $G_L = (V, E)$ on the set of agents:

$$V = C, \quad E = \{(c, c') \mid \exists t : (c, c', t) \text{ is a helper event}\}.$$

Multiple helper events between the same ordered pair (c, c') — possibly at different time steps and along different beam paths — collapse into a single edge. The time dimension is otherwise discarded; a separate scalar, the **synchronous width**, records the maximum number of distinct helpers active

at the same time step and is exposed alongside the profile label without entering the classification decision.

5.6.5 Profile Families

The profile label is the output of a small decision procedure applied to G_L , with the binary cooperation requirement as a hard precondition. The priority order over the cooperation-required labels is **fully coupled** $\succ$ **mutual** $\succ$ **distributed** $\succ$ **chain** $\succ$ **asymmetric** $\succ$ **cooperative**, so a level matching several criteria is labelled by the strongest one. The two non-cooperative labels (unsolvable and independent) short-circuit before this order is considered. We describe each label below with a short geometric example.

Independent. The binary detector returns non-cooperative: $\Phi_{\text{strict}}(L, T_{\text{max}})$ is satisfiable, so no agent ever has to truncate a beam. The profile analyzer short-circuits and returns independent. **Example.** A grid with two agents whose direct paths to their respective exits do not cross any beam at all (Figure 3).

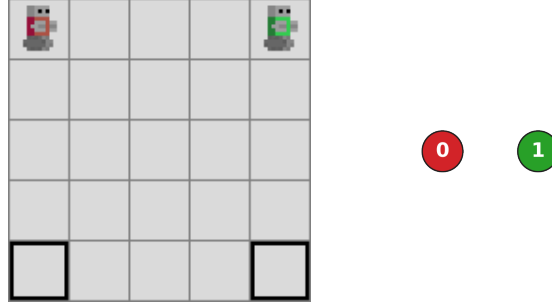


Figure 3: independent: two agents with disjoint direct paths to two exits. No laser is present, so the strict-SAT encoding admits the same trajectory as the standard one. The dependency graph (right) has no edges.

Cooperative (no observed helper). Binary cooperation holds — strict-SAT is UNSAT — yet the extracted plan contains no helper event. This label is a **defensive fallback**: the helper-event extractor only records an event when a helper sits on its own beam path with a foreign-colour agent strictly downstream at the same time step, and a few edge cases (e.g. a helper truncating at the last cell of its own beam path) escape that criterion. We have not observed this fallback fire on any LLE level encountered in this thesis; the label is listed for completeness. The same word cooperative reappears in Chapter 6 as a **filter target** meaning “any level satisfying the binary cooperation criterion, regardless of finer profile”. The two uses are distinct: the classifier returns cooperative only as the fallback above, while the generator filter cooperative accepts levels of any cooperative profile.

Asymmetric. The residual category: the dependency graph has at least one edge $(c, c') \in E$ and matches none of the stronger profiles above. The label is therefore the **fallback** applied whenever cooperation is required but the graph fits neither **fully coupled**, **mutual**, **distributed**, nor **chain**; a typical asymmetric graph is a small set of one-way edges, but sparser shapes that escape the chain criterion (e.g. two short paths that do not span every participating agent) also fall here. **Example.** Two agents of distinct colours; only the red beam blocks the blue agent’s path to its exit, so the red agent must step into its own beam at some moment to let the blue agent pass. The reciprocal situation never arises since there is no blue beam. The dependency graph has the single edge $0 \rightarrow 1$ (Figure 4).

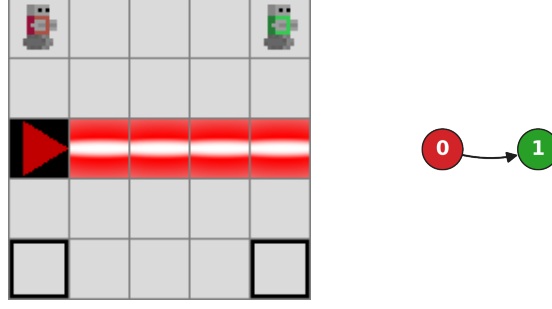


Figure 4: **asymmetric**: one red laser splits the grid horizontally. The red agent (top-left) crosses its own beam unscathed; the blue agent (top-right) requires red to truncate the beam so it can reach the bottom-right exit. Edges: $\{(0, 1)\}$.

Mutual. The dependency graph contains a mutual pair, i.e. both edges $(c, c'), (c', c) \in E$ for some pair $c \neq c'$. **Example.** Two laser sources of distinct colours, each crossing the other agent's path: each agent must truncate its own beam to shield the other at some point in the plan (Figure 5). Both edges are present, and a level is classified as **mutual** even when additional one-way edges between other pairs also exist, as long as the agents do not all belong to a single strongly connected component. Because **fully coupled** takes priority in the decision order, the **mutual** label is only reachable with $|C| \geq 3$ agents: when $|C| = 2$, a reciprocal pair is itself a strongly connected component spanning the whole agent set (size $2 = |C| > 1$), so the level is labelled **fully_coupled** and the **mutual** branch is never reached. The example in Figure 5 therefore includes a third agent on an independent path, outside the reciprocal pair, which keeps the largest strongly connected component at size two while $|C| = 3$.



Figure 5: **mutual**: two stacked beams of distinct colours. Each agent is immune to its own beam but must wait for the other to truncate the foreign beam before crossing. A third agent on an independent path is required to keep the profile distinct from **fully_coupled**: with only the reciprocal pair the two agents would form a strongly connected component spanning the whole agent set. It participates in no helper relationship, hence appears as an isolated vertex in the dependency graph. Edges: $\{(0, 1), (1, 0)\}$.

Chain. The dependency graph is a directed path: every vertex has in-degree and out-degree at most one, the longest chain has length at least two (i.e. at least two consecutive edges), and that longest chain visits every participating vertex. **Example.** Three agents arranged so that agent 0 must shield agent 1 across the red beam, agent 1 must shield agent 2 across the blue beam, and neither agent 2 nor agent 0 has any further helping role (Figure 6). The graph is $0 \rightarrow 1 \rightarrow 2$.

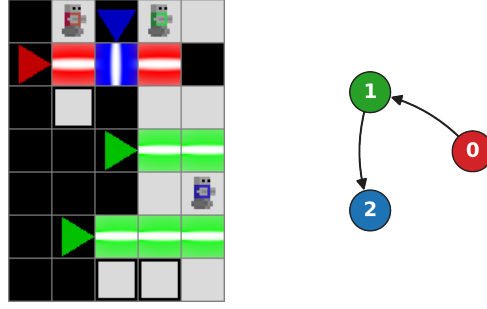


Figure 6: chain: three agents and two beams. Walls confine each agent to a separate region so helping flows in one direction only — red helps blue across the red beam, blue helps the third agent across the blue beam. Edges: $\{(0, 1), (1, 2)\}$.

Distributed. At least one agent has in-degree ≥ 2 in the dependency graph. **Example.** Three agents in which two distinct same-colour helpers (agents 0 and 1) must each truncate their respective beams to free agent 2's path to its exit (Figure 7). The edges are $\{(0, 1), (0, 2), (1, 2)\}$, so agent 2 has in-degree two and the level is distributed. (The extra edge $(0, 1)$ does not promote the level to mutual because the reciprocal edge $(1, 0)$ is absent.)



Figure 7: distributed: agent 2 must traverse both the red and the blue beam to reach its exit, requiring truncation from both other agents. In-degree of agent 2 is two. Edges: $\{(0, 1), (0, 2), (1, 2)\}$.

Fully coupled. The largest strongly connected component of the dependency graph spans the entire agent set: its size equals the number of agents $|C|$ (with $|C| > 1$). **Example.** Three agents in which every pair of agents helps each other, yielding a directed cycle on all three agents and a strongly connected component of size three $= |C|$ (Figure 8). This is the strictest profile and is rare on small grids.

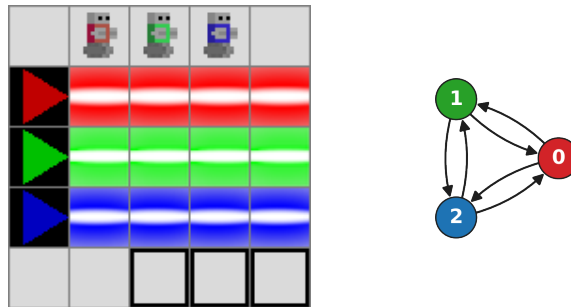


Figure 8: fully_coupled: three agents and three beams stacked, so every agent must shield, and be shielded by, both others to reach the bottom row of exits. The complete dependency graph has all six edges between distinct agents. Edges: $\{(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)\}$.

The analyzer also exposes the auxiliary scalars used by the decision procedure (longest chain length, largest SCC size, synchronous width) so downstream code can re-classify levels under a different policy without re-running any SAT calls.

Scope note. The cooperation notion defined here is intentionally specific: it captures same-colour beam-truncation as the relevant cooperative act. A level that requires two agents to coordinate their movements for unrelated geometric reasons — without any laser blocking being involved — would not be identified as cooperative by this detector. This definition is not claimed to exhaust every possible interpretation of cooperation in multi-agent environments; it is the specific mechanism studied in this benchmark, and the formal guarantee is scoped accordingly.

Taxonomy scope. The eight labels above are designed to be informative **in practice** on small LLE grids, not to partition every directed graph one could imagine. Graphs that combine weaker structures without satisfying any stronger criterion fall into the residual `asymmetric` class. For instance, a 3-cycle among three of four agents (a strict 3-SCC that does not span every agent) is labelled `asymmetric`, not `fully_coupled`, because the SCC is smaller than $|C|$. Such pathological shapes have not appeared in our experiments, but the catch-all behaviour of `asymmetric` is by design rather than an oversight.

Model-dependence of finer-grained analyses. The binary detector above is a property of the **level**: the satisfiability of $\Phi(L, T_{\max})$ and $\Phi_{\text{strict}}(L, T_{\max})$ does not depend on which satisfying assignment the SAT solver happens to return. The necessary-helper set of Section 5.6 is likewise an invariant of the level, being a sequence of one-shot satisfiability checks. The full profile label, however, depends on the helper events extracted from a single standard model, which the SAT solver chooses without any preference among satisfying assignments. A cooperative level typically admits many valid joint plans, and different plans may exhibit different helping patterns: an agent that helps another in one solution may be passive in another. The profile label therefore reflects the structure of the extracted plan, not an intrinsic invariant of the level. This is acceptable for the generation use case considered here — the analyzer still acts as a sound filter that certifies the extracted plan exhibits the targeted profile — but two solver runs on the same level can in principle yield different profile labels.

Chapter 6

Procedural Generators

6.1 Design Pattern

All generators follow a common architecture built around three principles:

1. **SAT as oracle:** the solver is not post-hoc; it is embedded in the generation loop. A candidate level is accepted only if the solver confirms the desired property (solvability, cooperation, or a specific cooperation profile).
2. **Separation of concerns:** level construction and property verification are decoupled. Generators build candidate levels using domain-specific heuristics; the solver decides acceptance.
3. **Extensibility:** every generator shares a common base class and registers itself in a name-keyed registry, so a new generator can be added without modifying core code.

Each generator repeatedly performs the same loop: sample or construct a candidate layout, reject it if it violates generator-specific structural constraints, build the corresponding LLE world instance, and run the appropriate SAT-based acceptance test. A single standard SAT call certifies solvability. When the user additionally requires cooperation, the generator also runs the strict-beam counterfactual test from Chapter 5; when a specific cooperation profile is requested, the profile analyzer of Section 5.6 acts as a soft filter on top. The cooperation requirement and the profile filter are parameters rather than separate generator classes — **what** the generator constructs and **which extra properties** are checked are independent axes.

6.2 Generation Targets

Viewed through the solvability and cooperation definitions of Section 3.3, the generator family targets the three level categories shown in Figure 9. Every accepted level is solvable — categories (b) or (c); unsolvable levels in category (a) are always rejected. When the user additionally requires cooperation, levels in (b) are rejected as well, so only category (c) survives.

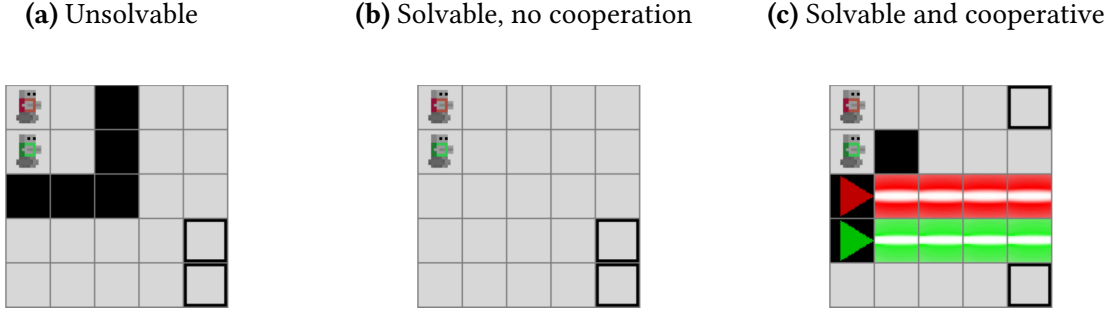


Figure 9: Target level categories for the generator family.

6.3 Cooperation Profile Targeting

Every generator in this chapter certifies **solvability** for every accepted level — that is the role of the SAT oracle and it is always on. On top of that always-on guarantee, the user can stack two additional acceptance requirements drawn from Section 5.6:

- **Cooperation** (optional). The candidate must additionally satisfy the strict-UNSAT counterfactual of Theorem 5.1, so every accepted level provably requires the same-colour beam-truncation mechanism. Without this flag the cooperation profile is not inspected.
- **Profile filter** (optional, requires cooperation). The candidate must additionally match one of the requested cooperation-profile labels (e.g. mutual, chain, distributed). The accepted set is the corresponding sub-class of category (c) above.

Solvability is therefore the floor; cooperation and profile filtering are layered on top. All other generator parameters — grid size, number of agents n_a , number of lasers n_l , wall budget, horizon $T_{\max}$ — are independent knobs and can be combined freely with any choice on the cooperation axis.

6.4 Random Generator

The random generator samples every layout component uniformly: agent start positions, exits, wall positions, and laser source positions are drawn pairwise-distinct from the grid, and each laser source is given a random direction. The resulting candidate is submitted to the SAT oracle under whichever acceptance setting (solvability only, plus cooperation, or plus cooperation with profile filter) was requested. When a lower-bound horizon $T_{\min}$ is provided — a user-supplied difficulty knob, separate from the encoding horizon $T_{\max}$ — the generator also requires the candidate to be unsatisfiable for $T_{\min} - 1$, selecting levels that fall inside a difficulty window.

This generator is deliberately simple. Its main value is methodological: it gives an unbiased sampling baseline against which more structured generators can be compared. Its main weakness is rejection rate. As the grid grows and the number of interacting entities increases, purely random layouts quickly become dominated by unsolvable or trivial instances.

6.5 Constrained Random Generator

A structured variant that biases generation toward well-formed, solvable configurations before any SAT call is made. Relative to the random generator, it rejects candidates that are already geometrically

degenerate — for example a laser that points immediately outside the grid, a laser with zero beam length, a laser on the bottom edge oriented downward, or an exit lying on an unavoidable beam segment.

These degeneracies are not merely cosmetic. A laser whose beam has zero length, or that points off the grid, emits **no active beam**: its source then behaves like an inert wall tile, and the level presents none of the beam-crossing structure the laser was meant to introduce. Crucially, the base random generator still **accepts** such a level whenever it is solvable — a level reduced to plain navigation usually is — so its solvable output is dominated by these laser-free instances. The geometric filters do not themselves prove solvability or cooperation, but by discarding inert-laser and beam-stranded layouts they ensure that accepted levels actually contain the laser interactions the experiments depend on; and in the cooperative setting, where an active beam is a **prerequisite** for the same-colour truncation mechanism, they also avoid a large number of SAT calls that could never have yielded a cooperative level. The generator otherwise offers the same formal guarantees as the random generator (solvable, optionally cooperative with profile filter), and is the default we use when the experiments call for “random sampling” of solvable or cooperative levels. The effect is visible by comparing the unfiltered pools of Section A.14 with the filtered pools of Section A.15 at the same grid, agent, laser, and wall settings.

6.6 Constructive Generator

The constructive generator replaces blind sampling with a partial-by-construction layout. On a grid of H rows and W columns with n_a agents, it picks a random orientation, samples a set of n_a distinct **lane indices** without replacement on the orientation axis (rows for the horizontal orientation, columns for the vertical one), places one agent start at one end of each lane and the corresponding exit at the other end, and reserves every cell of every lane as non-buildable. Lane indices are sampled **without** a contiguity constraint, so the lane band can be split anywhere on the orientation axis — this is the main source of within-pool diversity.

When the generator is run in cooperative mode, the laser-placement step plants a deliberate cooperation dependency for every requested laser, rather than relying on the solver to **discover** one after the fact. For each of the n_l lasers, the generator picks

- a **perpendicular column** (or row, in the vertical orientation) from a pool of distinct values in the interior of the perpendicular axis;
- an axis position strictly before or after the lane band, so the beam crosses the lane geometry;
- a direction perpendicular to the lane axis, pointing toward the lane band.

Each laser is given a **distinct colour** in $\{0, \dots, n_l - 1\}$. We require $n_l \leq n_a$ so that each laser colour corresponds to a unique agent of that colour, in line with the at-most-one-source-per-colour assumption of Definition 3.1. Distinctness of the perpendicular columns guarantees that the resulting beams are parallel straight lines on disjoint columns, so no two laser sources or beam segments coincide.

The full unblocked beam segment of every laser, from source to grid edge, is added to the reserved cell set before walls are placed; without this step a wall could land between two non-adjacent lanes and silently break the cooperation requirement. The remaining free cells are then shuffled uniformly and the first n_w of them are placed as walls, where n_w is the requested wall budget — shuffling rather than taking a row-major prefix is what gives within-pool wall-mask diversity. The finished candidate is verified by the SAT oracle exactly as in the random and constrained random generators.

The distinct-colour multi-laser construction is the key qualitative change relative to a single-structural-laser template. With $n_l = 1$ the geometry yields a single helper-beneficiary pair and the cooperation profile is **asymmetric** by construction. With $n_l = 2$ the two lasers cross each other’s lanes and the profile is **mutual** — every helper is also a beneficiary. With $n_l \geq 3$ on grids large enough to accommodate the construction the profile mix shifts toward **distributed** as one agent benefits from multiple distinct helpers. Example pools in solvable and cooperative settings are shown in Section A.16 and Section A.17; pools generated under an explicit profile filter (mutual, distributed) are shown in Section A.18.

6.7 Level-6-Style Generator

The level-6-style generator is a specialised constructive variant targeting the layout shape of LLE Level 6 (see Section 3.1): clustered agent starts and clustered exits placed on opposing sides of the grid, with a corridor of lasers between them. Agents are placed in a small rectangular cluster (a 2×2 block for four agents) in one third of the grid; exits are placed in a matching cluster in the opposite third. The orientation (vertical with start above exit, or horizontal with start left of exit) is chosen at random per call, and the perpendicular position of each cluster is also randomised. Lasers are then placed inside the corridor between the clusters, oriented perpendicular to the cluster axis so that each beam crosses the corridor; remaining free cells receive walls up to the requested wall budget.

This geometry consistently produces **mutual** cooperation profiles that mirror the structure of Level 6, and it is the generator we recommend for producing training instances intended to transfer to Level 6 (see the curriculum-transfer experiment in Chapter 7). The profile filter from Section 6.3 still applies — the default target is `mutual`, but other profile labels can be requested when the parameter combination supports them. Example pools at three grid sizes (up to the 12×13 Level 6 footprint) are shown in Section A.19.

6.8 Summary

Generator	Construction Bias
Random	Uniform random sampling
Constrained Random	Random + geometric rejection
Constructive	Reserved agent lanes + planted dependency
Level-6-Style	Clustered starts/exits + corridor lasers

Table 2: Generator families and their construction bias before the SAT oracle is consulted.

Chapter 7

Empirical Evaluation

7.1 Evaluation Protocol

The first benchmark implemented in this project isolates the SAT solver rather than the generators. Its goal is to compare the two movement formulations used in the encoding: the **local** formulation, which combines neighbourhood-based exclusivity with backward consistency, and the **global** formulation, which enforces uniqueness by pairwise exclusion over the whole grid.

7.1.1 Metrics

For each run, the benchmarking code records:

- the total number of clauses in the generated CNF;
- the clause count contributed by each major constraint family;
- the CNF generation time;
- the SAT solving time;
- the total time obtained by summing generation and solving.

The implementation also stores per-constraint method profiles for the movement constraint, which allows us to inspect the part of the final CNF generated inside the movement-constraint module.

7.1.2 Solver and Level Sets

All benchmark runs use the same SAT backend as the main solver implementation, namely `Minisat22` through the `PySAT` interface. The benchmarking script can evaluate the six hand-crafted benchmark levels distributed with the environment, each paired with a horizon known to be sufficient for solvability. It can also benchmark custom levels constructed programmatically.

The SAT encoding comparison reported in the following sections uses four representative levels: three synthetic instances of increasing size and one original benchmark level. This combination exposes both scaling behaviour and the behaviour of the solver on a realistic cooperative puzzle.

7.1.3 Protocol

For each level and each movement formulation, the benchmark performs one profiled run to extract the exact clause counts and the full constraint breakdown. It then repeats the same solver invocation for 100 runs, each time on a fresh copy of the world, and reports the mean and standard deviation of generation time and solve time. Using fresh world copies avoids benchmark contamination by mutable environment state.

No timeout or parallel speedup is introduced in this protocol. The measurements should therefore be read as direct comparisons between the two SAT formulations on the same machine, rather than as hardware-independent absolute performance claims.

7.1.4 Outputs

The benchmarking pipeline produces a console summary table, a JSON file containing aggregated benchmark statistics, and a set of plots for clause counts, per-constraint clause breakdowns, and timing statistics. The results reported in the following sections draw on these outputs to interpret the trade-off between the two movement formulations.

The empirical evaluation proceeds in six parts. The first measures the effect of one modelling choice inside the SAT solver — the local versus global formulation of the agent-uniqueness constraint — on four benchmark levels. The second characterises the generator family’s **efficiency** through per-generator rejection rates, and the third its **output diversity** through the cooperation-profile distribution of accepted levels. The fourth asks whether off-the-shelf value-decomposition MARL agents can **learn** cooperative levels emitted by the generator on a small grid, and the fifth shows that enlarging the generated training pool restores generalisation. The sixth turns to curriculum learning: we first isolate the curriculum **mechanism** on a target that direct training can already reach (Section 7.11), then ask whether a curriculum can cross the **mutual**-cooperation boundary up to the hand-crafted LLE Level 6 (Section 7.12) — and find that it cannot, for reasons we trace to the base task rather than to the curriculum itself. Software versions and seed conventions for every experiment in this chapter are summarised in Section A.2.

7.2 Experimental Question

The experiment asks whether the **global** and **local** movement formulations introduced in Chapter 4 differ materially in CNF size and runtime on levels of increasing complexity.

This is an important baseline question because the uniqueness constraint appears at every time step for every agent. A poor formulation therefore affects the full reduction, not only a negligible subpart of it. The protocol itself is defined in Section 7.1; the present chapter reports only the level set, the observed results, and their interpretation.

7.3 Benchmark Instances

Four levels of increasing size and complexity were used:

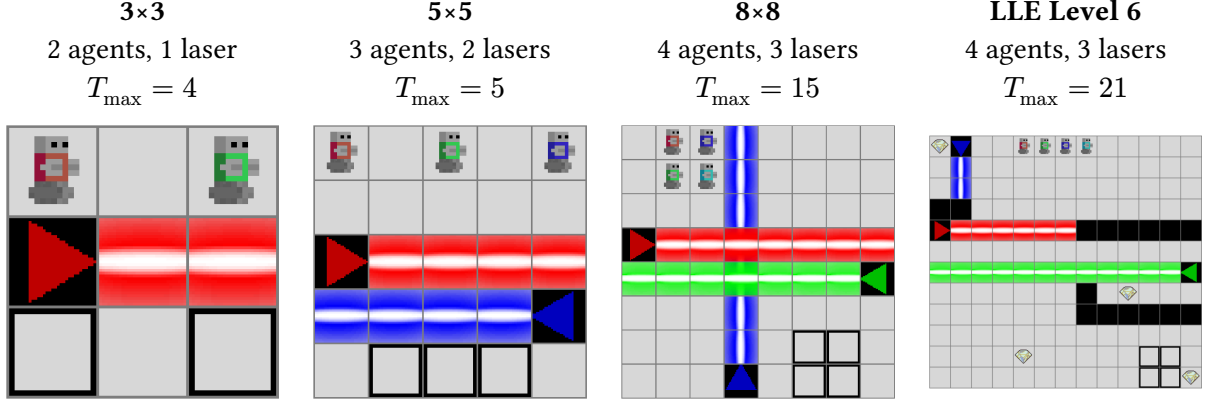


Figure 10: The four test levels used for benchmarking, ordered by increasing complexity. Each level is paired with the horizon $T_{\max}$ used to encode it in the SAT formulation; the horizon is set to the minimum number of joint steps known to be sufficient for solvability of the level.

The first three levels were constructed to expose scaling behaviour as grid size and agent count increase. The fourth is LLE Level 6, one of the hand-crafted benchmark levels distributed with LLE, included to assess solver behaviour on a realistic cooperative instance. Per-level parameters (grid, agent and laser counts, horizon) are tabulated in Section A.1.

7.4 Results

7.4.1 Clause Count

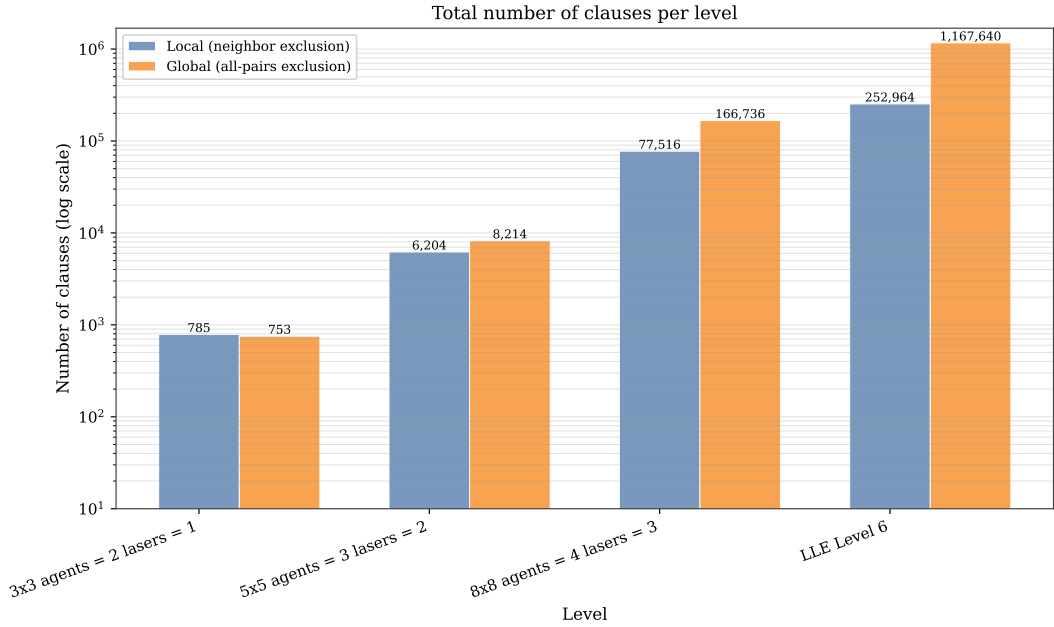


Figure 11: Number of clauses generated by the global and local uniqueness formulations across the four benchmark levels. The vertical axis is logarithmic because clause counts span more than three orders of magnitude.

The clause-count results closely follow the theoretical expectation developed in Chapter 4. On the smallest 3×3 level, the global formulation is slightly more compact than the local one (753 clauses

versus 785), which is consistent with the fact that the instance lies below the expected crossover regime. From the 5×5 level onward, the trend reverses: the local formulation generates 6,204 clauses versus 8,214 for the global one, then 77,516 versus 166,736 on the 8×8 level, and finally 252,964 versus 1,167,640 on LLE Level 6.

This widening gap is the main structural result of the experiment. The local formulation scales much better because its exclusivity clauses are tied to neighbourhood-sized successor sets, while the global formulation introduces pairwise exclusions across the full position set. The per-constraint-family decomposition behind these totals is reported in Section A.4.

7.4.2 Solving Time

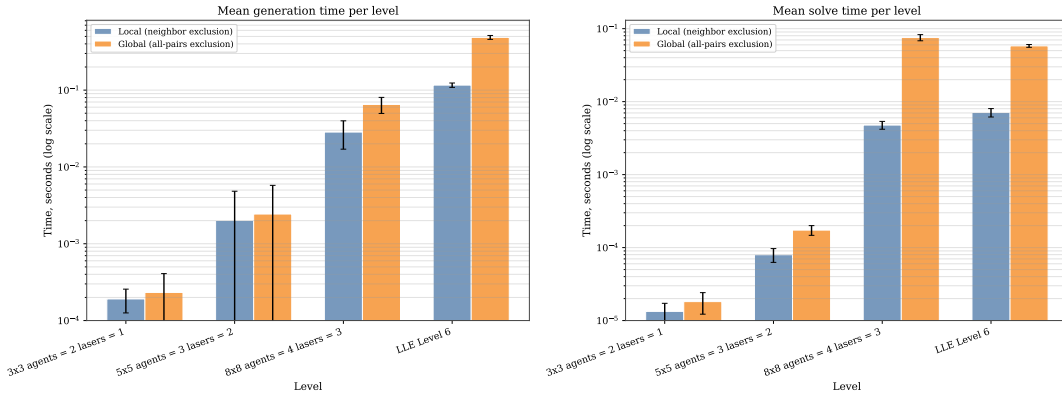


Figure 12: SAT solver time (seconds) for the global and local uniqueness formulations across the four benchmark levels. Both axes are logarithmic because solve times span more than three orders of magnitude across the four levels.

The timing results show the same qualitative pattern. On the smallest synthetic instances, both methods solve essentially instantaneously and the difference is negligible. Once the levels become moderately large, however, the local formulation is consistently faster both to generate and to solve. The contrast is especially visible on the 8×8 level and on LLE Level 6, where the much larger global CNF induces a clear runtime penalty.

One subtlety deserves comment. On LLE Level 6 the global formulation produces about 7 times more clauses than on the synthetic 8×8 instance, yet its mean solve time is **lower**. This is not a contradiction: SAT solver runtime is driven by the search structure, not strictly by formula size. Hand-crafted levels such as Level 6 admit propagation-friendly satisfying assignments that modern CDCL solvers find quickly, while randomly structured large instances can expose pathological search behaviour even at smaller clause counts.

The practical conclusion is therefore the same as the analytical one: the local formulation is not merely a cleaner theoretical encoding, but the preferable default implementation choice for the solver developed in this thesis. Raw per-(level, method) gen/solve timings are reported in Section A.5.

7.5 Interpretation and Scope

The current results support one claim strongly and several others only weakly.

What is established is that the internal SAT formulation matters. The choice between local and global uniqueness has a first-order effect on both CNF size and runtime, and that effect is already substantial on realistic LLE instances.

What is **not** yet established empirically is the broader quality of the generator framework. The present experiment does not measure generator acceptance rates, diversity of accepted levels, distribution of cooperation profiles, or any downstream training effect on MARL agents. The following two sections address the first two of these open questions; the downstream MARL question is addressed in Section 7.9 and Section 7.12.

7.6 Generator Rejection Rates

7.6.1 Experimental Question

Acceptance rates determine the practical cost of the rejection-sampling strategy used by the generators. A generator with a 1 % acceptance rate requires approximately one hundred solver calls per usable level, which directly affects generation throughput. This experiment measures, for each generator and grid size, how many attempts are needed to find one accepted level and how much wall-clock time those attempts consume.

7.6.2 Protocol

For each generator and grid-size combination, a fixed number of independent trials is executed: 200 trials for the 3×3 and 5×5 grids, and 20 trials for the 8×8 grid. Each trial searches for one accepted level by repeatedly drawing a fresh single-attempt candidate until one is accepted. A trial ends when one level is accepted or a per-trial budget is exhausted (500 attempts for the small grids; 100 attempts or 30 seconds for the 8×8 grid). Failed trials are excluded from the mean attempt count and reported separately.

Each attempt samples a candidate layout, validates geometric constraints (where applicable), constructs the corresponding LLE world, and runs the SAT-based acceptance test of the generator. The mean number of attempts per accepted level is the direct measurement of the rejection cost: it is approximately the reciprocal of the acceptance rate.

Five generator settings are evaluated, drawn from the four families of Chapter 6: the Constrained Random generator in solvable and cooperative settings, the Constructive generator in solvable and cooperative settings, and the Level-6-Style generator. Three grid configurations are used: 3×3 with 2 agents and 1 laser, 5×5 with 3 agents and 2 lasers, and 8×8 with 4 agents and 3 lasers.

7.6.3 Results

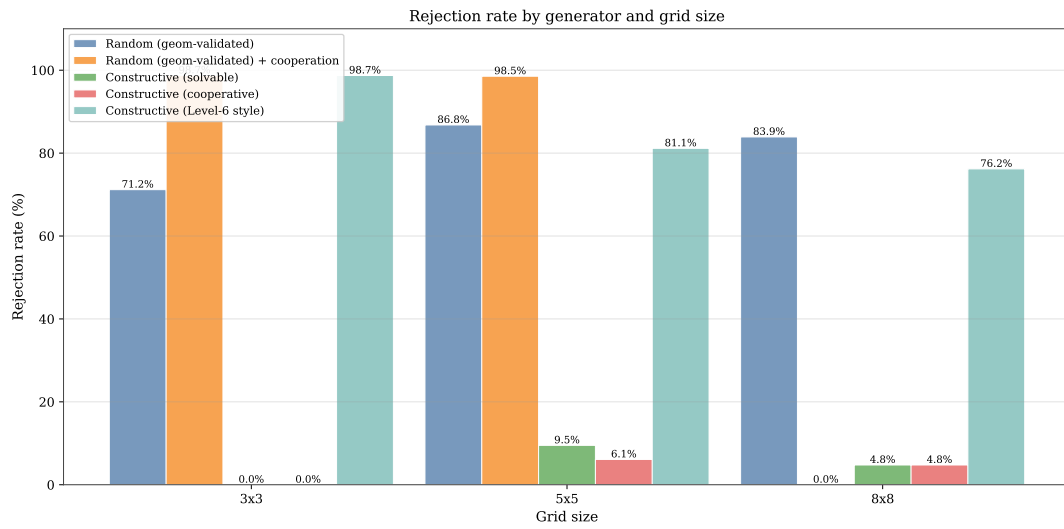


Figure 13: Rejection rate (%) per generator and grid size, derived from the mean number of attempts per accepted level.

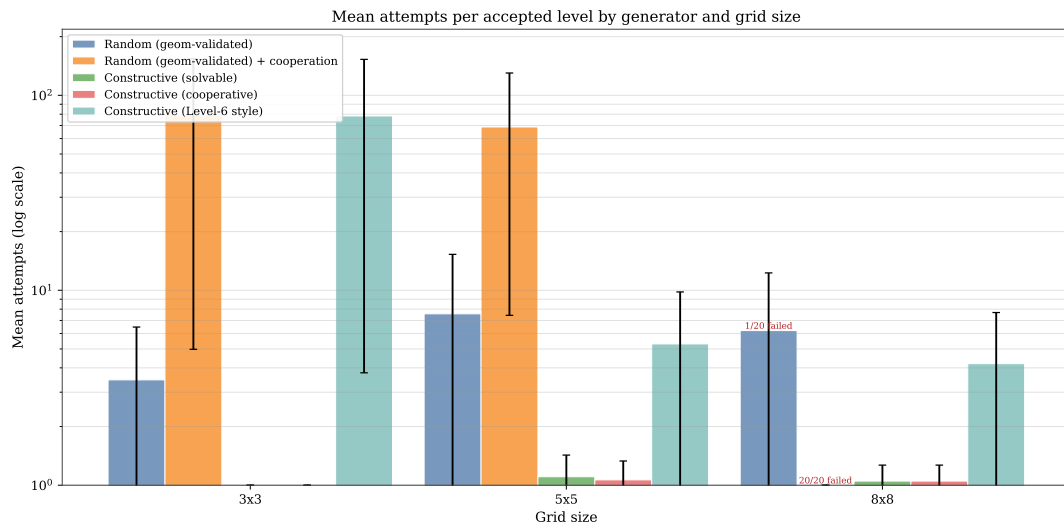


Figure 14: Mean number of attempts needed to find one accepted level, per generator and grid size. Each attempt is one independent generation call; the mean is taken over successful trials only.

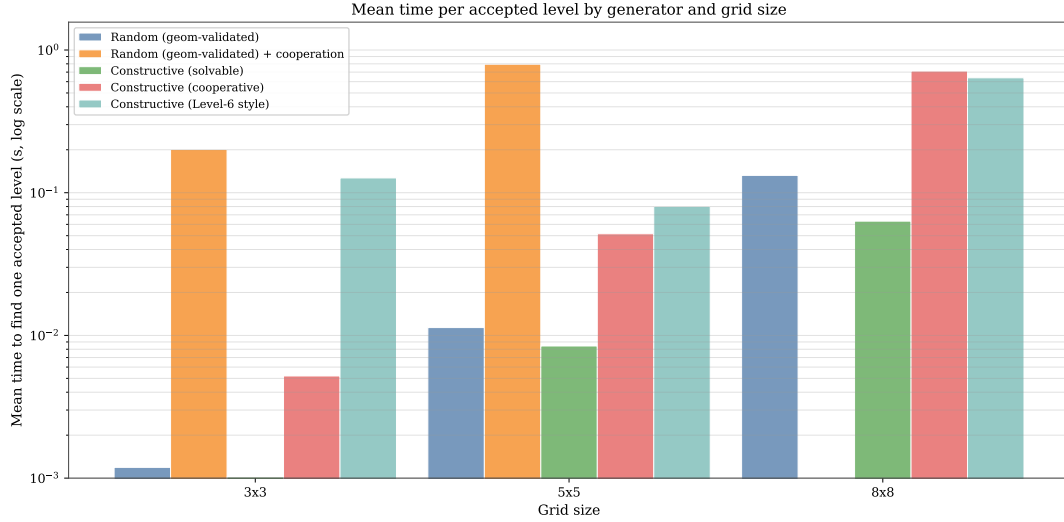


Figure 15: Mean wall-clock time (seconds) to obtain one accepted level, per generator and grid size. Time includes all rejected attempts that precede each accepted level.

7.6.4 Interpretation

The results split the five generators into three regimes.

The Constrained Random generator in **solvable** mode has rejection rates of 71.2 %, 86.8 %, and 83.9 % across the three grid sizes, with mean attempts of 3.5, 7.6, and 6.2. Rejection here is driven by the fact that random layouts rarely admit a valid joint trajectory within the requested horizon. On the 8×8 grid, one trial out of twenty exhausted its budget without finding an accepted level.

The two Constructive variants are the cheapest in the family. The Constructive generator in **solvable** mode runs at 0.0 %, 9.5 %, and 4.8 % rejection across the three grid sizes (1.00, 1.10, and 1.05 mean attempts), and in **cooperative** mode at 0.0 %, 6.1 %, and 4.8 % (1.00, 1.06, and 1.05 mean attempts). The two settings match almost exactly: the multi-colour structural-laser construction of Chapter 6 guarantees that every accepted candidate satisfies the binary cooperation criterion by construction, so the SAT acceptance step rejects only the small fraction of candidates whose random wall sample happens to produce an unsolvable or trivial layout.

The Constrained Random generator in **cooperative** mode has 98.7 % rejection at 3×3 (79.5 mean attempts) and 98.5 % at 5×5 (68.8 mean attempts). The high cost reflects the strict subset structure: cooperation-requiring layouts are a strict subset of solvable layouts, and a uniformly sampled candidate is unlikely to require beam-blocking.¹

The Level-6-Style generator sits between the two regimes. On the 3×3 grid its rejection is 98.7 % (78.3 mean attempts), because the cluster template requires more grid cells than the small grid can offer without geometric conflicts. From 5×5 upward the cluster geometry becomes feasible: rejection drops to 81.1 % and 76.2 % at 5×5 and 8×8 (5.3 and 4.2 mean attempts respectively). The Level-6-Style template trades a moderate efficiency cost for the richer cooperation profile reported in Section 7.7. Detailed per-(generator, grid) numbers are tabulated in Section A.6.

¹The 8×8 configuration of the Constrained Random generator in cooperative mode could not be measured here: the LLE C extension crashed during trial 6 of 20 (SIGSEGV) on a randomly-sampled world. We report this row as missing rather than inferring a value, and excluded it from the plots.

7.7 Cooperation Profile Distribution

7.7.1 Experimental Question

The binary cooperation detector of Chapter 5 certifies that every accepted cooperative level requires at least one same-colour beam-truncation step, but it does not distinguish **which** structural pattern of inter-agent dependency the level exhibits. This experiment measures the distribution of cooperation profiles among accepted levels, using the cooperation profile analyzer also introduced in Chapter 5.

7.7.2 Protocol

For each of three cooperative generators (Constrained Random, Constructive, and Level-6-Style, all in cooperative mode) and two grid configurations (5×5 with 2 agents and 1 laser, 8×8 with 3 agents and 2 lasers), the script accepts 100 cooperative levels on the small grid and 50 on the large grid, then classifies each into one of the analyzer’s profile families: `asymmetric`, `mutual`, `chain`, `distributed`, `fully_coupled`, or `cooperative`.

7.7.3 Results

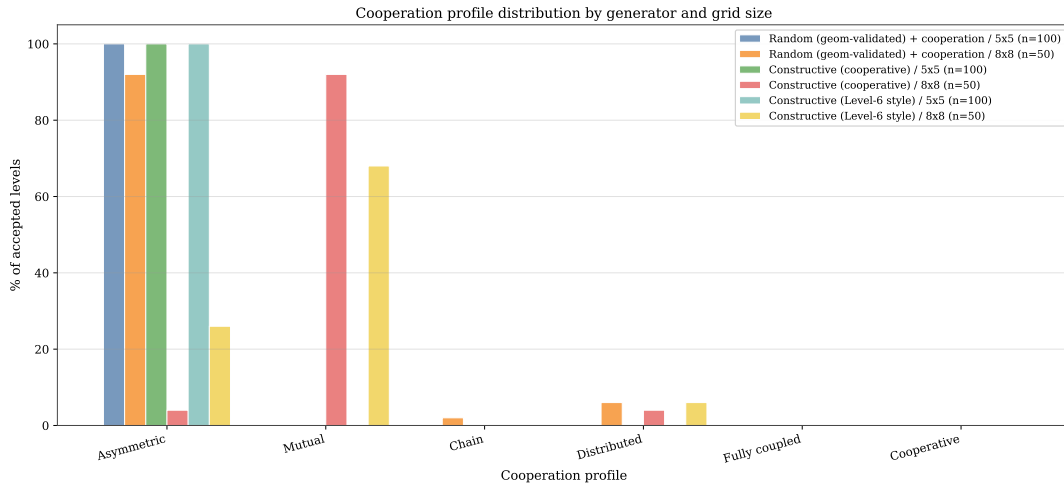


Figure 16: Distribution of cooperation profiles among accepted cooperative levels, grouped by generator and grid size. Each bar shows the fraction of accepted levels assigned to a given profile family.

7.7.4 Interpretation

On the 5×5 grid every accepted level from all three generators is classified as `asymmetric` (100 out of 100 in each case). With only 2 agents, the families `chain`, `distributed`, and `fully_coupled` are **structurally impossible** by construction — they require at least 3 agents — and the `mutual` family is in principle reachable but does not appear: with a single laser the cooperation pattern is restricted to a one-way blocking action by the laser’s colour-matched agent.

The 8×8 configuration with 3 agents and 2 lasers exposes the contrast between the three generators. The Constructive generator produces 46 `mutual`, 2 `asymmetric`, and 2 `distributed` levels out of 50,

i.e. a 92 % majority of mutual cooperation. This reflects the multi-colour structural-laser construction: with two distinct-colour lasers each crossing the entire lane band, the agent of colour 0 must block laser 0 for the other agents to pass, and symmetrically agent 1 must block laser 1, so two helpers act on their own beams in turn — the canonical mutual pattern.

The Constrained Random generator produces 46 asymmetric, 3 distributed, and 1 chain level out of 50 on the same 8×8 configuration. Random layouts that happen to require beam-blocking do so most often through a one-way dependency; richer profiles arise only incidentally and the generator does not reach the mutual family here.

The Level-6-Style generator produces 34 mutual, 13 asymmetric, and 3 distributed levels out of 50 on the same configuration. The cluster-and-corridor geometry forces all agents through a shared corridor and tends to produce multiple agents helping each other simultaneously, much like the canonical LLE Level 6 itself (which the analyzer also classifies as mutual). The two Constructive families therefore split their cooperative output as follows: the Constructive generator favours the mutual pattern by design (92 % mutual), and the Level-6-Style generator favours it by geometric pressure (68 % mutual).

Across the three generators, no level was classified as `fully_coupled` in this benchmark. That family requires a strongly connected dependency graph of size at least three, which neither the lane template nor the cluster template targets by construction; reliably generating fully-coupled instances would require either a larger laser count or the profile-targeted generation mode, in which the generator’s acceptance filter explicitly rejects non-matching profiles. The raw per-(generator, grid) profile counts are tabulated in Section A.7.

7.8 Discussion

The two benchmarks above characterise complementary aspects of the generation framework.

The rejection-rate experiment measures **efficiency**. The headline finding is that the Constructive generator in cooperative mode runs at near-zero rejection (0–6 % across the three grid sizes, comparable to the solvable setting), thanks to the multi-colour structural-laser construction that guarantees cooperation by construction rather than by rejection sampling. This makes the generator practical for producing large pools at scale.

The profile-distribution experiment measures **output diversity**. The headline finding is that the same Constructive cooperative generator produces predominantly mutual levels (92 % mutual on 8×8 with two lasers). Together with the Level-6-Style generator (68 % mutual on the same grid), the family spans the asymmetric / mutual / distributed range of the analyzer’s classification.

What these experiments do not establish is whether certified levels are useful for training. The generator framework guarantees formal properties of accepted levels, but whether those properties translate into better or faster learning for MARL agents compared to uncertified baselines remains an open empirical question, addressed in Section 7.9 and Section 7.12.

7.9 Learnability of Generated Cooperative Levels

7.9.1 Experimental Question

We now ask whether off-the-shelf MARL agents can learn to solve levels produced by the Constructive generator in cooperative mode, and whether the choice of MARL algorithm matters as the cooperation requirement scales up in grid size and agent count. The cooperative generator certifies every accepted level under the binary criterion of Chapter 5: standard SAT and strict UNSAT, so every training and evaluation level requires at least one same-colour beam-truncation step. The learnability question is therefore not whether the levels are solvable in principle (they are, by construction), but whether the joint policy that solves them is reachable by common value-decomposition methods within a modest training budget.

We restrict ourselves to three baseline cooperative-MARL algorithms: independent Q -learning (IQL), value-decomposition networks (VDN), and QMIX. They form a natural progression in the strength of the credit-assignment assumption: from none (IQL), to additive team-value (VDN), to a monotonic mixing network (QMIX).

7.9.2 Protocol

The experiment fixes a single small-grid configuration that exercises the cooperation pressure end-to-end while remaining within reach of a 200,000-step budget: a 5×5 grid with two agents and one laser, horizon $T_{\max} = 10$, and the Constructive cooperative generator. The configuration is summarised in Table 3. An earlier 8×8 , three-agent, two-laser configuration produced 0% success for every algorithm and seed we ran, leaving open whether the algorithms are fundamentally unable or whether the task simply exceeds the budget. The 5×5 rerun isolates the former question.

Grid	Agents	Lasers	$T_{\max}$	Generator	Steps
5×5	2	1	10	Constructive (cooperative)	200,000

Table 3: Learnability-experiment configuration.

The pre-flight script generates two disjoint level pools from independent seeded streams: a training pool of $|\mathcal{D}_{\text{train}}| = 20$ levels (split seed 20260618) and a held-out test pool of $|\mathcal{D}_{\text{test}}| = 20$ levels (split seed 20260619). The same pools are reused for all algorithms and all training seeds, so any cross-algorithm difference is attributable to optimisation rather than to a pool resample. Per-level renderings of both pools are reproduced in the appendix (Section A.9, Section A.10).

We train $|\mathcal{A}| \times |\mathcal{S}| = 3 \times 20 = 60$ independent agent instances:

$$\mathcal{A} = \{\text{IQL}, \text{VDN}, \text{QMIX}\}, \quad \mathcal{S} = \{0, 1, \dots, 19\}$$

The seed $s \in \mathcal{S}$ controls the Python, NumPy, and PyTorch random streams as well as the pool-sampling RNG. Hyperparameters are identical across algorithms (Section A.3): an Adam optimiser at learning rate 5×10^{-4} , batch size 64, discount factor $\gamma = 0.95$, a gradient update every 5 environment steps, gradient-norm clipping at 10, an ε -greedy training policy decaying linearly from 1.0 to 0.05 over the first 100,000 environment steps, and a fully greedy evaluation policy.

Within each training run, every episode samples one level $L \in \mathcal{D}_{\text{train}}$ uniformly with replacement; the episode horizon equals the generator’s T_{max} . Every 10,000 environment steps we run a greedy evaluation of 50 episodes on $\mathcal{D}_{\text{train}}$ and 50 episodes on $\mathcal{D}_{\text{test}}$, sampling levels uniformly from each pool. After the final training step we run a longer evaluation of 200 episodes per pool; the success rate $\hat{s}$ reported below is this final greedy estimate, with one estimate per (algorithm, seed) cell:

$$\hat{s}_{\text{train}}(a, s) = \frac{1}{200} \sum_{i=1}^{200} \mathbb{1}[\text{episode } i \text{ reaches all exits}]$$

and analogously for $\hat{s}_{\text{test}}(a, s)$.

The headline aggregate per algorithm $a \in \mathcal{A}$ is the mean and standard deviation over the twenty seeds; we also report a 95 % confidence interval $\pm t_{19,0.025} \frac{\sigma}{\sqrt{20}}$ ($t_{19,0.025} \approx 2.093$) in the per-seed appendix tables.

7.9.3 Results

Figure 17 reports the train- and test-pool success rates as a function of environment steps, averaged over the twenty seeds with a 95 % confidence band. All three algorithms reach a non-trivial success rate on the training pool well before the 200,000-step budget is exhausted, confirming that the 5×5 cooperative task is learnable in principle. The held-out test pool is markedly harder for every algorithm: the test curves plateau below the corresponding training curves, with a train-test gap of 0.37 for IQL, 0.39 for QMIX, and 0.52 for VDN at the end of training (Table 4, and the per-seed numbers in Table 15).

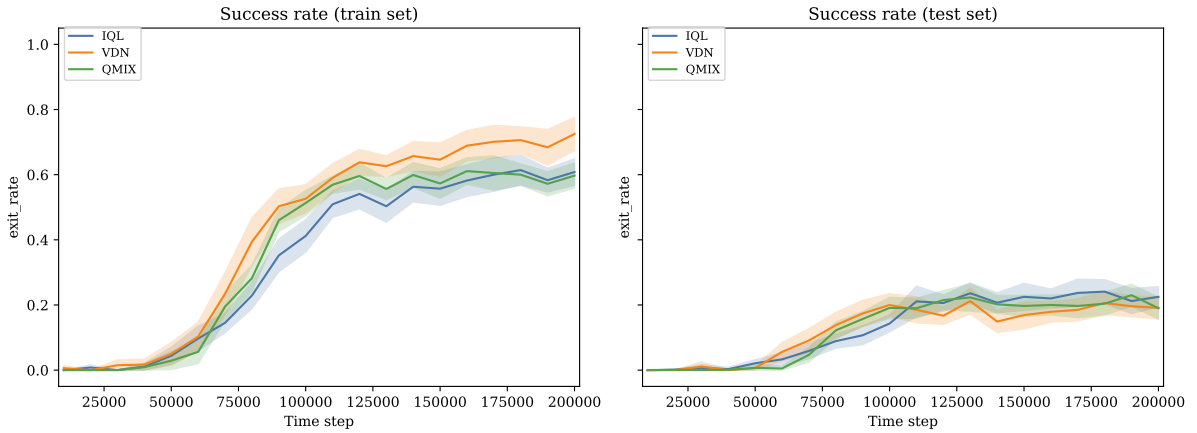


Figure 17: Mean success rate (greedy exit rate) on the training and held-out test pools as a function of environment steps, for IQL, VDN, and QMIX. Shaded bands are 95 % confidence intervals over the twenty training seeds, clipped to the per-step minimum and maximum.

Figure 18 shows the per-algorithm final success rates from the longer 200-episode evaluation. The per-algorithm aggregates are summarised in Table 4: VDN achieves the strongest training-pool success (0.70 ± 0.05), followed by IQL (0.60 ± 0.03) and QMIX (0.59 ± 0.03); on the held-out pool the ranking flips slightly, with IQL at 0.23 ± 0.03 , QMIX at 0.20 ± 0.03 , and VDN at 0.18 ± 0.03 . The cross-algorithm spread is small relative to the train/test gap, so the dominant signal in this configuration is generalisation, not credit assignment: every algorithm overfits the twenty training levels rather than abstracting the cooperation pattern across the pool.

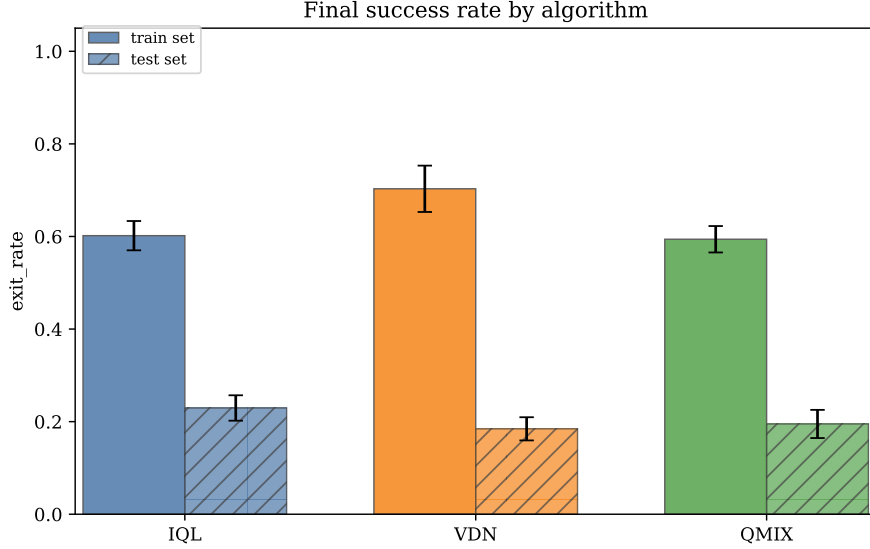


Figure 18: Final-evaluation success rate per algorithm (200 episodes per pool), train versus test. Error bars are 95 % confidence intervals over the twenty training seeds.

Algorithm	n	Train mean $\pm$ CI95	Test mean $\pm$ CI95	Train — test gap
IQL	20	0.60 ± 0.03	0.23 ± 0.03	0.37
VDN	20	0.70 ± 0.05	0.18 ± 0.03	0.52
QMIX	20	0.59 ± 0.03	0.20 ± 0.03	0.39

Table 4: Per-algorithm final success rates on the 5×5 cooperative pools, aggregated over twenty training seeds. The 200-episode greedy evaluation is summarised as a mean and a 95 % confidence interval ($\pm t_{19,0.025} \frac{\sigma}{\sqrt{20}}$, $t_{19,0.025} \approx 2.093$).

7.10 Training-Pool Scaling Restores Generalisation

7.10.1 Experimental Question

The learnability experiment of Section 7.9 isolated **generalisation**, rather than credit assignment, as the dominant failure mode: all three algorithms reached a non-trivial training-pool success rate but overfit the twenty training levels, with a train–test gap of up to 0.52. Because the constructive cooperative generator produces certified levels at near-zero marginal cost (Section 7.6), the natural remedy is simply more **training data**. We therefore ask whether enlarging the training pool — holding the task, the budget regime, and the held-out test pool fixed — closes the generalisation gap.

7.10.2 Protocol

The task, generator, horizon, and hyperparameters are exactly those of Section 7.9 (a 5×5 grid, two agents, one laser, $T_{\max} = 10$, constructive cooperative generator). The only manipulated variable is the training-pool size $|\mathcal{D}_{\text{train}}| \in \{20, 100, 500\}$. The three training pools are **nested**: they are drawn

as prefixes of a single seeded generation stream, so the 100-level pool contains the 20-level pool and the 500-level pool contains both. A single held-out test pool of $|\mathcal{D}_{\text{test}}| = 50$ levels, drawn from an independent seed, is **identical** across all three conditions, so any change in held-out success is attributable to the training-pool size alone.

For each pool size we train $|\mathcal{A}| \times |\mathcal{S}| = 3 \times 20 = 60$ independent agents ($\mathcal{A} = \{\text{IQL}, \text{VDN}, \text{QMIX}\}$, $\mathcal{S} = \{0, \dots, 19\}$) for 300,000 environment steps, and report the final 200-episode greedy success rate on each pool. Aggregates are taken over all sixty runs per pool size, summarised as a mean and a 95% confidence interval.

7.10.3 Results

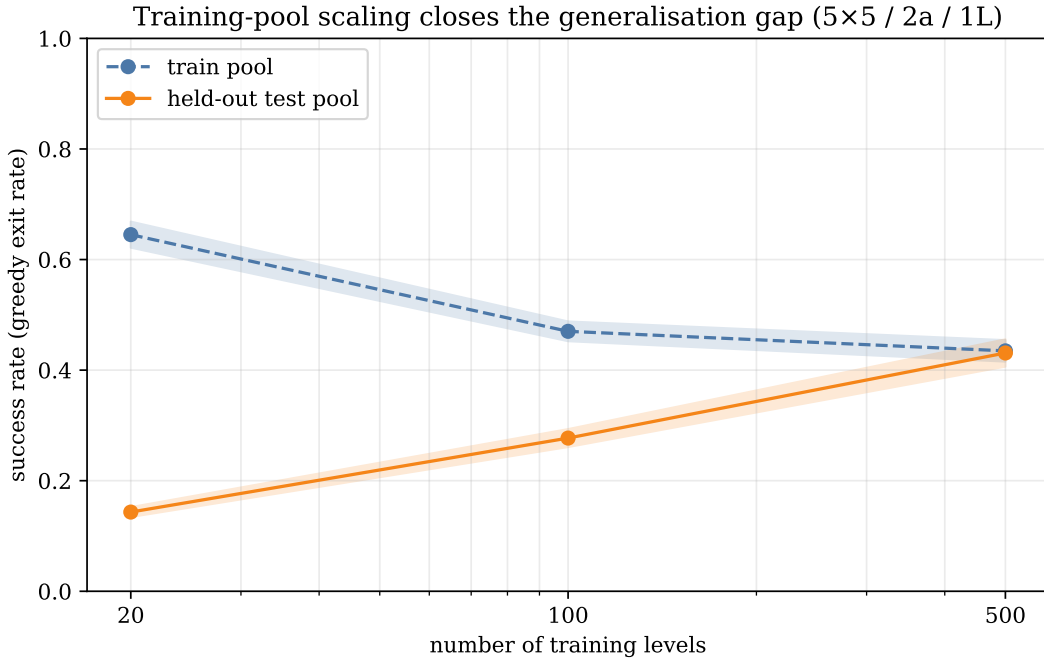


Figure 19: Final greedy success rate on the training and held-out test pools as a function of the number of training levels (log scale), for the 5×5 / 2-agent / 1-laser cooperative task. Each point aggregates the runs over three algorithms and twenty seeds; shaded bands are 95% confidence intervals. As the pool grows, training success falls while held-out success rises, and the two meet.

Figure 19 and Table 5 show a clear monotone effect. Held-out test success rises from 0.14 at twenty levels to 0.28 at one hundred and 0.43 at five hundred — roughly a threefold improvement — while training success **falls** from 0.65 to 0.43 over the same range. The train-test gap therefore collapses from 0.50 to 0.19 to approximately zero: at five hundred levels the agent performs essentially identically on training and held-out levels, i.e. it has stopped memorising the training pool and learned a policy that transfers across the level distribution. The monotone increase in held-out success holds for each algorithm individually (IQL $0.16 \rightarrow 0.26 \rightarrow 0.38$; VDN $0.14 \rightarrow 0.26 \rightarrow 0.42$; QMIX $0.13 \rightarrow 0.31 \rightarrow 0.49$), so the effect is a property of the data regime rather than of any single credit-assignment scheme.

$ \mathcal{D}_{\text{train}} $	n	Train mean $\pm$ CI95	Test mean $\pm$ CI95	Train — test gap
20	60	0.65 ± 0.02	0.14 ± 0.01	0.50
100	60	0.47 ± 0.02	0.28 ± 0.02	0.19
500	60	0.43 ± 0.02	0.43 ± 0.03	0.00

Table 5: Final success rates on the 5×5 cooperative task as the training-pool size grows, with a fixed 50-level held-out test pool. Each row aggregates the runs over three algorithms and twenty seeds.

7.10.4 Interpretation

This experiment isolates the practical payoff of the generator. The learnability ceiling reported in Section 7.9 is not an intrinsic limit of the value-decomposition algorithms on cooperative levels; it is a consequence of training on too few distinct instances. Because the constructive cooperative generator emits an effectively unlimited supply of certified — solvable and cooperation-requiring (Chapter 5) — levels at near-zero rejection cost, it directly supplies the quantity of training data needed to close the gap, something a fixed library of hand-crafted levels cannot. The generator’s formal guarantees thus translate into a concrete downstream benefit: scaling generated training data converts overfitting into generalisation. This also delimits the role of the curriculum studied next: curricula target either the **ordering** of staged exposure on a target that is already reachable (Section 7.11) or levels too **hard** to learn directly (Section 7.12), whereas data scaling targets levels that are learnable but not yet **generalised**.

7.11 Curriculum Ordering on a Reachable 6×6 Cooperative Target

7.11.1 Experimental Question

Section 7.10 showed that the learnability ceiling on **reachable** generated levels is a data-quantity effect rather than an intrinsic algorithmic limit. RQ6 (Section 7.12) asks the harder question of whether a staged curriculum transfers to the hand-crafted LLE Level 6, a target on which direct training is expected to fail outright. As a controlled precursor we first isolate the curriculum **mechanism itself** on a target that direct training can already reach. Holding the task, the difficulty ladder, the total training budget, and the held-out evaluation fixed, we vary only **how** the budget is allocated across a ladder of growing geometric and cooperative complexity, and ask whether the **ordering** of staged exposure changes final performance or sample efficiency on the target. Because the target is reachable by the direct baseline, any difference between conditions is attributable to scheduling alone rather than to the target being unreachable.

7.11.2 Protocol

The difficulty ladder has three rungs, each supplied by the generators of Chapter 6: a 4×4 / 2-agent / 0-laser navigation warm-up (random generator, $T_{\max} = 8$), a 5×5 / 2-agent / 1-laser cooperative rung ($T_{\max} = 10$), and the 6×6 / 2-agent / 1-laser cooperative **target** rung ($T_{\max} = 12$). The agent count is

fixed at two across all rungs so the Q -network input shape is constant; smaller observations are zero-padded to the target geometry. Each rung draws 100 distinct certified levels for training, and the target rung additionally holds out an independent pool of 50 levels for the generalisation metric.

We compare four budget-matched scheduling conditions, each trained for a total of 400,000 environment steps:

- **direct** spends the entire budget on the 6×6 target rung;
- **forward** walks the ladder easy-to-hard, allocating $(50, 150, 200) \times 10^3$ steps to the three rungs in order;
- **reverse** uses the same per-rung budgets but visits the rungs hard-to-easy, so it reaches the target early and ends on the navigation rung, arriving at the held-out evaluation cold;
- **mixed** draws a rung uniformly at random each episode (domain randomisation) for the whole budget.

Each condition is run with all three algorithms ($\mathcal{A} = \{\text{IQL}, \text{VDN}, \text{QMIX}\}$) and eight seeds, for 96 runs in total. The exploration rate is reset at every stage boundary and decayed independently within each rung’s budget, so that early rungs are not starved of exploitation nor the target of exploration. Evaluation is always performed on the 6×6 target: every 10,000 steps on 50 episodes during training, and a final 200-episode greedy success rate on both the training and held-out test pools at the end. Aggregates are reported as a mean and a 95% confidence interval over seeds.

7.11.3 Results

Table 6 summarises the final greedy success rate of each scheduling condition, pooled over the three algorithms and the eight seeds per cell ($n = 24$ runs per condition). On the held-out 6×6 target the ranking is **mixed** (0.22 ± 0.04), then **direct** (0.17 ± 0.03), then **forward** (0.15 ± 0.03), and last **reverse** (0.07 ± 0.02). The ordered easy-to-hard curriculum (**forward**) does not beat the **direct** baseline; the only condition that improves on direct is **mixed**, which exposes all three rungs in random order throughout training.

Condition	n	Train mean $\pm$ CI95	Test mean $\pm$ CI95	Train — test gap
direct	24	0.41 ± 0.05	0.17 ± 0.03	0.24
forward	24	0.39 ± 0.06	0.15 ± 0.03	0.24
mixed	24	0.39 ± 0.07	0.22 ± 0.04	0.17
reverse	24	0.10 ± 0.02	0.07 ± 0.02	0.03

Table 6: Final greedy success rate on the 6×6 / 2-agent / 1-laser target per scheduling condition, aggregated over three algorithms and eight seeds ($n = 24$ runs per condition). The 200-episode greedy evaluation is summarised as a mean and a 95% confidence interval ($\pm t_{23,0.025} \frac{\sigma}{\sqrt{24}}$, $t_{23,0.025} \approx 2.07$).

Per-(condition, algorithm) numbers are in Section A.12.

The learning curves and the per-algorithm final success rates are shown in Figure 20, Figure 21, and Figure 22; they confirm that the ordering above holds within each algorithm, not only in the pooled aggregate.

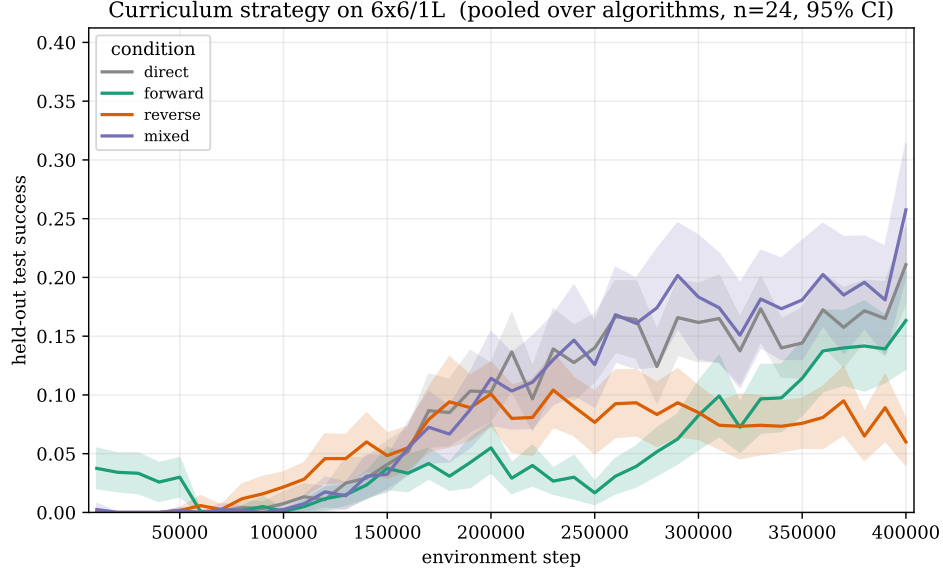


Figure 20: Held-out test success on the $6 \times 6 / 2\text{-agent} / 1\text{-laser}$ target as a function of environment step, one line per scheduling condition (direct, forward, reverse, mixed) pooled over the three algorithms and the eight seeds. Shaded bands are 95% confidence intervals.

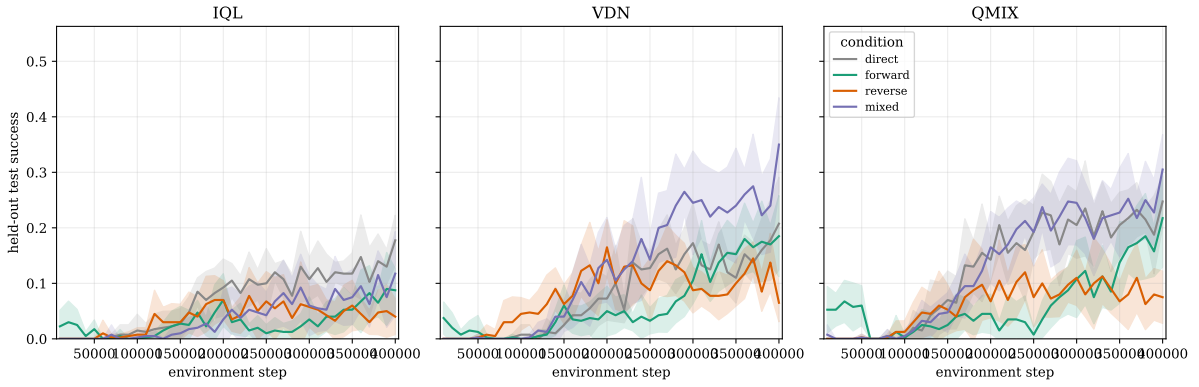


Figure 21: The same held-out test learning curves as Figure 20, shown per algorithm (IQL, VDN, QMIX) rather than pooled.

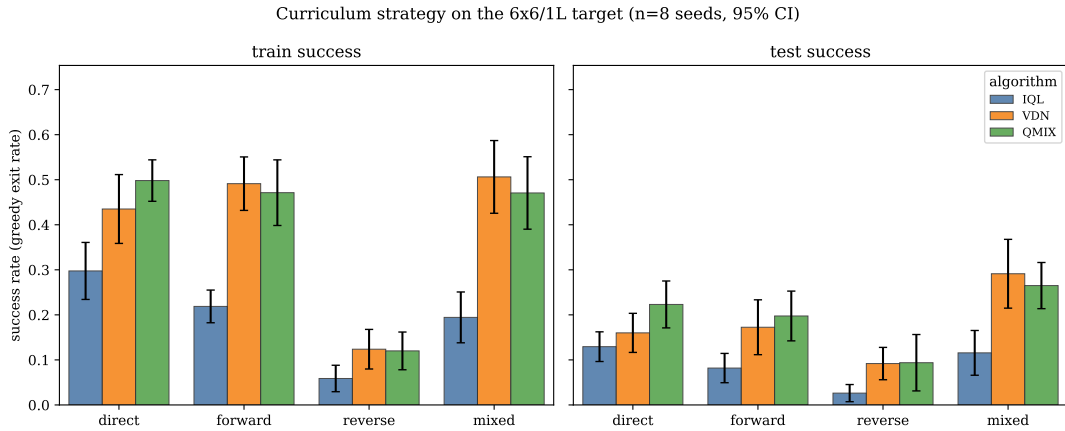


Figure 22: Final greedy success rate on the training and held-out test pools at the end of the 400,000-step budget, grouped by scheduling condition, with one bar per algorithm and 95% confidence-interval error bars over the eight seeds.

7.11.4 Interpretation

Two mechanisms explain why ordering does not help on a reachable target. First, under a fixed budget every step **forward** spends on the easier rungs is a step it does not spend on the target; when the target is directly learnable, that target experience dominates any transfer benefit, so moving budget to warm-up rungs can only match or cost final performance. Second, **reverse** — which reaches the target early and then trains on the easier rungs — collapses to near the floor: its training-pool success (0.10) is far below the others, the signature of catastrophic forgetting, since the network overwrites the target competence acquired early while fine-tuning on the navigation rung. The one condition that helps, **mixed**, is domain randomisation rather than a curriculum: its benefit comes from the **diversity** of exposure, not from any easy-to-hard ordering, and it is the same data-variety effect already isolated in Section 7.10.

The practical conclusion is that on a target the agent can already reach, scheduling the **order** of staged exposure offers no advantage over training directly on the target, and a poorly chosen order (reverse) is actively harmful. This leaves open the case that curriculum learning is actually designed for — a target the agent **cannot** reach directly — which Section 7.12 addresses.

7.12 The Limits of Curriculum Learning on Mutually-Cooperative Targets

This section addresses RQ6 of Chapter 1.

7.12.1 Experimental Question

Section 7.11 isolated the curriculum **mechanism** on a target that direct training can already reach, and found that ordering confers no advantage. The motivating promise of curriculum learning is different, however: that staged exposure can make **reachable** a target that direct training cannot solve at all. The canonical such target in this thesis is the hand-crafted LLE Level 6 — a 12×13 map with four agents and three lasers whose solution requires **mutual** cooperation, the dependency pattern in which several agents must each truncate a beam for the others before anyone can exit (Chapter 5). We therefore ask whether a curriculum of generated levels of growing cooperative complexity enables value-decomposition agents to solve Level 6, and, more generally, whether such agents can cross the mutual-cooperation boundary at all.

7.12.2 Protocol

We approached the question in three stages of increasing ambition, each reusing the trainer, hyperparameters (Section A.3), and greedy 200-episode evaluation of the preceding experiments.

1. **Frontier probe.** Direct training from scratch on a fixed 6×6 grid with two agents and two lasers — the smallest instance whose solution requires **mutual** cooperation — for IQL, VDN, and QMIX. This isolates whether the mutual target carries any learnable signal at all before a curriculum is layered on top, holding the grid fixed so that the laser count is the only change from the reachable asymmetric target of Section 7.11.

2. **Two-laser curriculum.** The four scheduling conditions of Section 7.11 (direct, forward, reverse, mixed) on a fixed 6×6 grid, with a three-rung ladder that ramps only the cooperation requirement — zero, one, then two lasers — toward the mutual two-laser target, under a budget-matched 600,000-step budget.
3. **Level-6 transfer.** A four-stage curriculum of generated levels growing in both geometry and cooperation ($6 \times 6/1$ laser $\rightarrow 8 \times 8/2$ lasers $\rightarrow 10 \times 10/3$ lasers $\rightarrow$ a 12×13 Level-6-style stage), with four agents throughout, evaluated on the hand-crafted Level 6. The curriculum condition (CURR) is compared against three baselines that isolate the contribution of staging: training only on the hardest stage (B1), an anti-curriculum hard-to-easy ordering (B2), and direct training on Level 6 (B3). Runs use up to 2,000,000 environment steps — an order of magnitude beyond the reachable-target budget.

Because every preliminary condition returned zero success (below), and because a single run at the Level-6 budget is computationally expensive, we deliberately limited each sweep to a small number of seeds rather than expanding to the eight or twenty seeds used for the reachable targets. As the results show, the conclusion rests on the **uniformity** of the zero across grids, pool sizes, algorithms, budgets, and schedules, not on seed-level precision. Full per-run numbers and the exact stage configurations are tabulated in Section A.13.

7.12.3 Results

Across all three stages, no configuration learned to solve a mutually-cooperative target. Table 7 places these outcomes next to the **reachable** results of the preceding sections on a single axis of increasing cooperation difficulty. The contrast is the result: success is non-trivial as long as cooperation is **asymmetric** — a one-way blocking action — and collapses to zero the moment the target requires **mutual** blocking, and it stays at zero even when the training budget is increased tenfold.

Task (grid / agents / lasers)	Cooperation	Budget	Train	Test
$5 \times 5 / 2 / 1$	asymmetric	200k	0.63	0.20
$6 \times 6 / 2 / 1$	asymmetric	400k	0.41	0.17
$6 \times 6 / 2 / 2$	mutual	600k	0.08	0.00
$5 \times 5 / 2 / 2$	mutual	200k	0.00	0.00
$12 \times 13 / 4 / 3$ (Level 6)	mutual	2M	0.00	0.00

Table 7: Greedy success rate as the cooperation requirement increases, pooled over algorithms and seeds. “Train” is success on the levels seen during training (the held-out **generated** pool for the Level-6 row); “Test” is success on the held-out target (Level 6 for the last row). The first two rows are the reachable asymmetric targets of Section 7.9 and Section 7.11; the last three are the mutual targets of this section. For the mutual rows the value is the mean over the available algorithm and seed runs (the $6 \times 6 / 2$ -laser row is the direct-training frontier probe); for the Level-6 row it is the best achieved over the four conditions (B1, B2, B3, CURR). In every case held-out success is zero. Per-run numbers are in Section A.13.

On the **frontier probe**, all three algorithms occasionally solved a **training** level (a peak greedy success of 0.06–0.09) but never generalised: held-out success was exactly zero for every algorithm and seed. Shrinking the grid to 5×5 did not help — two lasers suffice to drive success to zero regardless of

grid size. Two diagnostic checks confirm that this is a learnability wall rather than a budget shortfall: training on a **single** fixed mutual level (removing the need to generalise entirely) still plateaued below 0.12 success, and extending the budget to 1,500,000 steps left success flat, with the success curve peaking early and then **decaying** rather than climbing.

On the **two-laser curriculum**, every scheduling condition — including forward staging — finished at essentially zero held-out success (the best condition reached 0.01). Ordering a curriculum toward a target that carries no learnable signal does not create one.

On the **Level-6 transfer**, every condition reached 0.0 success on Level 6 and, tellingly, 0.0 success on the in-distribution held-out **generated** pool as well, even after 2,000,000 steps. The full four-stage curriculum (CURR) was no better than direct training on Level 6 (B3), and the agents’ mean episode return stayed negative throughout — they accumulated step penalties without ever assembling the coordinated double-blocking sequence the level requires.

7.12.4 Why Curriculum Learning Does Not Help Here

The two regimes of Section 7.11 and this section fail for complementary reasons. On a **reachable** target, a curriculum is unnecessary: ordering trades budget away from the target and, hard-to-easy, induces catastrophic forgetting, so it can only match or underperform direct training. On an **unreachable** mutual target, a curriculum is powerless, for three reasons that the experiments above separate.

- **There is no signal to amplify.** A curriculum works by transporting a policy with non-zero value from an easy stage to a harder one, where it stumbles onto reward more often and bootstraps. The mutual target returns reward only after a long, jointly-gated sequence — both lasers must be truncated before any agent can exit — so a from-scratch or warm-started policy receives identically zero reward (the negative returns above). A curriculum can make a faint signal louder; it cannot manufacture one from zero.
- **The missing skill is absent from every easier stage.** The easier rungs teach navigation and **asymmetric** cooperation, in which one agent blocks and the other passes — a pattern with a partial-credit path, since the passing agent can be rewarded before coordination is perfect. The mutual target requires a **deadlock-style** interdependency in which neither agent can make progress first. This is a discontinuity, not a ramp: no easy-to-hard ordering of asymmetric stages gradually constructs the mutual behaviour, because that behaviour is qualitatively absent below the final rung.
- **The algorithm cannot represent the solution.** Value-decomposition methods (VDN, QMIX) assume the joint action-value factorises monotonically into per-agent utilities. Mutual cooperation requires both agents to take individually-costly actions simultaneously — standing in a beam so the other may pass — the canonical case that monotonic factorisation cannot represent. The nonzero **training** but zero **held-out** success on the frontier probe is the fingerprint: the rare successes are memorised joint trajectories, not a generalisable coordination policy. Reordering the training data does not change the function class, so the representational bottleneck is left untouched.

The decisive observation is the zero held-out **generated**-pool success in the Level-6 transfer: the failure is not one of **transfer** or distribution shift between generated levels and the hand-crafted target — the agents never learned the base mutually-cooperative task at all. The failure of curriculum learning here is therefore not a failure of curriculum design but a consequence of the target lying beyond the reach of the underlying MARL method.

7.12.5 Scope and Threats to Validity

We are careful not to overclaim. These experiments establish that the value-decomposition algorithms evaluated here (IQL, VDN, QMIX), under a fixed set of hyperparameters and a sparse joint-exit reward, do not solve mutually-cooperative LLE targets, and that no curriculum over generated levels changes this — **not** that curriculum learning can never succeed on such targets. Three caveats bound the claim. First, the algorithm family is narrow; centralised-critic actor-critic methods and approaches with explicit coordinated exploration were not tested and might cross the wall (Chapter 8). Second, the reward is sparse and unshaped; a dense or intrinsic signal that credits partial coordination would change the very signal whose absence we identify as the bottleneck. Third, the seed counts on the mutual targets are small — a deliberate choice given the cost of each run and the uniformity of the zero outcome; they suffice to establish that success is zero, but not to measure small effects between equally-failing conditions. Directions that target the actual bottleneck — base learnability rather than curriculum ordering — are discussed in Chapter 8.

Conclusion

8.1 Summary

This thesis developed a SAT-based framework for reasoning about solvability and cooperation in a modeled subset of the Laser Learning Environment. We first formalised bounded-horizon solvability as a decision problem and reduced it to satisfiability of a CNF formula. On top of that reduction, we introduced a strict beam-antics counterfactual that turns the blocking-based cooperation mechanism of LLE into a second decision problem on the same level, and refined the binary outcome into a profile taxonomy that classifies the dependency structure of a cooperative level.

These decision procedures were then embedded inside a family of procedural generators. The resulting framework does not guarantee that generated levels are pedagogically optimal for MARL, but it does guarantee that accepted levels satisfy the formal properties checked by the solver. This is the central contribution of the thesis: procedural generation coupled to **explicit certification** rather than procedural generation guided only by heuristic plausibility.

8.2 Answering the Research Questions

We restate the six research questions of Chapter 1 and assess what each chapter delivers.

- **RQ1 — formal verification of solvability.** Answered by the bounded-horizon SAT reduction of Chapter 4, whose correctness is established by Proposition 4.1. The encoding is polynomial-time constructible, places bounded-horizon LLE solvability in NP, and proves the problem is at most as hard as SAT.
- **RQ2 — formal verification of the cooperation requirement.** Answered by the strict-counterfactual encoding of Chapter 5, whose correctness is established by Theorem 5.1: a level requires cooperation if and only if the standard formula is SAT and the strict formula is UNSAT. The criterion is decidable on the same finite horizon used for solvability.
- **RQ3 — embedding verification inside a generator.** Answered by the generator family of Chapter 6. Each generator uses the SAT solver as an acceptance oracle, so every emitted level carries a solver certificate of the advertised property.
- **RQ4 — controlling the cooperation profile.** Answered constructively. The cooperation profile analyzer of Section 5.6 labels every certified cooperative level, and the generators expose a profile filter that restricts acceptance to a chosen subset of the taxonomy. The profile-distribution experiment of Section 7.7 shows that the constructive generator reliably produces **mutual** profiles when $n_l \geq 2$ and admits **distributed** profiles when $n_l \geq 3$ on grids large enough. The **fully coupled** label

did not arise in the tested parameter range and remains a target for the future-work parameter-sensitivity study below.

- **RQ5 — learnability of certified cooperative levels.** Answered positively on a small grid. The learnability experiment of Section 7.9 shows that IQL, VDN, and QMIX all reach non-trivial greedy success rates (0.59–0.70 on the training pool, 0.18–0.23 on the held-out pool) on the 5×5 cooperative pool within a 200,000-step budget. The dominant difficulty observed is generalisation to the held-out pool rather than credit assignment. The data-scaling experiment of Section 7.10 shows this gap is a data-quantity effect rather than an algorithmic limit: enlarging the certified training pool to five hundred levels closes the train–test gap from 0.50 to approximately zero. The certification framework thus delivers a concrete downstream payoff — more certified data converts overfitting into generalisation.
- **RQ6 — curriculum transfer to LLE Level 6.** Answered negatively, within scope. The experiments of Section 7.11 and Section 7.12 show that, for the value-decomposition algorithms evaluated here, a curriculum of generated levels confers no advantage. On a reachable asymmetric target, ordering the stages does not beat direct training, and only data **diversity** helps; on the mutually-cooperative LLE Level 6, no condition — including a four-stage curriculum trained for two million steps — exceeds zero success, and neither does direct training. Because the agents also fail on the in-distribution generated pool, the obstacle is the base task being unlearnable by these methods rather than a failure of curriculum **transfer**. We therefore read RQ6 as a scoped negative result and identify the learnability of mutual coordination — not curriculum design — as the lever for future work.

8.3 Future Work

Several directions extend the present work naturally.

- **Complexity classification of bounded-horizon LLE solvability.** This thesis shows that bounded-horizon LLE solvability is in NP and is polynomial-time reducible to SAT, but we do not settle whether the problem is NP-hard. The exact placement has both theoretical and practical consequences: if the problem is NP-hard, the SAT-based oracle is optimal up to constant factors under the standard $P \neq NP$ assumption; if it lies in P, then in principle the SAT solver could be replaced by a polynomial-time algorithm tailored to the LLE structure, with a corresponding speed-up for every generator in the family. Establishing the lower bound is therefore both a theoretical complement to the present work and an engineering lever for the generator pipeline.
- **Model extension to gems and void tiles.** The formal model deliberately omits gems and treats void tiles as walls. Incorporating gem tiles would add an additional reachability condition — every agent must visit a designated gem cell at some point before reaching its exit — and would extend the property certified by the SAT oracle accordingly. Distinguishing void tiles from walls would require a revised movement semantics (agents cannot enter void cells under any condition, but the cell still propagates beams differently). Both extensions preserve the SAT-based architecture; only the encoding and the formal definitions need to be updated.
- **Parameter sensitivity and output diversity.** The acceptance-rate experiments fix the agent and laser counts at each grid size. A fuller characterisation would vary these parameters along independent axes to locate the practical frontier at which rejection rates become prohibitive, would measure within-pool diversity of accepted levels at a fixed parameter setting beyond the wall-Hamming statistics reported here, and would attempt to surface the **fully coupled** profile by raising n_l and the grid size.

- **Richer cooperation metrics.** The cooperation-profile taxonomy of Section 5.6 captures five dependency-graph patterns. Extensions could promote the synchronous-width scalar already exposed by the analyzer into a first-class profile axis, and could add a chain-length difficulty axis on top of the qualitative chain label. These richer targets would let the generator family target **graded** cooperation difficulty rather than only qualitative profile families.
- **Algorithm-space breadth.** The learnability and curriculum-transfer experiments evaluate three value-decomposition algorithms (IQL, VDN, QMIX). Comparison against centralised-critic actor-critic methods such as MADDPG [6] and COMA [7], and against latent-variable variants such as MAVEN [8], would give a fuller picture of which algorithm families benefit most from certified training material on coordination-critical tasks.
- **Making mutual coordination learnable.** The curriculum experiments of Section 7.12 locate the obstacle to solving LLE Level 6 in the base learnability of **mutual** cooperation under a sparse joint-exit reward, not in the ordering of training levels: no curriculum can amplify a learning signal that is identically zero. The natural levers therefore act on the reward and the algorithm rather than on the curriculum. A dense or intrinsic reward that credits partial coordination — for instance, an agent successfully truncating a beam for another — would supply the signal whose absence we identify as the bottleneck, and the centralised-critic or coordinated-exploration methods listed above could supply the representational capacity that monotonic value decomposition lacks. With a base learner able to acquire mutual coordination in isolation, the curriculum question of RQ6 could be revisited on a foundation that the present value-decomposition methods do not provide.

The main open question is therefore not whether solver-based certification is possible in LLE; the present thesis answers that positively. The open question is how far that certification framework can be extended — in terms of richer mechanics, richer cooperation structures, downstream learning effects, and the exact complexity status of the underlying decision problem — before additional formal layers are required.

Appendix

A.1 Benchmark Levels for SAT Encoding Comparison

The four levels used in the SAT encoding comparison (Chapter 7) are shown below with their exact parameters.

Level	Grid	Agents	Lasers	Horizon $T_{\max}$
Synthetic 3×3	3×3	2	1	4
Synthetic 5×5	5×5	3	2	5
Synthetic 8×8	8×8	4	3	15
Benchmark Level 6	12×13	4	3	21

Table 8: Parameters of the four benchmark levels used in the SAT encoding comparison.

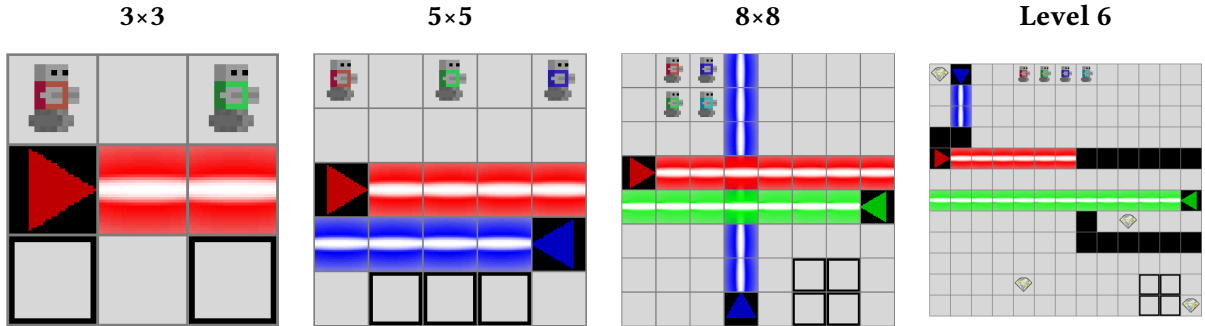


Figure 23: Visual representation of the four benchmark levels.

A.2 Reproducibility — Software and Seed Conventions

All experiments are reproducible from a single Python source tree. The relevant versions and seed conventions are summarised below.

Component	Version / Value
Python	3.12 or later (3.13 used for the runs reported here)
SAT solver	Minisat22 via the PySAT interface [33]
LLE engine	laser-learning-environment (Python + Rust bindings)
MARL trainer	marl framework, called from src/experiments/learnability/run_experiment.py
Plotting backend	matplotlib
Solver-comparison seeds	None (deterministic CNF; only timings vary across runs)
Rejection benchmark	Unseeded (seed=None); reported as means over 200 trials (3×3 , 5×5) or 20 trials (8×8)
Profile benchmark	Unseeded (seed=None); reported as counts over 100 (5×5) or 50 (8×8) accepted levels
Learnability training-pool seed	20260618
Learnability training seeds	$\mathcal{S} = \{0, 1, \dots, 19\}$ (twenty seeds per algorithm)
Data-scaling pool seeds	20260618 (train), 20260619 (test); pools nested, test size 50
Data-scaling training seeds	$\mathcal{S} = \{0, 1, \dots, 19\}$ per algorithm and pool size
Curriculum-strategy seeds	$\mathcal{S} = \{0, 1, \dots, 7\}$ (eight seeds per condition and algorithm)
Curriculum two-laser / Level-6 seeds	Small per-condition sets (1–6 seeds); see Section A.13
Appendix-gallery pool seeds	Distinct per pool, listed in each gallery parameter table

Table 9: Software components and seed conventions. Hardware specifications are deferred; the SAT benchmarks run on commodity laptops and the MARL runs on a CUDA workstation with sm_61 GPUs.

A.3 Learnability Hyperparameters

The learnability experiment (Section 7.9), the data-scaling experiment (Section 7.10), and the curriculum experiments (Section 7.11, Section 7.12) share the same trainer construction in src/experiments/learnability/run_experiment.py and the curriculum runners under src/experiments/. The hyperparameters listed in Table 10 are identical across the three algorithms (IQL, VDN, QMIX) and are reused unchanged for the data-scaling and curriculum runs; the only addition is the per-stage exploration reset used by the curriculum runs, described in Section 7.11.

Hyperparameter	Value	Source
Optimiser	Adam	<code>marl.algos.{DQN,VDN,QMix}</code> default
Learning rate	5×10^{-4}	<code>run_experiment.py</code> (<code>lr=5e-4</code>)
Batch size	64	<code>run_experiment.py</code> (<code>batch_size=64</code>)
Discount factor γ	0.95	<code>run_experiment.py</code> (<code>gamma=0.95</code>)
Train interval	1 update / 5 env steps	<code>run_experiment.py</code> (<code>train_interval=(5,"step")</code>)
Gradient-norm clipping	10.0	<code>run_experiment.py</code> (<code>grad_norm_clipping=10</code>)
ε schedule (train)	linear 1.0 $\rightarrow$ 0.05 over 100,000 steps	<code>EpsilonGreedy.linear(1.0, 0.05, 100_000)</code>
Evaluation policy	greedy ($\varepsilon = 0$)	<code>ArgMax()</code>
Q-network architecture	<code>marl.qnetworks.from_env</code> default	marl framework default
Mixer (QMIX)	<code>mixers.QMix.from_env</code>	<code>run_experiment.py</code>
Mixer (VDN)	sum of agent Q -values	<code>marl.algos.VDN</code> implicit mixer
Mixer (IQL)	none	<code>run_experiment.py</code> (<code>mixer=None</code>)
Independent Q -network heads (IQL, VDN)	yes	<code>qnetworks.from_env(..., independent=True)</code>
Independent Q -network heads (QMIX)	no (shared)	<code>qnetworks.from_env(...)</code> default

Table 10: Hyperparameters used in the learnability experiment (Section 7.9) and reused unchanged for the data-scaling (Section 7.10) and curriculum (Section 7.11, Section 7.12) experiments.

A.4 SAT Encoding — Per-Family Clause Counts

The figures of Chapter 7 report total CNF size for the four benchmark levels under the two movement formulations. Table 11 gives the per-constraint-family decomposition behind those totals.

Level	Method	Initialization	Movement	Laser	Total
3×3	local	23	537	225	785
3×3	global	23	505	225	753
5×5	local	87	4 233	1 884	6 204
5×5	global	87	6 243	1 884	8 214
8×8	local	304	54 076	23 136	77 516
8×8	global	304	143 296	23 136	166 736
Level 6	local	690	174 592	77 682	252 964
Level 6	global	690	1 089 268	77 682	1 167 640

Table 11: Clause counts per constraint family for the two SAT movement formulations on the four benchmark levels. Source: results/MLG-Student-Day/benchmark_results.json.

A.5 SAT Encoding — Generation and Solve Times

Table 12 reports the mean CNF generation time and mean SAT solve time for each (level, method) pair, averaged over the 100 timing runs of the benchmark protocol described in Section 7.1. Standard deviations follow the mean in parentheses.

Level	Method	Gen time (ms)	Solve time (ms)	Total (ms)
3×3	local	0.19 ± 0.07	0.013 ± 0.004	0.20
3×3	global	0.23 ± 0.18	0.018 ± 0.006	0.25
5×5	local	2.0 ± 2.8	0.08 ± 0.02	2.1
5×5	global	2.4 ± 3.3	0.17 ± 0.03	2.6
8×8	local	28 ± 11	4.8 ± 0.6	33
8×8	global	65 ± 15	76 ± 7	141
Level 6	local	116 ± 8	7.1 ± 0.9	123
Level 6	global	486 ± 29	58 ± 3	544

Table 12: Mean CNF generation, SAT solve, and total times (milliseconds) per level and movement formulation, averaged over 100 timing runs. Source: results/MLG-Student-Day/benchmark_results.json.

A.6 Generator Rejection — Detailed Numbers

The figures of Section 7.6 report per-generator rejection rates and mean attempts. Table 13 gives the full numbers behind those plots, including the number of successful trials, the number of failed trials (per-trial budget exhausted), and the mean number of attempts per accepted level.

Generator	Grid	Success	Fail	Mean attempts	Rejection (%)
Constrained Random (solvable)	3×3	200	0	3.5	71.2
Constrained Random (solvable)	5×5	200	0	7.6	86.8
Constrained Random (solvable)	8×8	19	1	6.2	83.9
Constrained Random (cooperative)	3×3	200	0	79.5	98.7
Constrained Random (cooperative)	5×5	200	0	68.8	98.5
Constrained Random (cooperative)	8×8	—	20	—	—
Constructive (solvable)	3×3	200	0	1.00	0.0
Constructive (solvable)	5×5	200	0	1.10	9.5
Constructive (solvable)	8×8	20	0	1.05	4.8
Constructive (cooperative)	3×3	200	0	1.00	0.0
Constructive (cooperative)	5×5	200	0	1.06	6.1
Constructive (cooperative)	8×8	20	0	1.05	4.8
Level-6-Style	3×3	200	0	78.3	98.7
Level-6-Style	5×5	200	0	5.3	81.1
Level-6-Style	8×8	20	0	4.2	76.2

Table 13: Detailed rejection-benchmark numbers per generator setting and grid size. “Success” is the number of successful trials and “Fail” the number of trials that exhausted their per-trial budget (500 attempts for the small grids; 100 attempts or 30 seconds for the 8×8 grid). The 8×8 Constrained Random cooperative row is missing because the LLE C extension segfaulted during the benchmark on a randomly-sampled world. Source: `results/rejection_benchmark/benchmark_results.json`.

A.7 Cooperation Profile Distribution – Detailed Counts

The figure of Section 7.7 shows the cooperation-profile breakdown of accepted cooperative levels for three generator settings and two grid sizes. Table 14 gives the raw counts.

Generator	Grid	n	asym.	mutual	chain	distr.	full
Constrained Random (cooperative)	5×5	100	100	0	0	0	0
Constrained Random (cooperative)	8×8	50	46	0	1	3	0
Constructive (cooperative)	5×5	100	100	0	0	0	0
Constructive (cooperative)	8×8	50	2	46	0	2	0
Level-6-Style	5×5	100	100	0	0	0	0
Level-6-Style	8×8	50	13	34	0	3	0

Table 14: Raw profile-count breakdown per generator and grid size. The columns **asym.**, **mutual**, **chain**, **distr.**, **full** correspond to the five cooperation-profile labels classified by the analyzer of Section 5.6. The 5×5 configuration uses 2 agents and 1 laser; the 8×8 configuration uses 3 agents and 2 lasers. Source: `results/profile_benchmark/benchmark_results.json`.

A.8 Learnability — Per-Seed Final Success Rates

Table 15 lists the final greedy success rate of every (algorithm, seed) cell in the learnability experiment of Section 7.9. Each row is one trained agent; “Train” is the success rate on the 20-level training pool and “Test” the success rate on the 20-level held-out pool, each estimated from 200 greedy evaluation episodes.

Algorithm	Seed	Train	Test
IQL	0	0.56	0.17
IQL	1	0.62	0.38
IQL	2	0.55	0.31
IQL	3	0.67	0.14
IQL	4	0.67	0.25
IQL	5	0.59	0.2
IQL	6	0.73	0.17
IQL	7	0.58	0.19
IQL	8	0.51	0.28
IQL	9	0.5	0.33
IQL	10	0.65	0.19
IQL	11	0.62	0.25
IQL	12	0.59	0.3
IQL	13	0.61	0.25
IQL	14	0.66	0.18
IQL	15	0.5	0.25
IQL	16	0.49	0.24
IQL	17	0.61	0.18
IQL	18	0.76	0.18
IQL	19	0.6	0.17
QMIX	0	0.54	0.19
QMIX	1	0.62	0.18
QMIX	2	0.55	0.23
QMIX	3	0.61	0.19
QMIX	4	0.64	0.08
QMIX	5	0.53	0.23
QMIX	6	0.59	0.32
QMIX	7	0.57	0.28
QMIX	8	0.56	0.24
QMIX	9	0.6	0.28
QMIX	10	0.48	0.26
QMIX	11	0.69	0.19
QMIX	12	0.73	0.13
QMIX	13	0.76	0.21
QMIX	14	0.56	0.16
QMIX	15	0.55	0.05
QMIX	16	0.6	0.07
QMIX	17	0.56	0.2
QMIX	18	0.57	0.23
QMIX	19	0.6	0.23
VDN	0	0.54	0.08
VDN	1	0.78	0.14
VDN	2	0.73	0.11
VDN	3	0.61	0.14
VDN	4	0.77	0.17

Table 15: Per-(algorithm, seed) final greedy success rates from the learnability experiment of Section 7.9. 60 rows total (3 algorithms $\times$ 20 seeds). Source: results/learnability_5x5/aggregated.json, produced by src/scripts/aggregate_learnability_results.py.

A.9 Learnability – Training Pool

Cooperative pool used as $\mathcal{D}_{\text{train}}$ for Section 7.9.

Field	Value
Pool path	results/learnability_5x5/levels/5x5_2a_1L_cooperative/train
Grid	5×5
Agents	2
Lasers	1
T_{max}	10
Generator	Constructive (cooperative mode)
Pool seed	20260618
Number of levels	20

Table 16: Parameters of the learnability training pool.

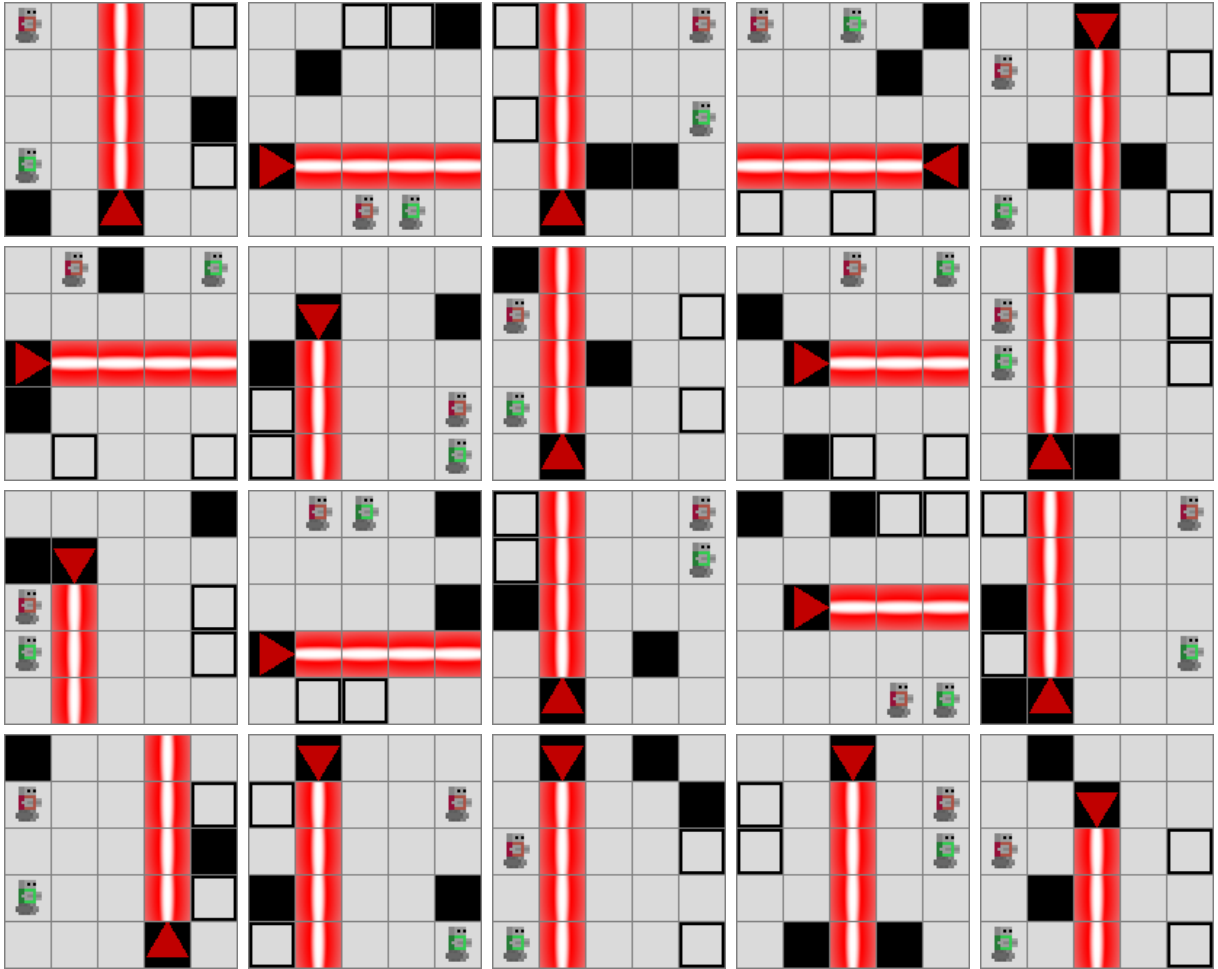


Figure 24: All 20 levels of the learnability training pool, in pool order.

A.10 Learnability – Test Pool

Held-out cooperative pool used as $\mathcal{D}_{\text{test}}$ for Section 7.9.

Field	Value
Pool path	results/learnability_5x5/levels/5x5_2a_1L_cooperative/test
Grid	5 × 5
Agents	2
Lasers	1
T_{max}	10
Generator	Constructive (cooperative mode)
Pool seed	20260619
Number of levels	20

Table 17: Parameters of the learnability test pool.

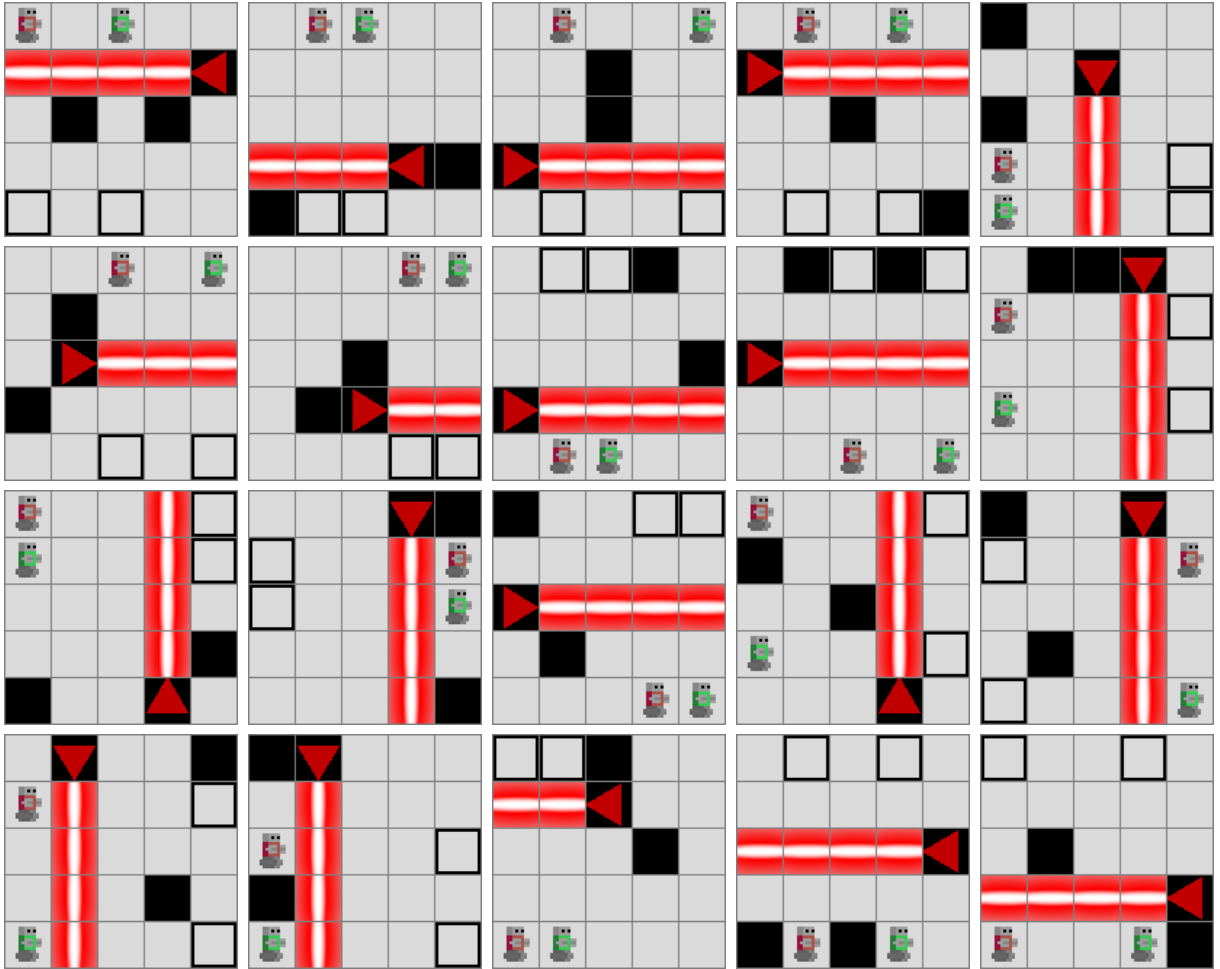


Figure 25: All 20 levels of the learnability test pool, in pool order.

A.11 Data-Scaling Experiment — Configuration

The data-scaling experiment (Section 7.10) reuses the learnability task, generator, and hyperparameters (Section A.3) unchanged, and varies only the training-pool size. Table 18 lists the full configuration.

Field	Value
Grid / agents / lasers	$5 \times 5 / 2 / 1$
Horizon $T_{\max}$	10
Generator	Constructive (cooperative mode)
Training-pool sizes $ \mathcal{D}_{\text{train}} $	20, 100, 500 (nested)
Held-out test-pool size $ \mathcal{D}_{\text{test}} $	50 (fixed across all conditions)
Training-pool seed	20260618
Test-pool seed	20260619
Environment steps per run	300,000
Algorithms	IQL, VDN, QMIX
Training seeds	$\mathcal{S} = \{0, 1, \dots, 19\}$
Runs per pool size	60 (3 algorithms $\times$ 20 seeds)

Table 18: Configuration of the data-scaling experiment (Section 7.10).

The three training pools are nested prefixes of a single seeded stream (seed 20260618): the 20-level pool is exactly the learnability training pool of Section A.9, the 100-level pool contains it, and the 500-level pool contains both. The 50-level held-out test pool (seed 20260619) is a superset of the 20-level learnability test pool of Section A.10. Per-level renderings of the 20-level pool therefore already appear in Figure 24 and Figure 25; the additional levels in the 100- and 500-level pools are further independent draws from the same constructive cooperative generator, representative samples of which are shown in Section A.17. They are not reproduced individually here.

The per-pool-size learning curves below mirror Figure 17: each panel pair shows the train- and test-pool success rate against environment steps for IQL, VDN, and QMIX, with 95% confidence bands over the seeds. Read together, they make the effect of Section 7.10 visible step by step — as the training pool grows from 20 to 100 to 500 levels, the test curves (right panel of each figure) climb toward the training curves (left panel), i.e. the generalisation gap closes.

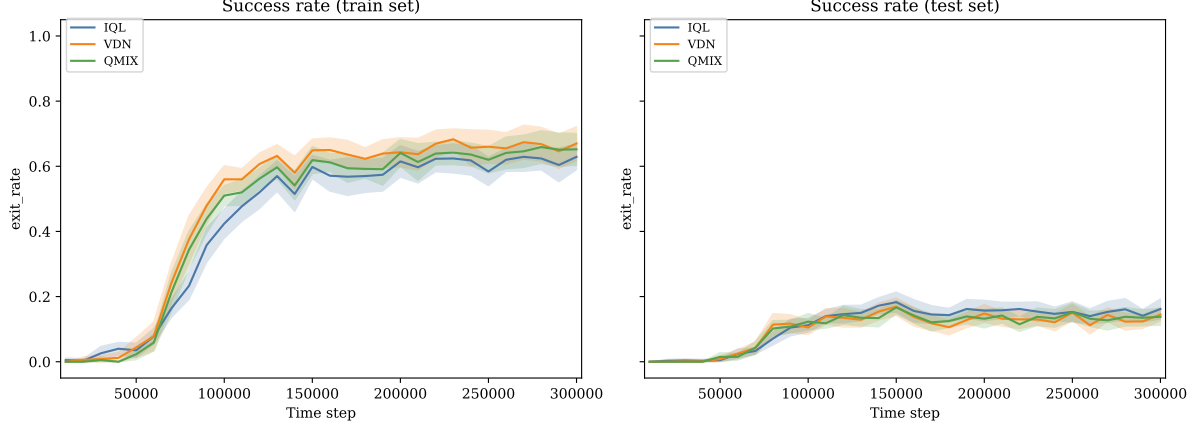


Figure 26: Data scaling with $|\mathcal{D}_{\text{train}}| = 20$ levels: mean train- and test-pool success rate versus environment steps, per algorithm, with 95% confidence bands over the seeds. The test curves plateau far below the training curves (large gap).

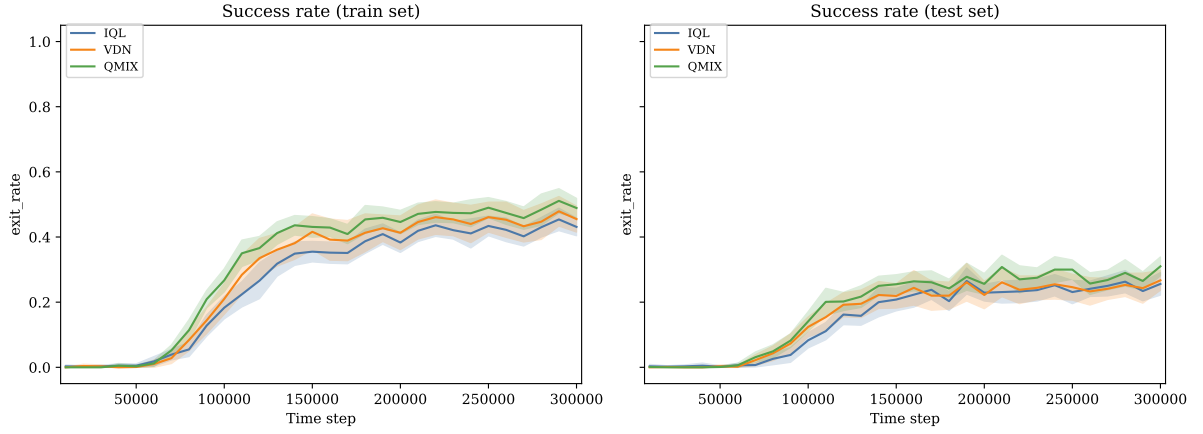


Figure 27: Data scaling with $|\mathcal{D}_{\text{train}}| = 100$ levels: the training curves drop and the test curves rise relative to the 20-level case, narrowing the gap.

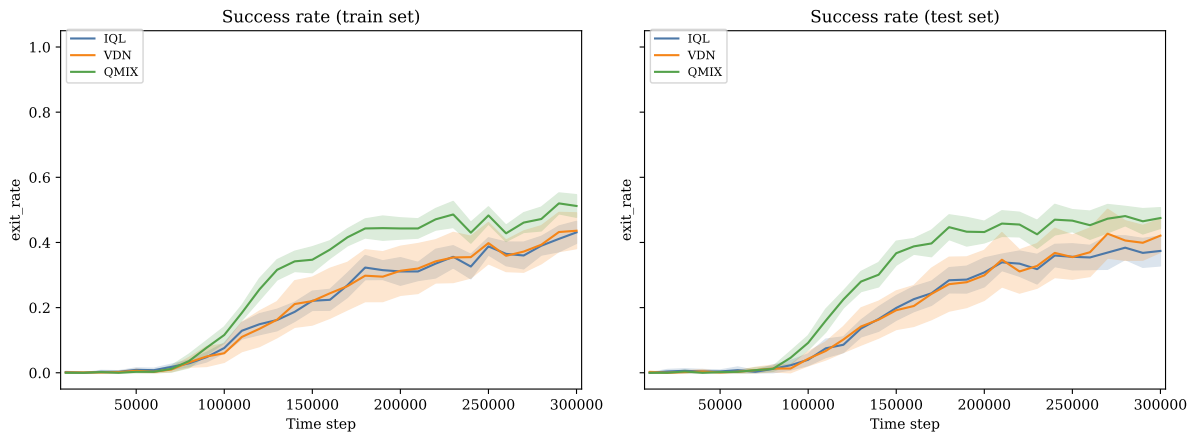


Figure 28: Data scaling with $|\mathcal{D}_{\text{train}}| = 500$ levels: the train and test curves nearly coincide — the agent generalises to held-out levels about as well as it fits the training pool.

A.12 Curriculum-Strategy Experiment — Configuration and Per-Cell Results

The curriculum-strategy experiment (Section 7.11) compares four budget-matched scheduling conditions on a 6×6 / 2-agent / 1-laser cooperative target reachable by the direct baseline. Table 19 gives the full configuration and Table 20 the per-(condition, algorithm) final success rates behind the pooled summary of Table 6.

Field	Value
Difficulty ladder	$4 \times 4/0L$ (random, $T_{\max} = 8$) $\rightarrow 5 \times 5/1L$ (cooperative, $T_{\max} = 10$) $\rightarrow 6 \times 6/1L$ target (cooperative, $T_{\max} = 12$)
Agents	2 (fixed across rungs; smaller observations zero-padded)
Train pool per rung	100 certified levels
Held-out target pool	50 levels
Conditions	direct, forward $(50, 150, 200) \times 10^3$, reverse (same per-rung budgets, reversed), mixed (uniform rung per episode)
Total budget per run	400,000 environment steps
Algorithms	IQL, VDN, QMIX
Seeds	$\mathcal{S} = \{0, 1, \dots, 7\}$ (eight per condition and algorithm)
Exploration	ε reset to 1.0 at each stage boundary, decayed to 0.05 over 30% of that rung’s budget
Evaluation	greedy, on the 6×6 target; 50 episodes every 10,000 steps and 200 episodes at the end

Table 19: Configuration of the curriculum-strategy experiment (Section 7.11).

Condition	Algorithm	n	Train mean $\pm$ CI95	Test mean $\pm$ CI95
direct	IQL	8	0.30 ± 0.06	0.13 ± 0.03
direct	VDN	8	0.43 ± 0.08	0.16 ± 0.04
direct	QMIX	8	0.50 ± 0.05	0.22 ± 0.05
forward	IQL	8	0.22 ± 0.04	0.08 ± 0.03
forward	VDN	8	0.49 ± 0.06	0.17 ± 0.06
forward	QMIX	8	0.47 ± 0.07	0.20 ± 0.06
mixed	IQL	8	0.19 ± 0.06	0.12 ± 0.05
mixed	VDN	8	0.51 ± 0.08	0.29 ± 0.08
mixed	QMIX	8	0.47 ± 0.08	0.27 ± 0.05
reverse	IQL	8	0.06 ± 0.03	0.03 ± 0.02
reverse	VDN	8	0.12 ± 0.04	0.09 ± 0.04
reverse	QMIX	8	0.12 ± 0.04	0.09 ± 0.06

Table 20: Final greedy success rate per (condition, algorithm) on the 6×6 / 2-agent / 1-laser target, each aggregated over eight seeds. Means with 95% confidence intervals ($\pm t_{7,0.025} \frac{\sigma}{\sqrt{8}}$, $t_{7,0.025} \approx 2.365$).

Source: results/curriculum_strategy/runs/.

A.12.1 Generated training pools

Each rung is SAT-generated once from the master seed `RNG_SEED = 20260521`; the per-rung training pool uses the derived seed `20260521 + 100  $\times$  stage` and the held-out target pool the same expression +1, so every draw is reproducible and the target’s train and eval pools never coincide. Unspecified wall budgets fall back to the generator default $\lfloor \text{grid area} / 10 \rfloor$. Table 21 lists the per-rung parameters; Figure 29, Figure 30, and Figure 31 show sixteen representative training levels from each rung.

Rung	Grid	Agents	Lasers	$T_{\max}$	Walls	Generator	Train	Eval
S1	4 $\times$ 4	2	0	8	1	Random (solvable)	100	0
S2	5 $\times$ 5	2	1	10	2	Constructive (cooperative)	100	0
S3	6 $\times$ 6	2	1	12	3	Constructive (cooperative)	100	50

Table 21: Per-rung generation parameters for the curriculum-strategy pools (Section 7.11).

Source: src/experiments/curriculum_strategy/configs.py; pools rendered by src/scripts/render_curriculum_strategy_pools.py.

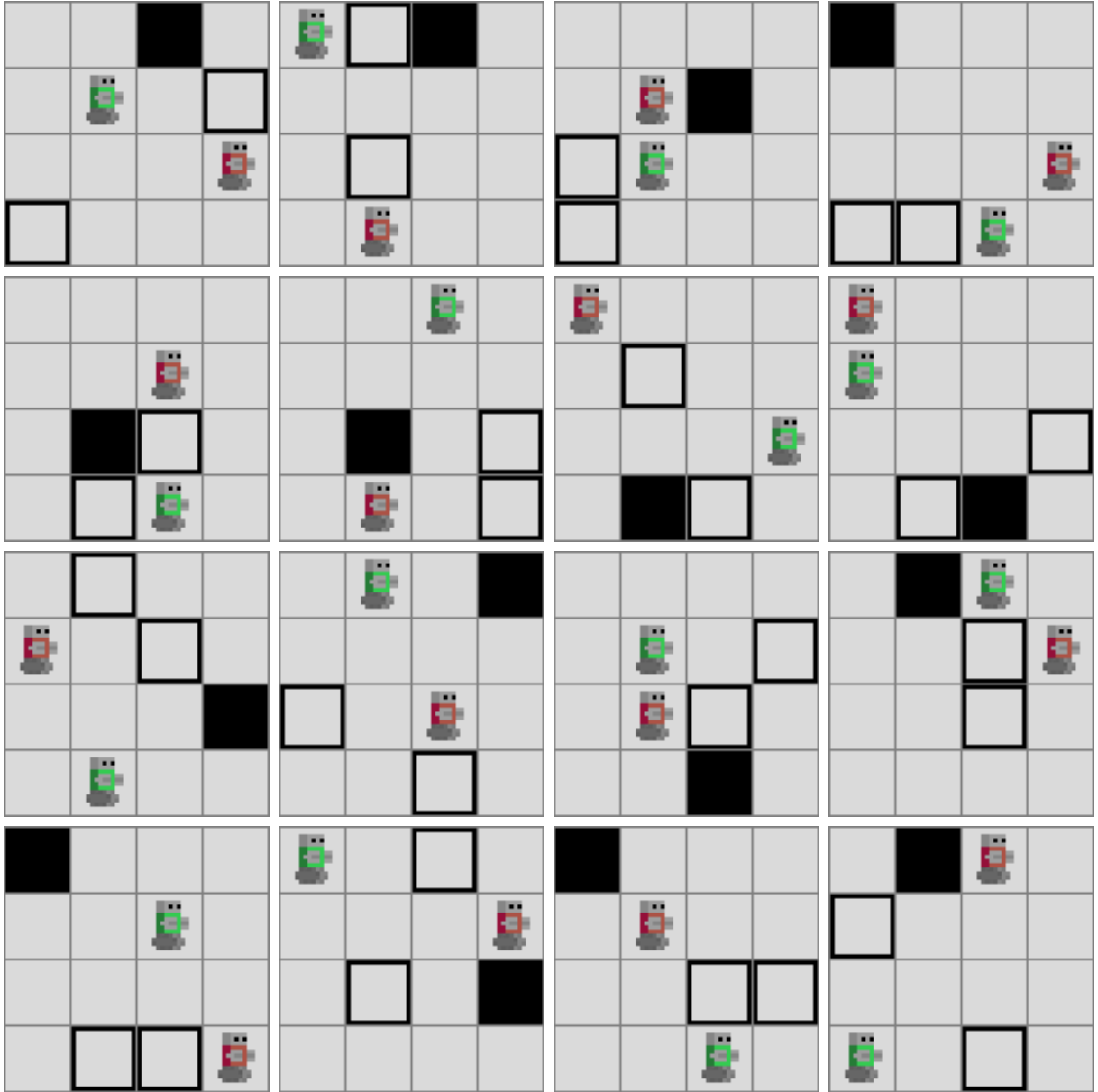


Figure 29: Sixteen S1 training levels: 4×4 , 2 agents, 0 lasers (navigation warm-up).

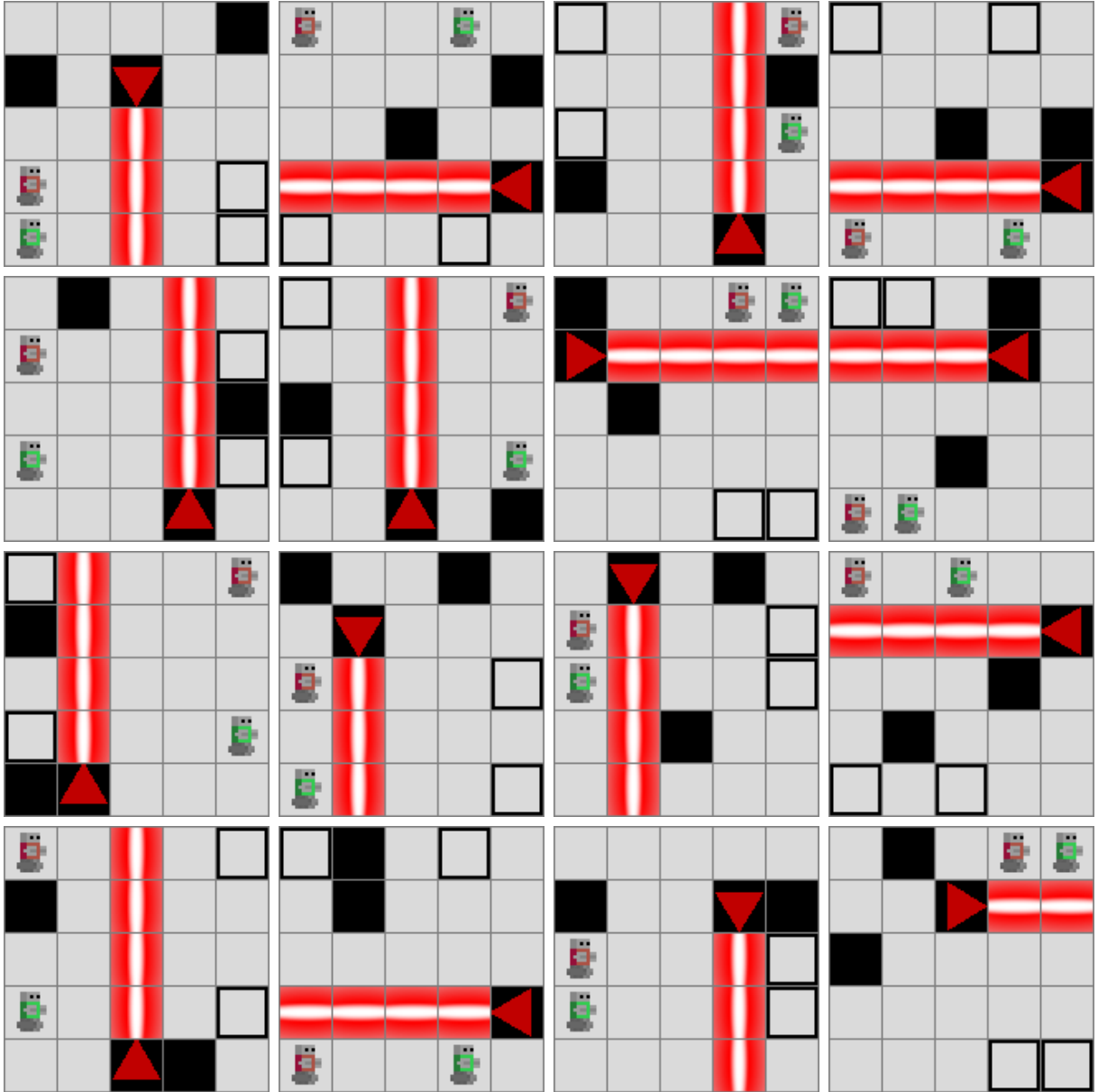


Figure 30: Sixteen S2 training levels: 5×5 , 2 agents, 1 laser (asymmetric cooperation).

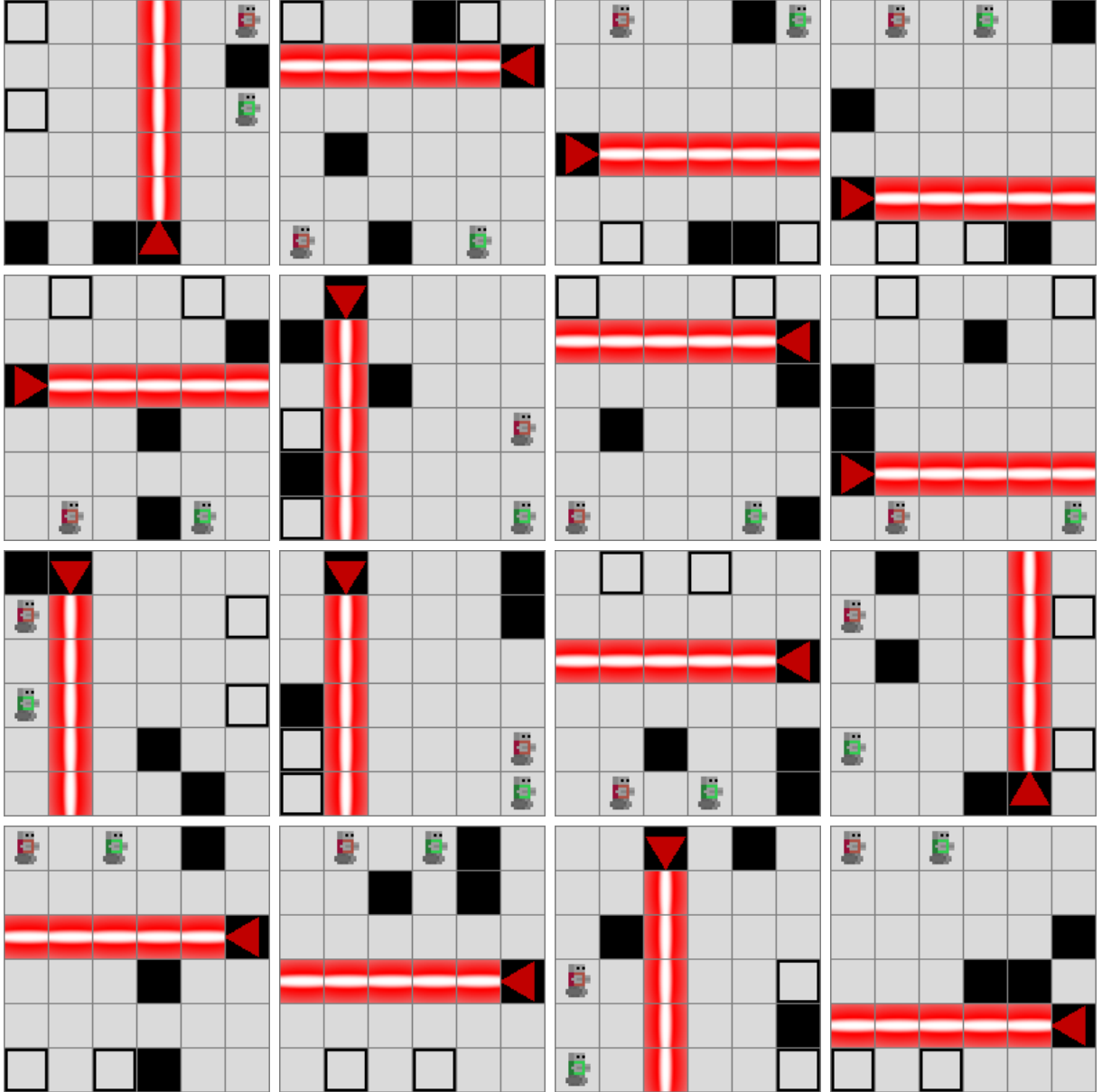


Figure 31: Sixteen S3 (target) training levels: 6×6 , 2 agents, 1 laser (asymmetric cooperation).

A.13 Curriculum-Transfer Experiments — Configurations and Results

This appendix collects the configurations and per-run results behind the three stages of Section 7.12: the frontier probe, the two-laser curriculum, and the Level-6 transfer. Across all three, held-out success on the mutually-cooperative target is uniformly zero; the tables below give the supporting numbers and the exact stage geometries.

A.13.1 Frontier Probe and Diagnostic Checks

Direct training from scratch on a fixed 6×6 / 2-agent / 2-laser **mutual** target ($T_{\max} = 18$, constructive cooperative generator constrained to the `fully_coupled` profile, a 20-level training pool and a 20-level held-out pool), 600,000 steps, three seeds per algorithm. Table 22 reports the final greedy success rates:

every algorithm reaches a small nonzero **training** success and exactly zero **held-out** success. Sixteen of the training levels are shown in Figure 32.

Algorithm	n	Train (mean)	Test (mean)
IQL	3	0.09	0.00
VDN	3	0.06	0.00
QMIX	3	0.08	0.00

Table 22: Frontier probe: final greedy success on the 6×6 / 2-agent / 2-laser mutual target, direct training from scratch. Source: results/learnability_6x6_2L/runs/.

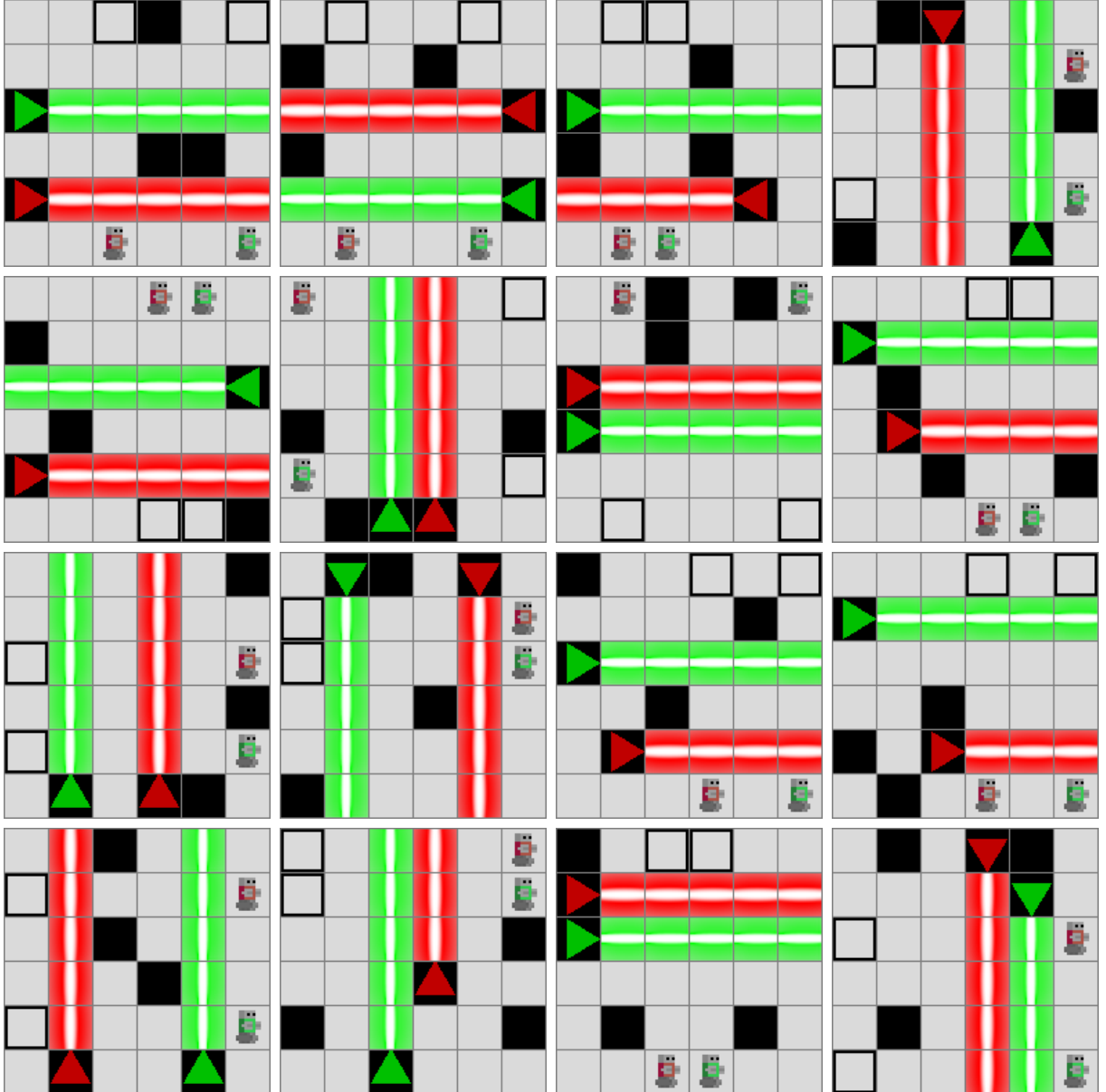


Figure 32: Sixteen of the twenty frontier-probe training levels: 6×6 , 2 agents, 2 lasers, fully_coupled cooperation profile. Source: results/learnability_6x6_2L/levels_png/train.

Two diagnostic checks rule out a budget shortfall rather than a learnability wall. An **overfit gate** trains a single algorithm (QMIX) on one fixed mutual level for 600,000 steps — removing the need to generalise entirely — and still plateaus below 0.12 training success with zero held-out success. A

budget-bump gate extends the budget to $1\{, \}500\{, \}000$ steps (VDN, 100-level pool): success stays flat, peaking early (around 0.10) and then decaying to zero rather than climbing. Finally, shrinking the grid to 5×5 / 2-laser (VDN, 200,000 steps) yields 0.00 on both train and test pools — two lasers drive success to zero independently of grid size.

A.13.2 Two-Laser Curriculum

The four scheduling conditions of Section 7.11, on a fixed 6×6 grid with a zero-, one-, then two-laser ladder to the mutual two-laser target, budget-matched at 600,000 steps. Owing to the cost of the runs and the uniformly zero outcome, coverage is partial: **direct** was run with VDN (six seeds) and QMIX (three seeds); **forward**, **reverse**, and **mixed** with VDN (three seeds each). Table 23 reports the final held-out success.

Condition	Algorithms (seeds)	Held-out test (mean)	Train (mean)
direct	VDN (6), QMIX (3)	0.00	0.08
forward	VDN (3)	0.00	0.05
mixed	VDN (3)	0.01	0.01
reverse	VDN (3)	0.00	0.00

Table 23: Two-laser curriculum: final greedy success on the 6×6 / 2-agent / 2-laser mutual target per scheduling condition. Source: results/curriculum_strategy_2L/runs/.

The fixed-grid ladder is SAT-generated once from RNG_SEED = 20260523 (per-rung seed 20260523 + $100 \times$ stage, +1 for the held-out target pool). Only the two-laser target rung (S3) is constrained to the fully_coupled cooperation profile; the lower rungs accept any solvable (S1) or cooperative (S2) level. Table 24 lists the per-rung parameters and Figure 33, Figure 34, and Figure 35 show sixteen representative training levels from each rung.

Rung	Grid	Agents	Lasers	$T_{\max}$	Walls	Generator	Profile	Train	Eval
S1	6×6	2	0	12	3	Random (solvable)	—	100	0
S2	6×6	2	1	14	3	Constructive (cooperative)	any	100	0
S3	6×6	2	2	18	3	Constructive (cooperative)	fully_coupled	100	50

Table 24: Per-rung generation parameters for the two-laser curriculum pools (Section 7.12). Source: src/experiments/curriculum_strategy_2L/configs.py; pools rendered by src/scripts/render_curriculum_strategy_2L_pools.py.

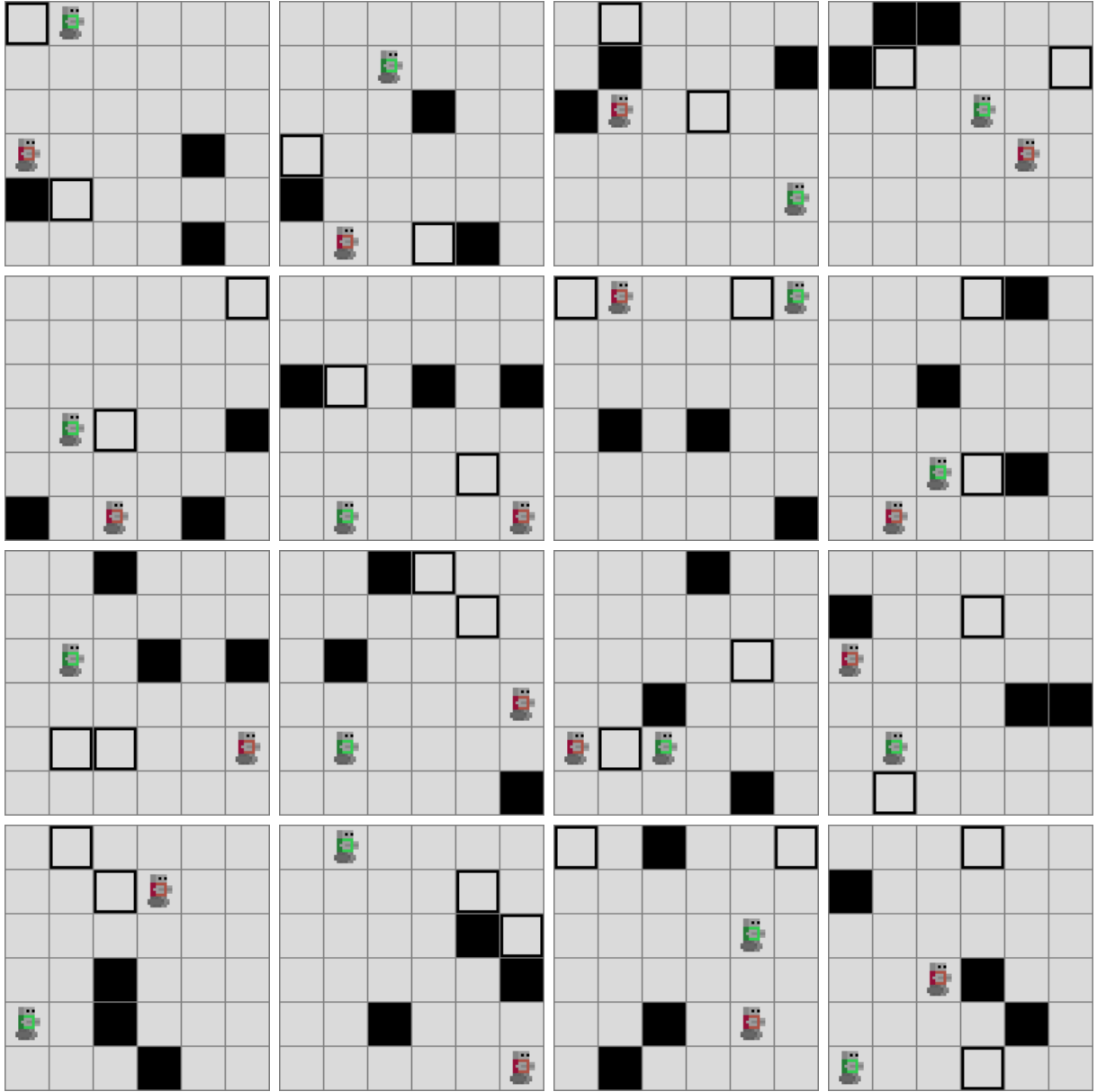


Figure 33: Sixteen S1 training levels: 6×6 , 2 agents, 0 lasers (navigation warm-up).

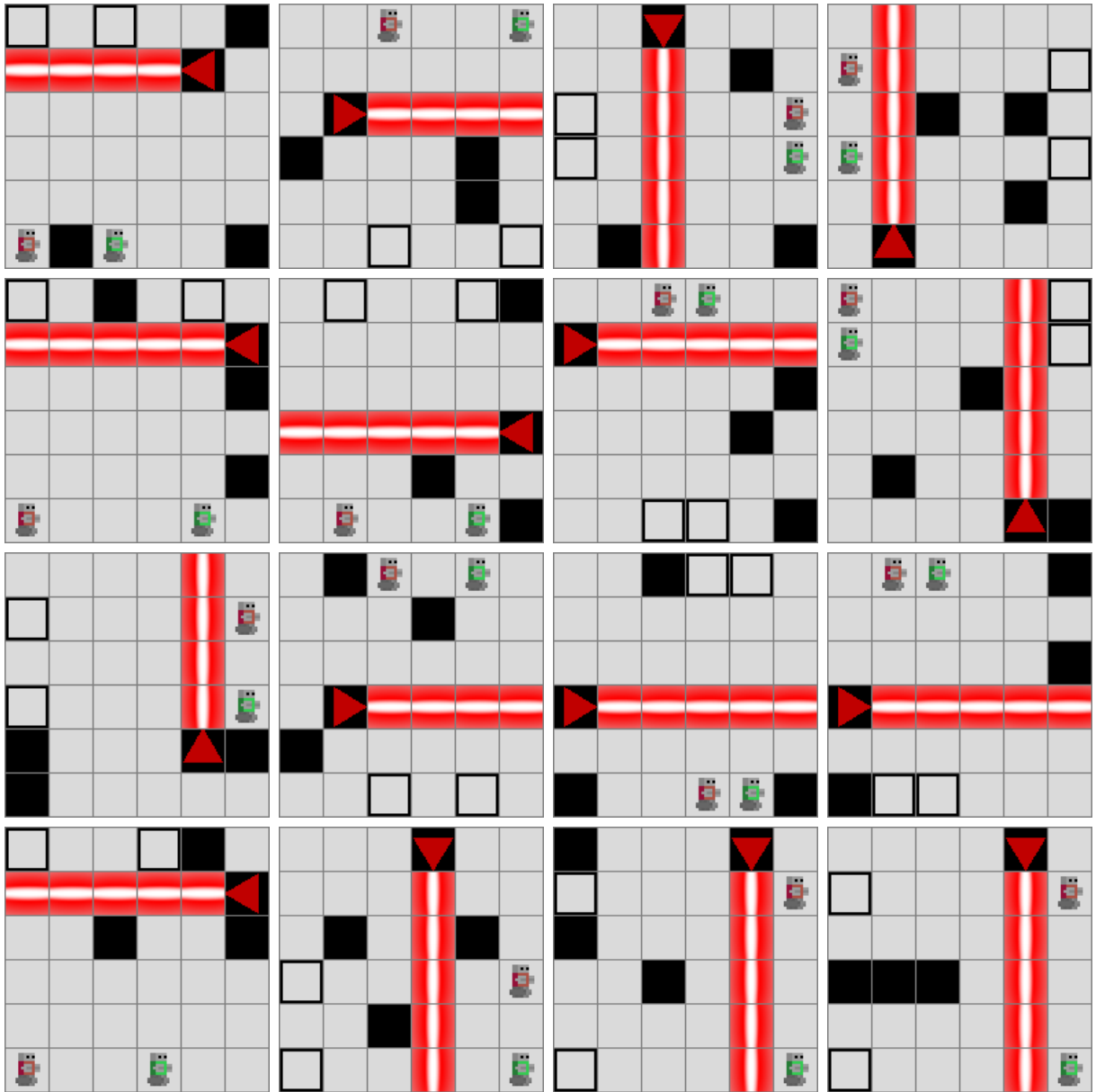


Figure 34: Sixteen S2 training levels: 6×6 , 2 agents, 1 laser (asymmetric cooperation).

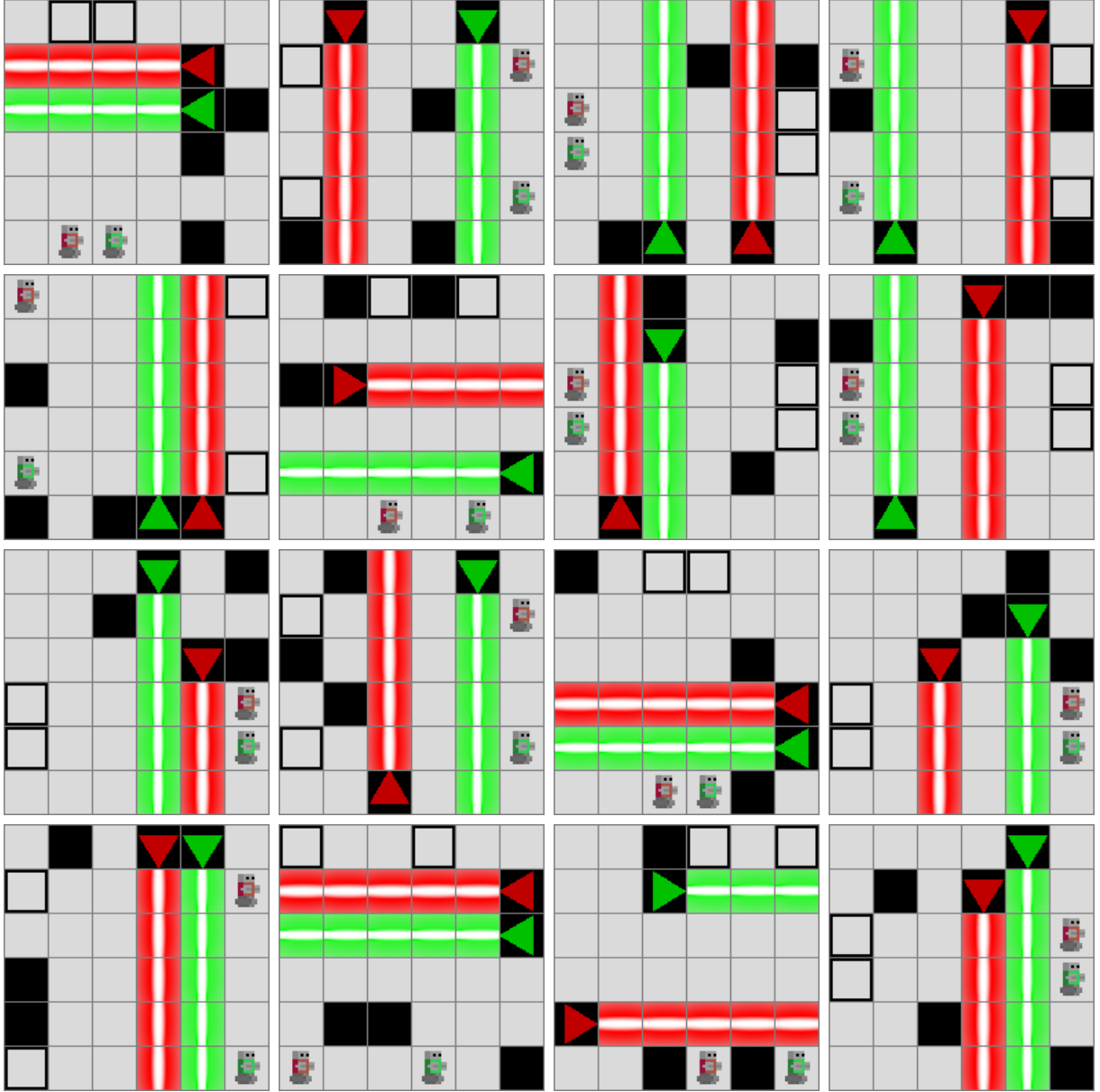


Figure 35: Sixteen S3 (target) training levels: 6×6 , 2 agents, 2 lasers, fully_coupled profile (the mutual two-laser target).

A.13.3 Level-6 Transfer

A four-stage curriculum of generated levels growing in geometry and cooperation, four agents throughout, with the gem reward set to zero in every environment so that the reward structure is consistent between the gem-free generated levels and the gem-bearing Level 6. Table 25 gives the stage ladder and Table 26 the per-condition outcomes. Every condition — including the full curriculum at two million steps — scores zero on Level 6 **and** on the in-distribution held-out generated pool; mean Level-6 return stays negative throughout. The stage pools are SAT-generated from master seed `RNG_SEED = 20260514`, and the runs were produced against `marl` commit 23c4d233. The stage pools themselves were not retained on disk; representative levels for the four stage generators at comparable grid sizes appear in Section A.15 (stage 1), Section A.17 (stages 2–3), and Section A.19 (stage 4, the 12×13 Level-6 footprint).

Stage	Grid	Lasers	Generator
1	6×6	1	Random (solvable)
2	8×8	2	Constructive (cooperative)
3	10×10	3	Constructive (cooperative)
4	12×13	3	Level-6-Style

Table 25: Level-6 transfer curriculum stages (condition CURR). Four agents throughout. Source: `src/experiments/curriculum/configs.py`.

Condition	Description	Steps	SR Level 6	SR generated pool
B1	Hardest stage only	750k	0.00	0.00
B2	Anti-curriculum (hard→easy)	750k–2M	0.00	0.00
B3	Direct on Level 6	750k	0.00	0.00
CURR	Four-stage curriculum	up to 2M	0.00	0.00

Table 26: Level-6 transfer: greedy success rate (200 episodes) on the hand-crafted Level 6 and on the in-distribution held-out generated pool, per condition. Runs use QMIX and VDN with one to four seeds per condition. Mean Level-6 return is negative for all trained conditions (agents accrue step penalties without exiting). Source: `results/curriculum_experiment/runs/`.

A.14 Generator Gallery — Base Random Generator

Pure random sampling with **no** geometric validation: the generator places agents, exits, walls, and laser sources uniformly at random and accepts any layout the SAT oracle certifies as solvable, regardless of laser geometry. Every level shown here is therefore solvable — the contrast with the Constrained Random generator of the next section is one of geometric **quality**, not of solvability.

The characteristic artefact is visible throughout the pools below: most levels contain a laser source that emits **no active beam**, because the source points immediately off the grid or sits flush against a wall or edge. Such a source is inert — it behaves like an ordinary wall tile rather than a laser — so the level presents none of the beam-crossing structure the laser was meant to introduce. Exits and agent starts may likewise fall on beam tiles (the LLE engine silently relocates an agent start that would be killed on spawn). These are exactly the degeneracies that the geometric filters of the Constrained Random generator (Section A.15) are designed to reject. The same grid, agent, laser, and wall parameters are used here as in that section, so the two galleries can be read side by side. Pools are generated by `src/scripts/generate_appendix_galleries.py`.

A.14.1 Base Random — 3×3, 2 agents, 1 laser

Grid	3 × 3
Agents	2
Lasers	1
Horizon $T_{\max}$	6
Wall budget	1
Seed	20260601
Samples in pool	16

Table 27: Parameters and seed for the Base Random 3×3 gallery pool.

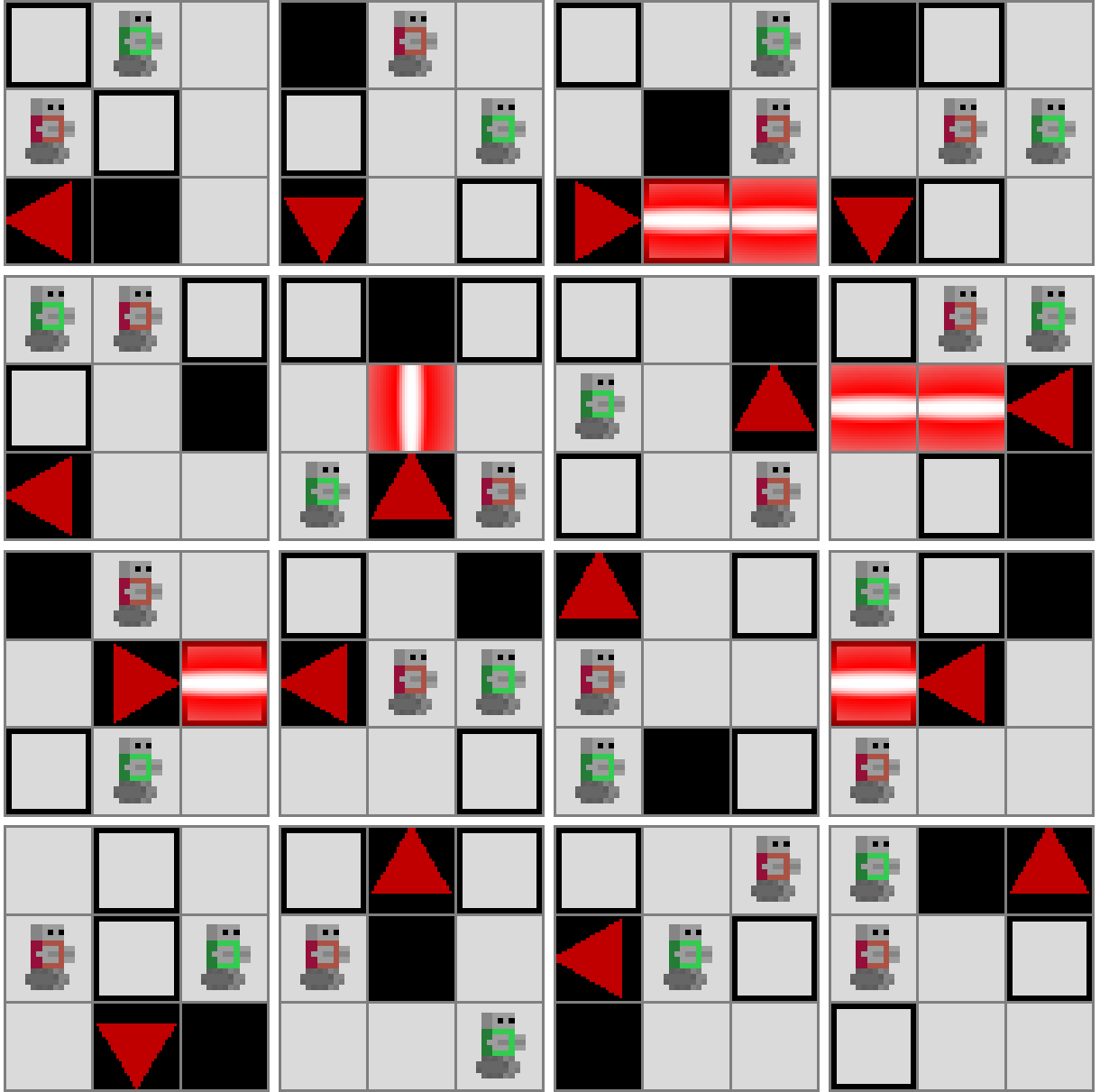


Figure 36: 16 Base Random 3×3 (2 agents, 1 laser) levels, no geometric validation.

A.14.2 Base Random — 5×5, 3 agents, 2 lasers

Grid	5 × 5
Agents	3
Lasers	2
Horizon $T_{\max}$	10
Wall budget	2
Seed	20260602
Samples in pool	16

Table 28: Parameters and seed for the Base Random 5×5 gallery pool.

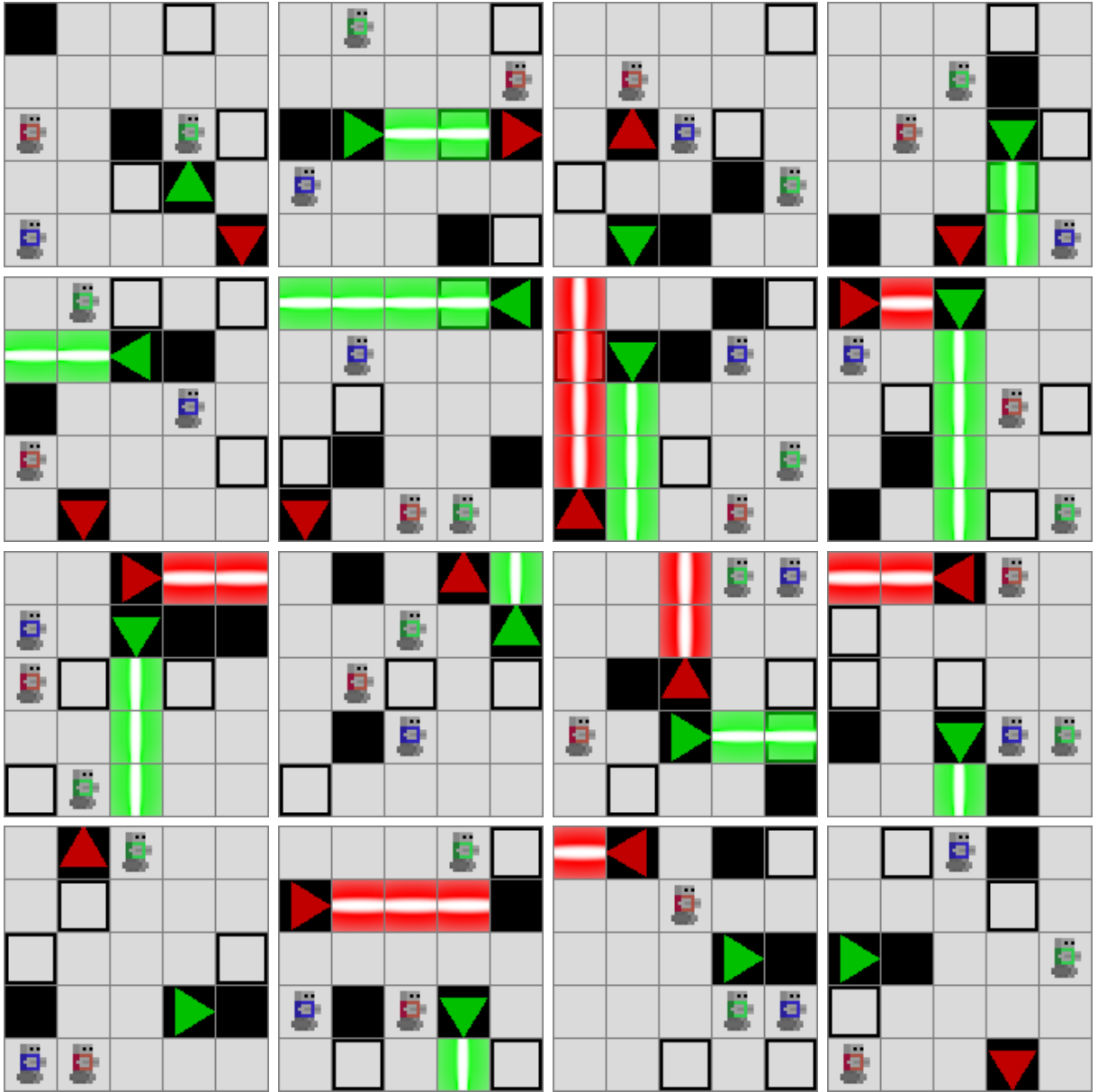


Figure 37: 16 Base Random 5×5 (3 agents, 2 lasers) levels, no geometric validation.

A.14.3 Base Random — 7×7, 4 agents, 2 lasers

Grid	7 × 7
Agents	4
Lasers	2
Horizon $T_{\max}$	14
Wall budget	4
Seed	20260603
Samples in pool	16

Table 29: Parameters and seed for the Base Random 7×7 gallery pool.

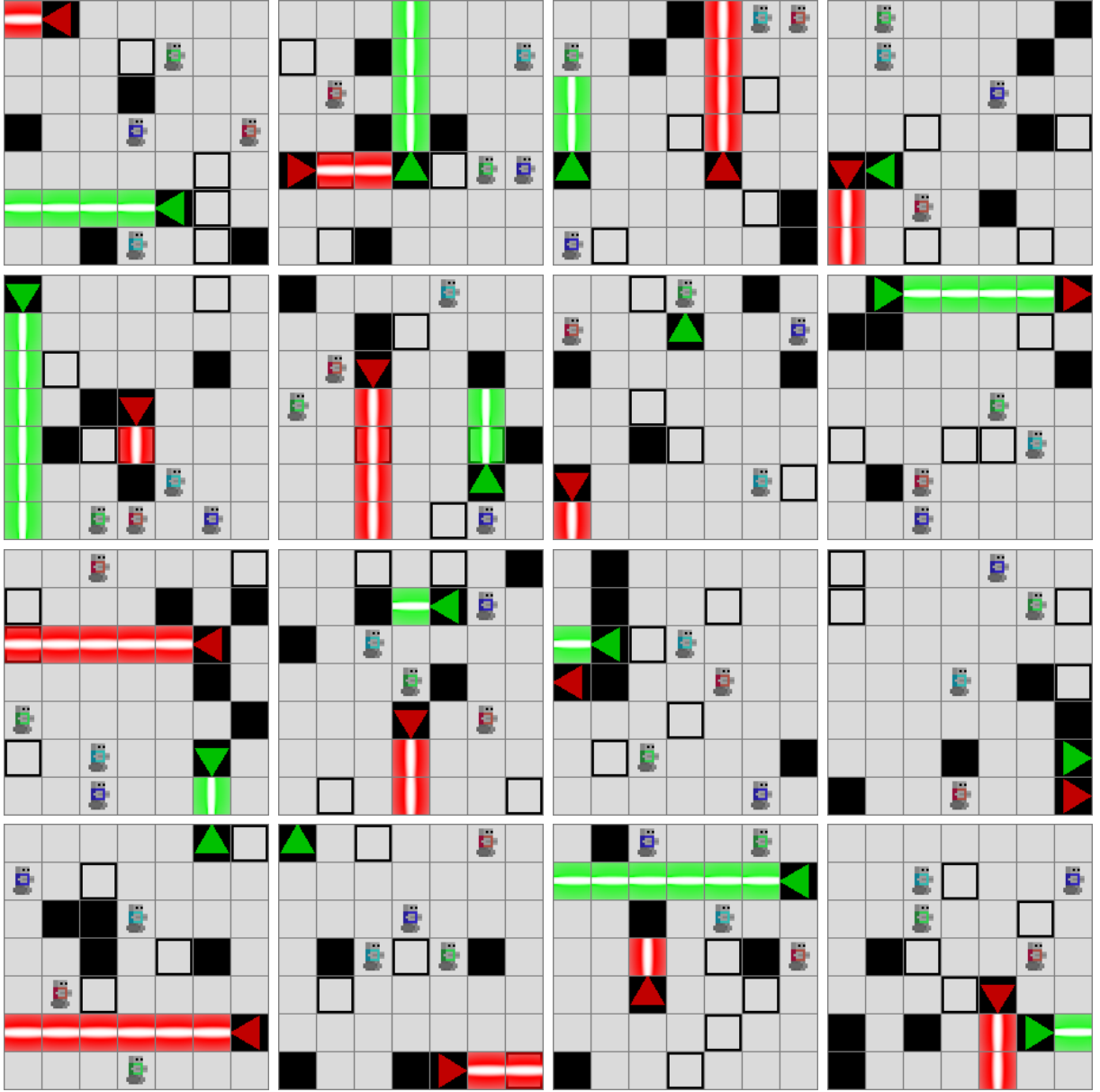


Figure 38: 16 Base Random 7×7 (4 agents, 2 lasers) levels, no geometric validation.

A.15 Generator Gallery — Constrained Random Generator

Random sampling plus geometric filters (no laser pointing immediately out of bounds, no zero-length beam, no exit on an unavoidable beam segment, etc.). Compared with the Base Random generator of Section A.14 — which uses the identical grid, agent, laser, and wall parameters but no geometric pre-filter — these filters remove the cosmetically degenerate layouts (inert laser sources that emit no beam, exits stranded on beam tiles) so that accepted levels actually exhibit the beam-crossing structure the experiments rely on. Pools are generated by `src/scripts/generate_appendix_galleries.py`.

A.15.1 Constrained Random — 3×3, 2 agents, 1 laser

Grid	3 × 3
Agents	2
Lasers	1
Horizon $T_{\max}$	6
Wall budget	1
Seed	20260611
Samples in pool	16

Table 30: Parameters and seed for the Constrained Random 3×3 gallery pool.

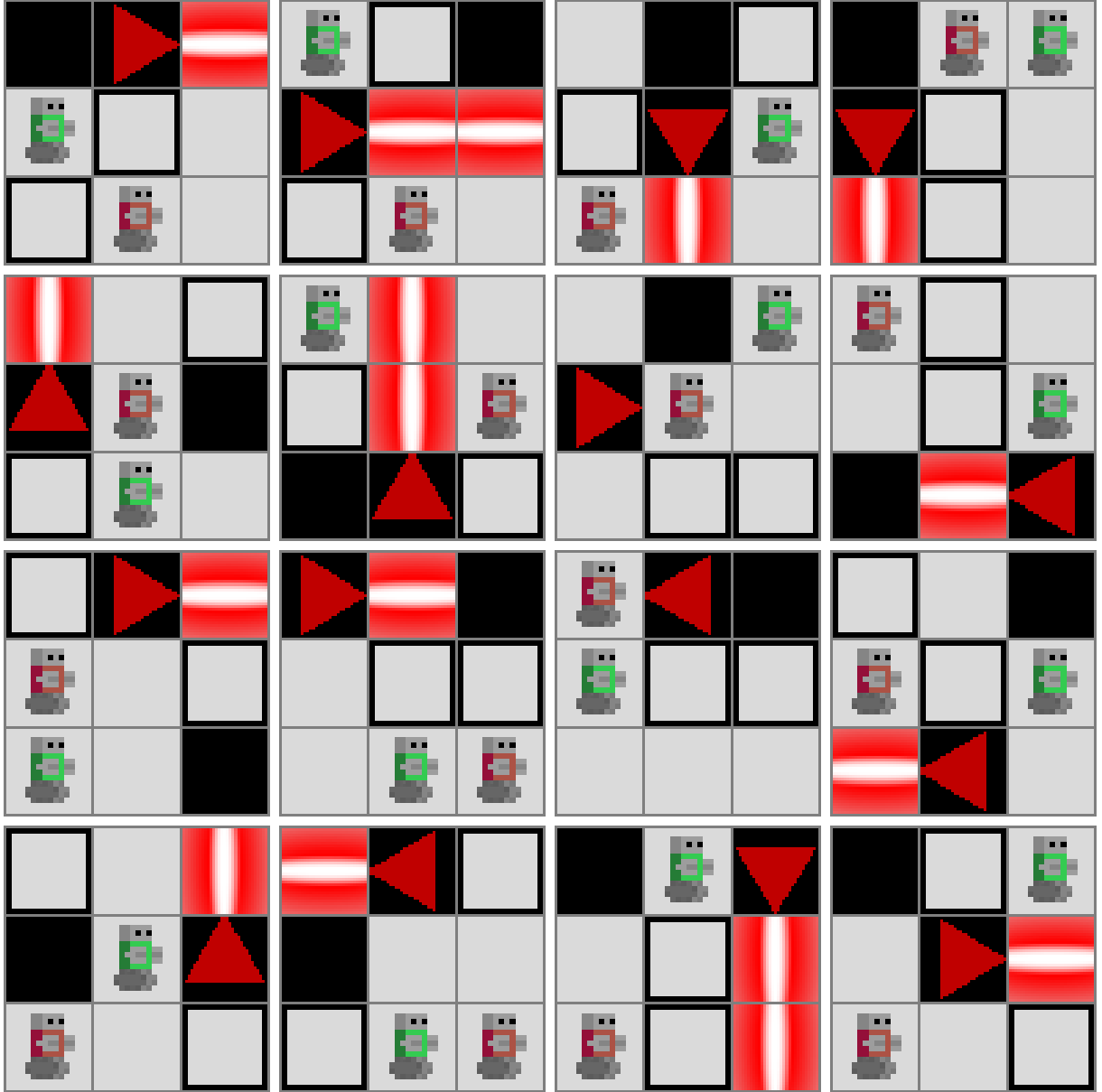


Figure 39: 16 Constrained Random 3x3 (2 agents, 1 laser) levels.

A.15.2 Constrained Random — 5x5, 3 agents, 2 lasers

Grid	5 × 5
Agents	3
Lasers	2
Horizon $T_{\max}$	10
Wall budget	2
Seed	20260612
Samples in pool	16

Table 31: Parameters and seed for the Constrained Random 5x5 gallery pool.

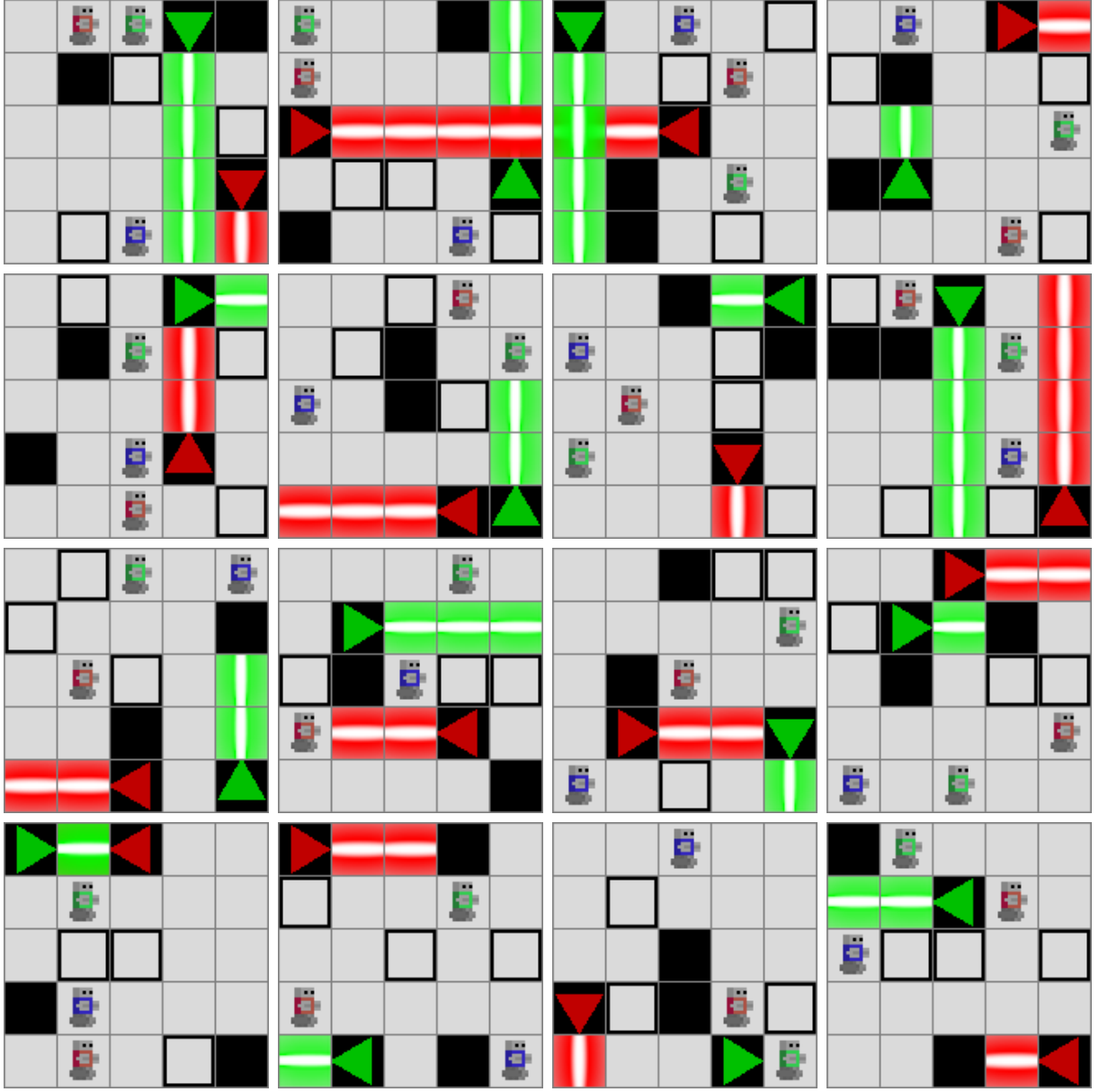


Figure 40: 16 Constrained Random 5×5 (3 agents, 2 lasers) levels.

A.15.3 Constrained Random — 7×7 , 4 agents, 2 lasers

Grid	7×7
Agents	4
Lasers	2
Horizon $T_{\max}$	14
Wall budget	4
Seed	20260613
Samples in pool	16

Table 32: Parameters and seed for the Constrained Random 7×7 gallery pool.

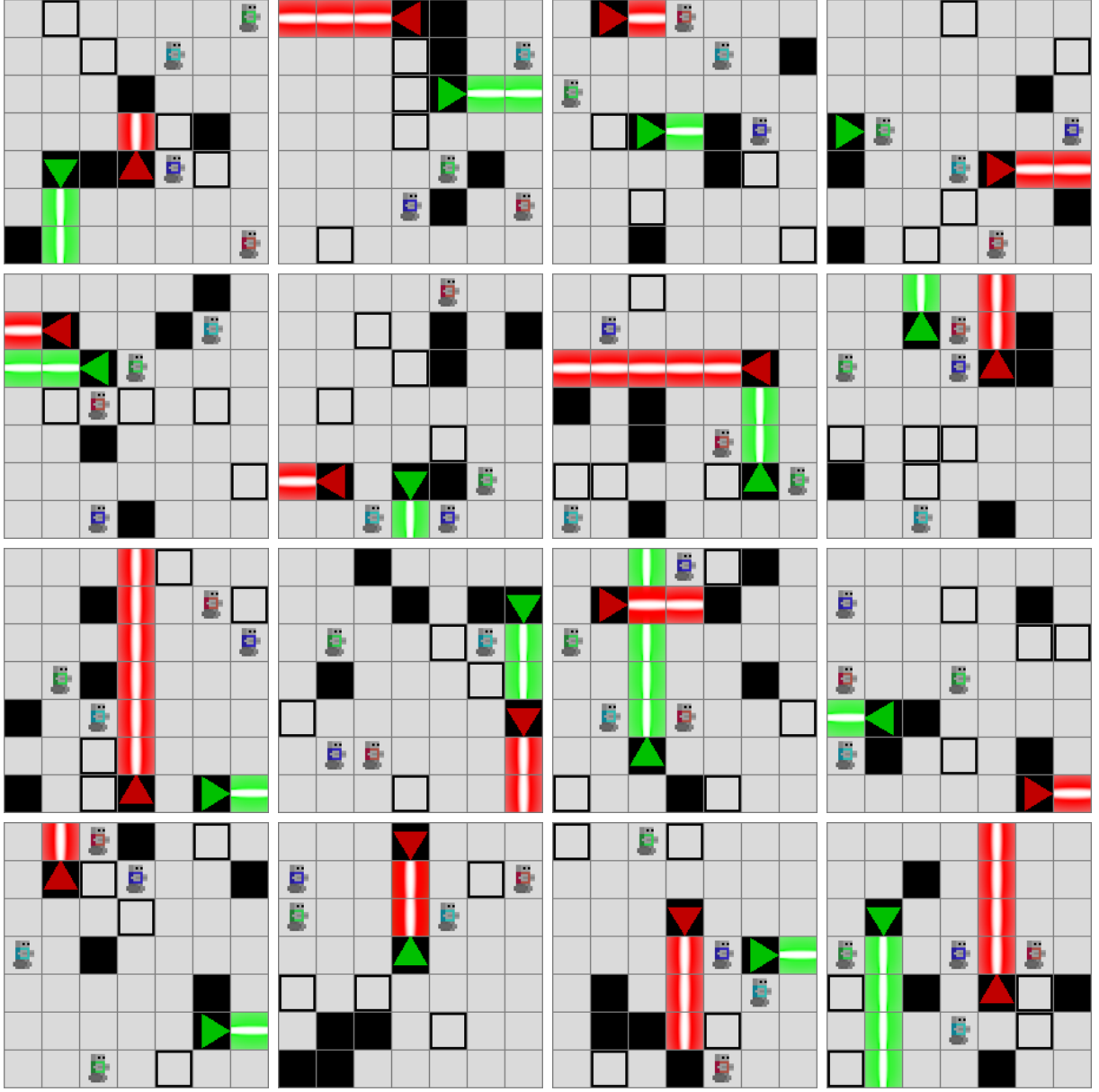


Figure 41: 16 Constrained Random 7×7 (4 agents, 2 lasers) levels.

A.16 Generator Gallery – Constructive Generator (Solvable)

Lane-based construction (one disjoint lane per agent) followed by random wall placement on the remaining cells. Cooperation is not required at generation time; only solvability is certified by the SAT oracle.

A.16.1 Constructive Solvable — 5×5, 3 agents, 1 laser

Grid	5 × 5
Agents	3
Lasers	1
Horizon $T_{\max}$	10
Wall budget	3
Seed	20260621
Samples in pool	16

Table 33: Parameters and seed for the Constructive Solvable 5×5 gallery pool.

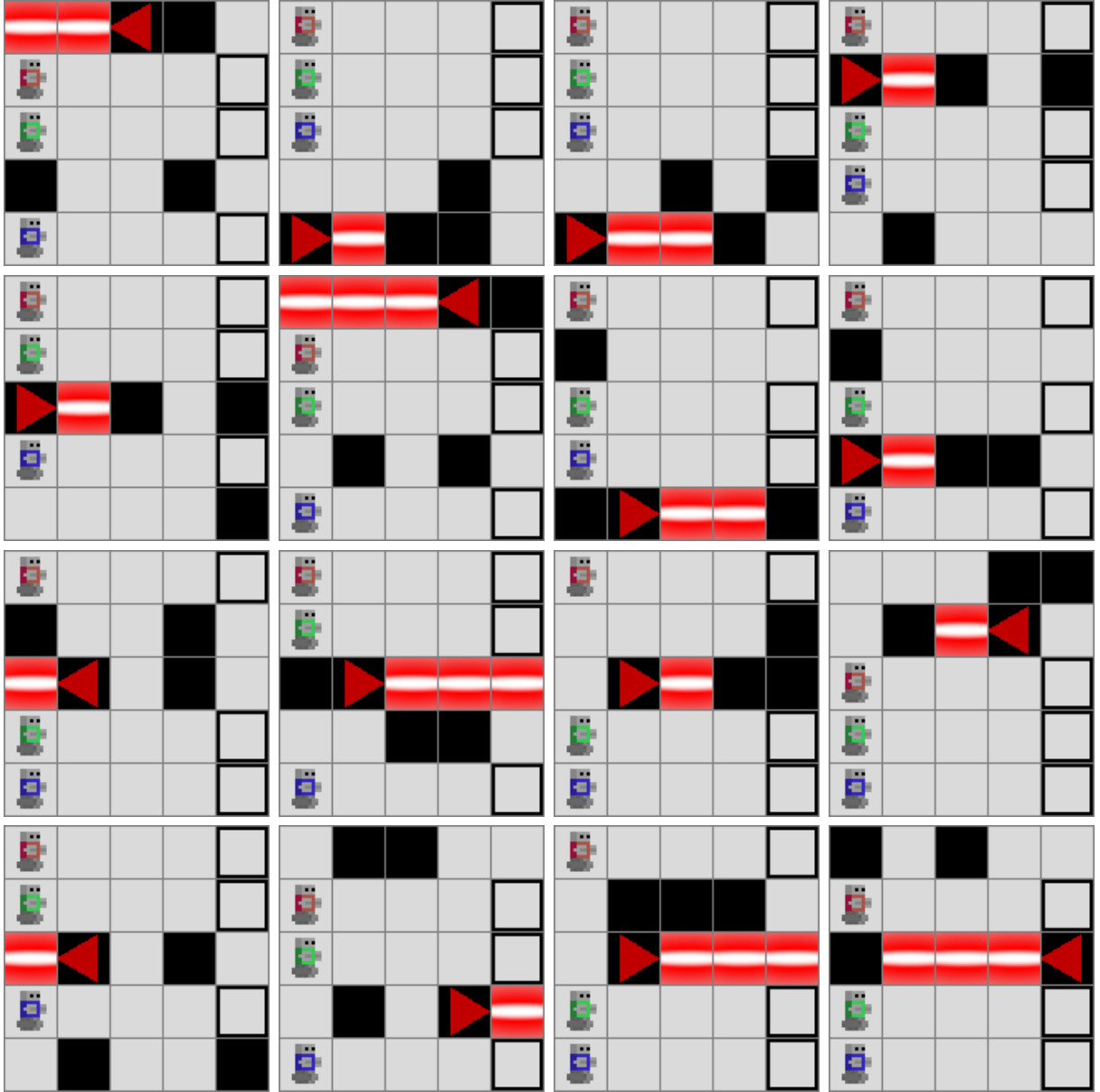


Figure 42: 16 Constructive Solvable 5×5 (3 agents, 1 laser) levels.

A.16.2 Constructive Solvable — 7×7, 4 agents, 2 lasers

Grid	7 × 7
Agents	4
Lasers	2
Horizon $T_{\max}$	14
Wall budget	5
Seed	20260622
Samples in pool	16

Table 34: Parameters and seed for the Constructive Solvable 7×7 gallery pool.

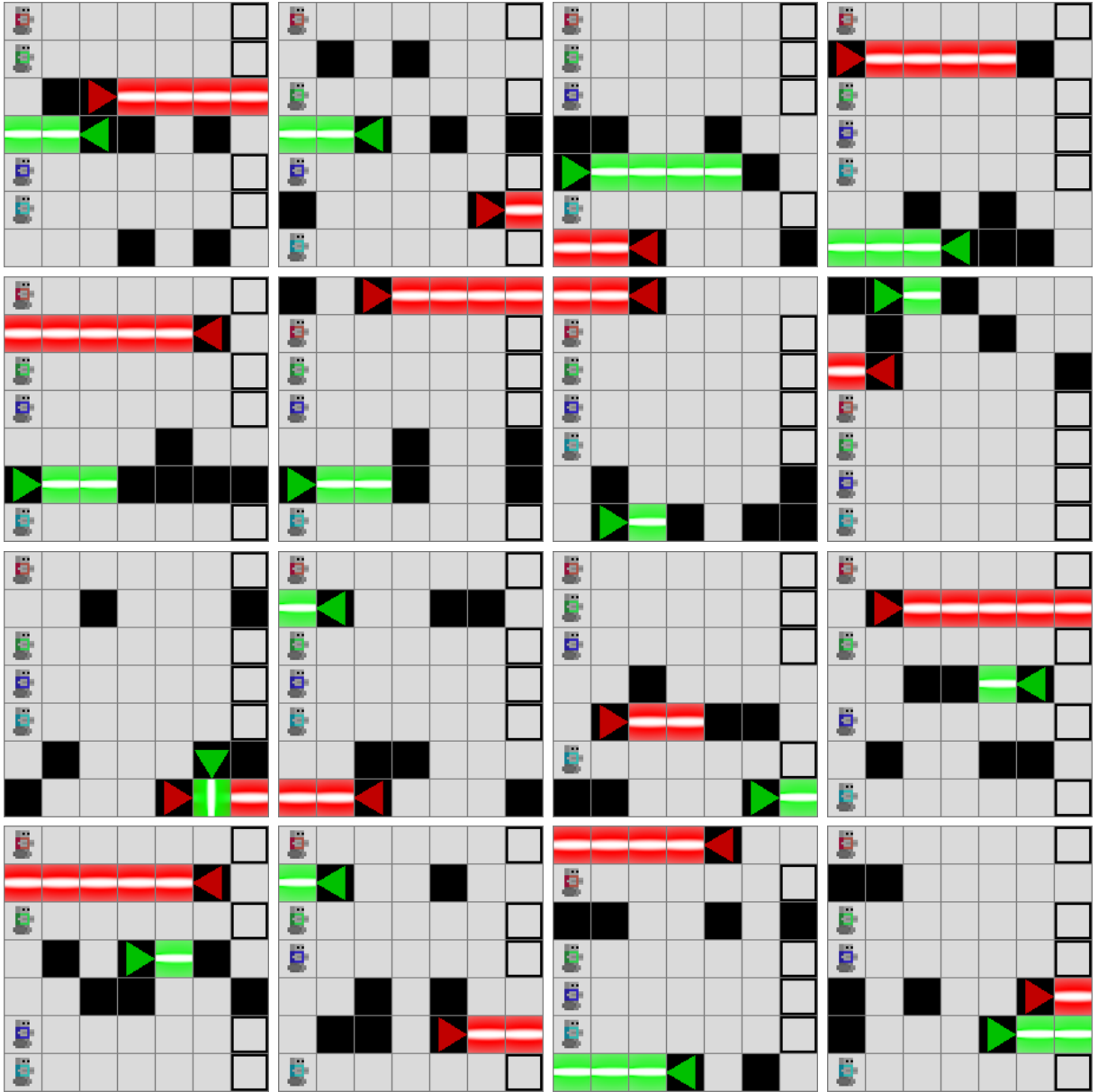


Figure 43: 16 Constructive Solvable 7×7 (4 agents, 2 lasers) levels.

A.16.3 Constructive Solvable — 9×9, 4 agents, 3 lasers

Grid	9 × 9
Agents	4
Lasers	3
Horizon $T_{\max}$	18
Wall budget	8
Seed	20260623
Samples in pool	16

Table 35: Parameters and seed for the Constructive Solvable 9×9 gallery pool.

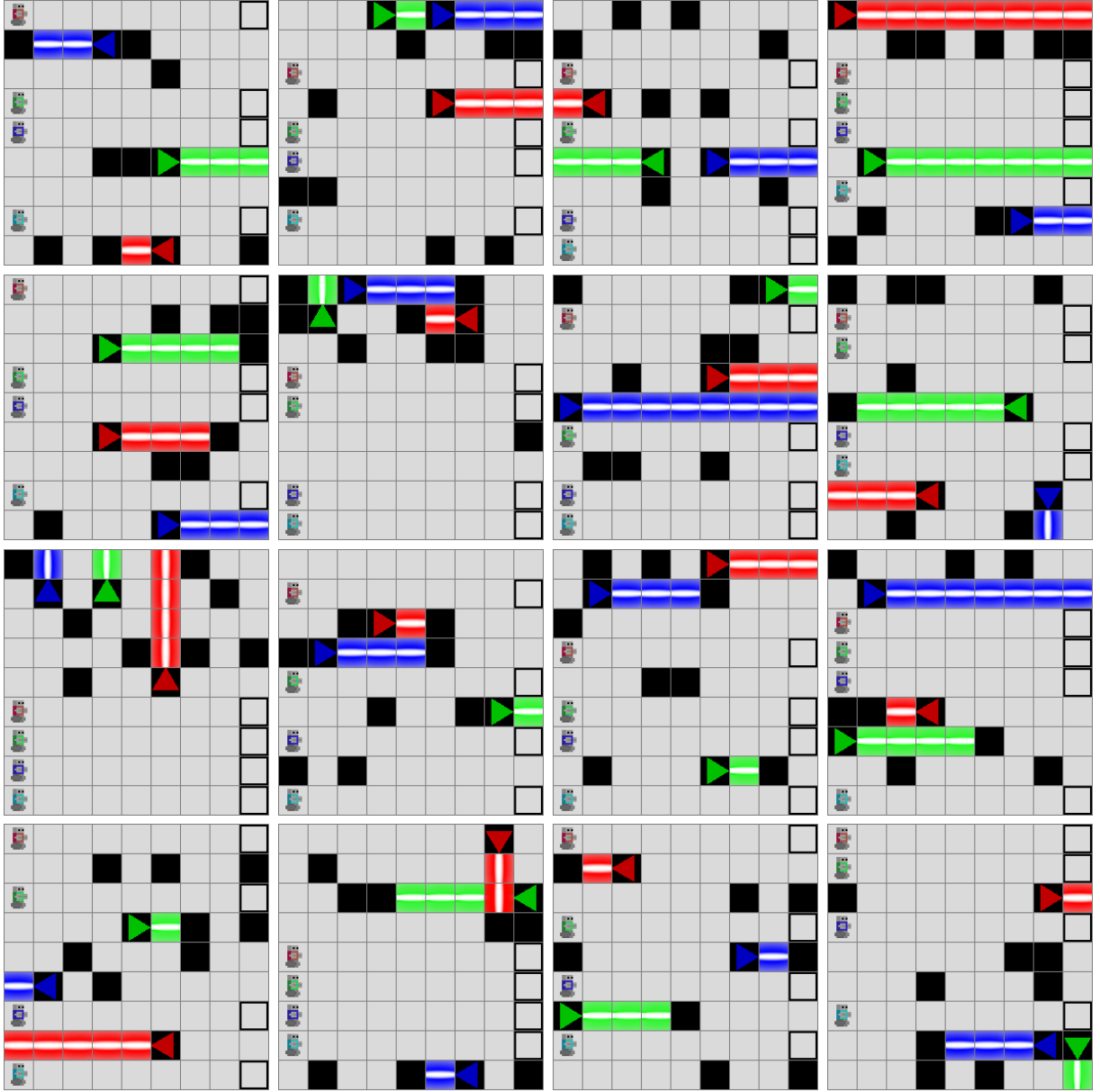


Figure 44: 16 Constructive Solvable 9×9 (4 agents, 3 lasers) levels.

A.17 Generator Gallery – Constructive Generator (Cooperative)

Lane-based construction with planted same-colour structural lasers, certified to satisfy the binary cooperation criterion of Theorem 5.1. No profile filter is applied; the gallery shows what the generator produces when any cooperation profile is admissible.

A.17.1 Constructive Cooperative – 5×5, 2 agents, 1 laser

Grid	5 × 5
Agents	2
Lasers	1
Horizon $T_{\max}$	10
Wall budget	2
Seed	20260631
Profile filter	cooperative
Samples in pool	16

Table 36: Parameters and seed for the Constructive Cooperative 5×5 gallery pool.

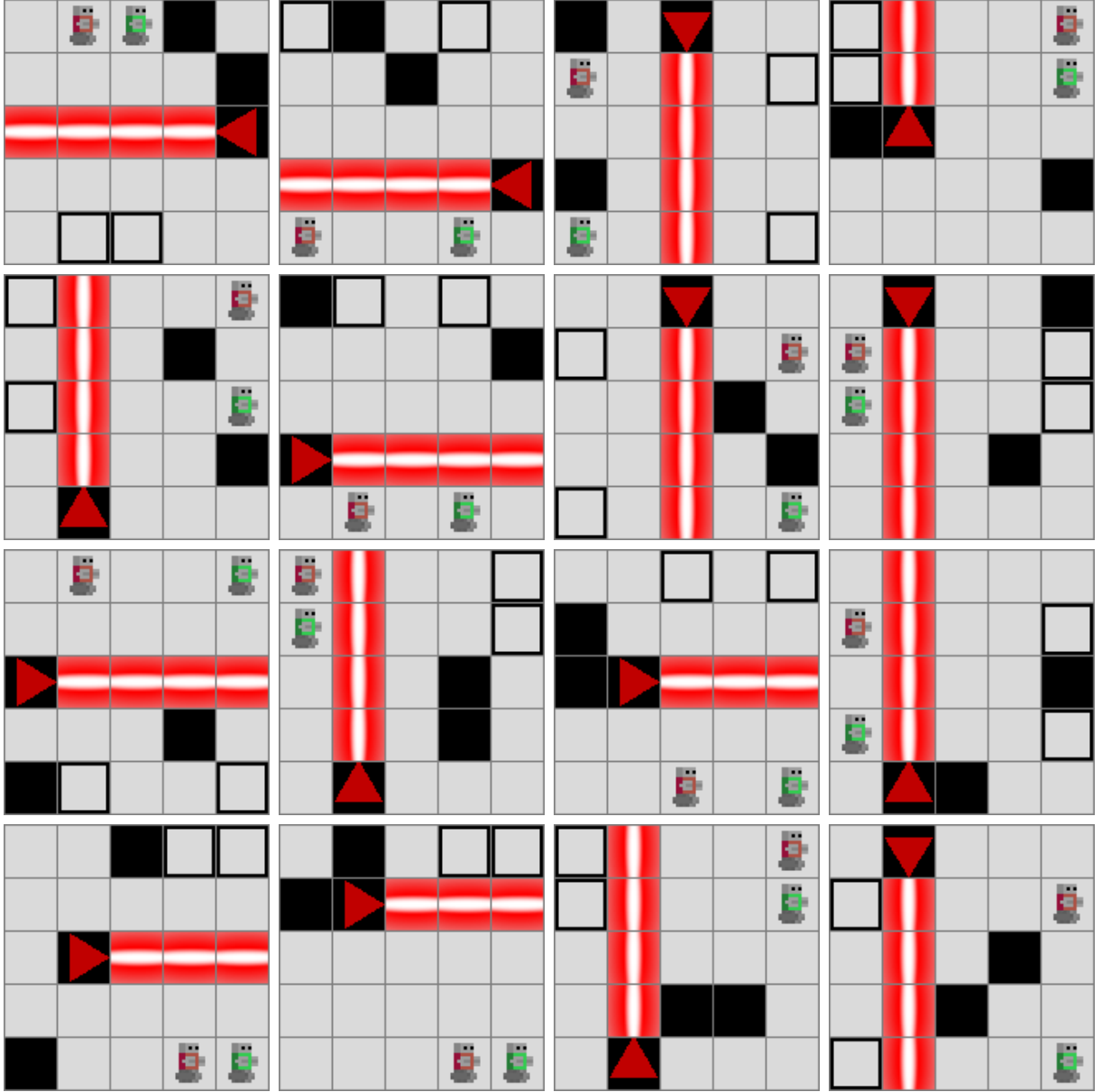


Figure 45: 16 Constructive Cooperative 5x5 (2 agents, 1 laser) levels.

A.17.2 Constructive Cooperative – 7x7, 3 agents, 2 lasers

Grid	7 × 7
Agents	3
Lasers	2
Horizon $T_{\max}$	14
Wall budget	4
Seed	20260632
Profile filter	cooperative
Samples in pool	16

Table 37: Parameters and seed for the Constructive Cooperative 7x7 gallery pool.

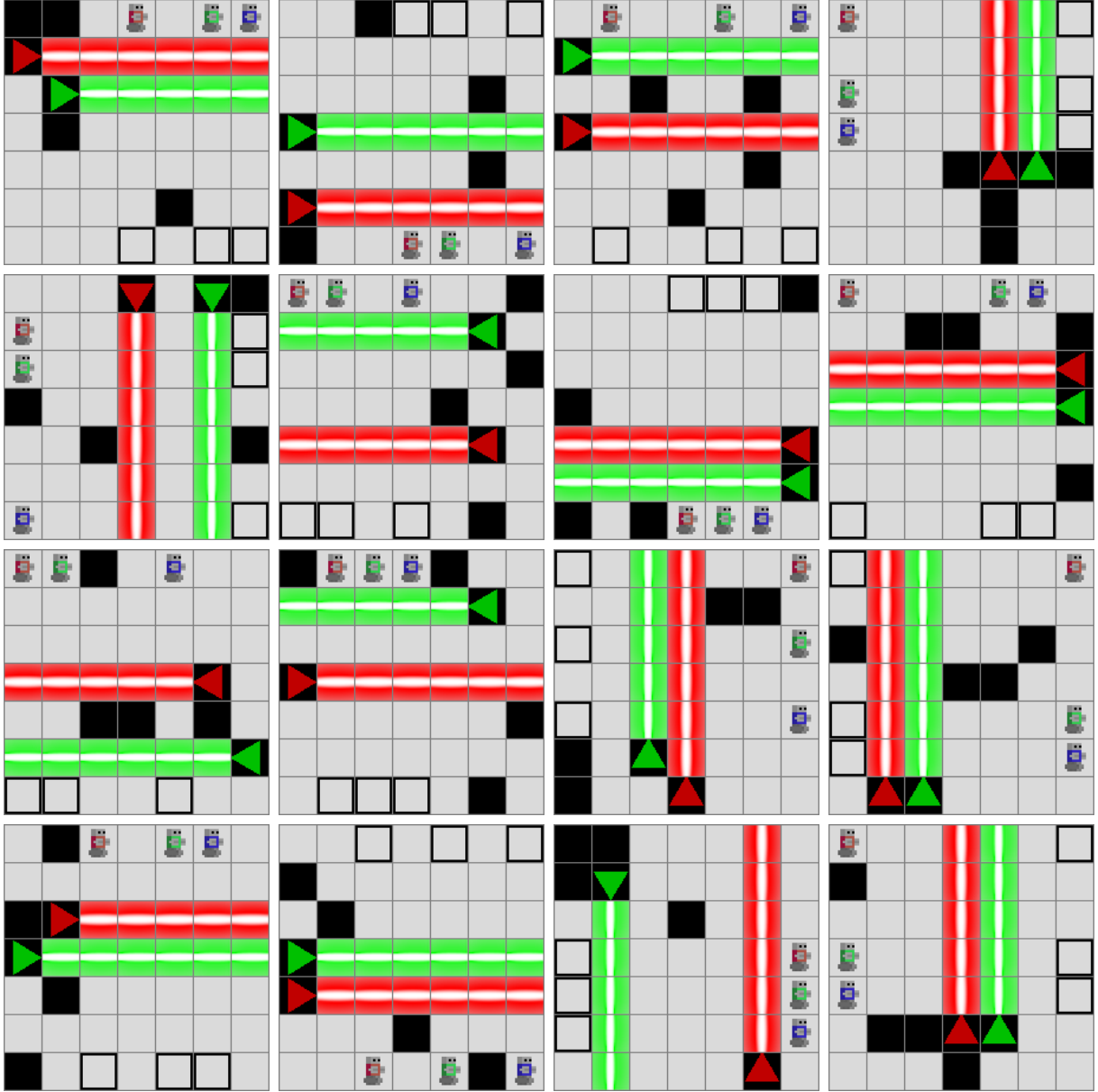


Figure 46: 16 Constructive Cooperative 7×7 (3 agents, 2 lasers) levels.

A.17.3 Constructive Cooperative – 9×9, 4 agents, 3 lasers

Grid	9 × 9
Agents	4
Lasers	3
Horizon $T_{\max}$	18
Wall budget	8
Seed	20260633
Profile filter	cooperative
Samples in pool	16

Table 38: Parameters and seed for the Constructive Cooperative 9×9 gallery pool.

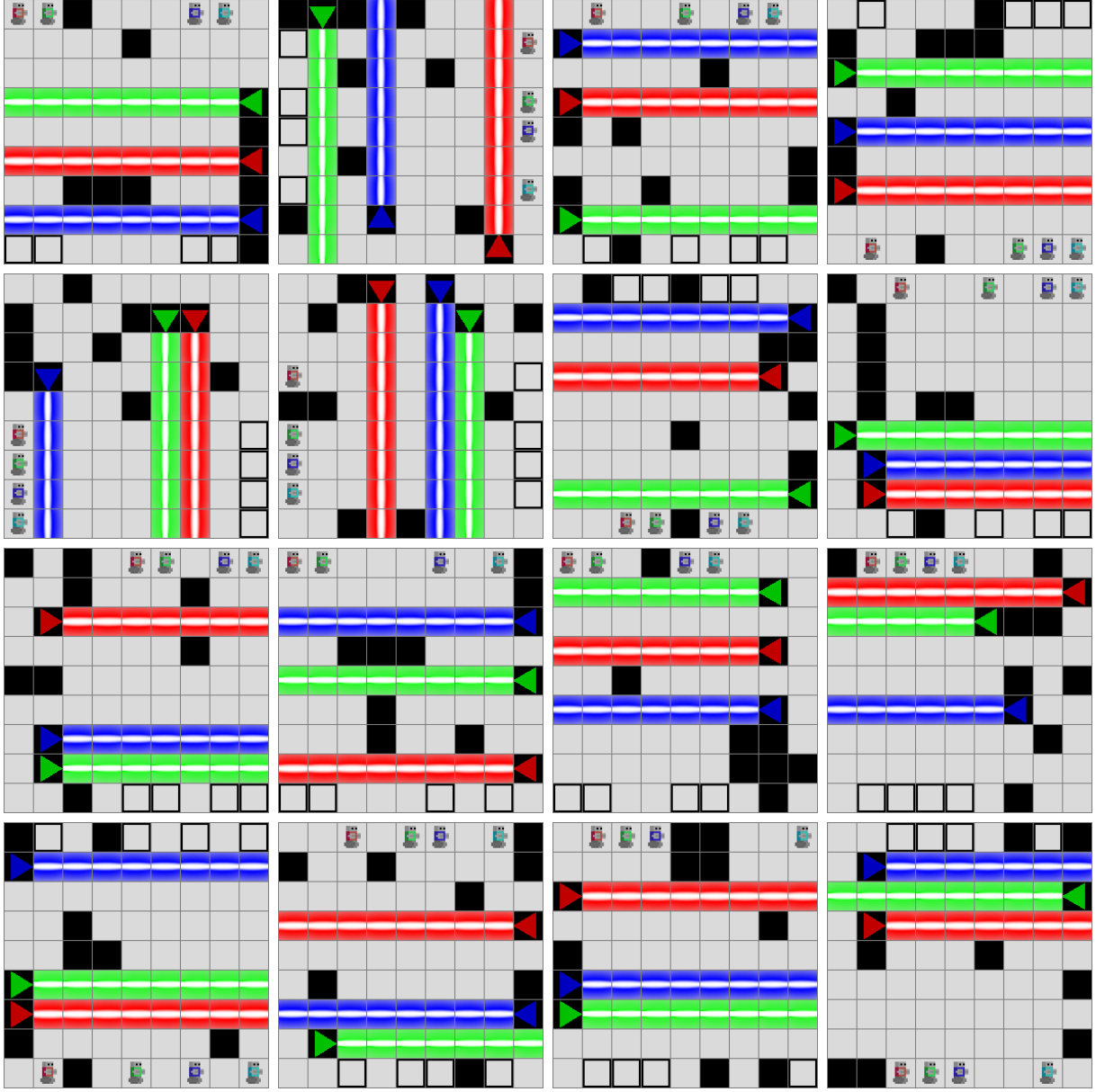


Figure 47: 16 Constructive Cooperative 9×9 (4 agents, 3 lasers) levels.

A.18 Generator Gallery – Constructive Cooperative with Profile Filter

Same Constructive Cooperative generator as the previous section, but with an explicit profile filter so accepted levels match a single label (e.g. **mutual** or **distributed**).

A.18.1 Constructive Cooperative – 8×8, 3 agents, 2 lasers, profile = mutual

Grid	8 × 8
Agents	3
Lasers	2
Horizon $T_{\max}$	16
Wall budget	5
Seed	20260641
Profile filter	mutual
Samples in pool	16

Table 39: Parameters and seed for the profile-filtered (mutual) 8×8 gallery pool.

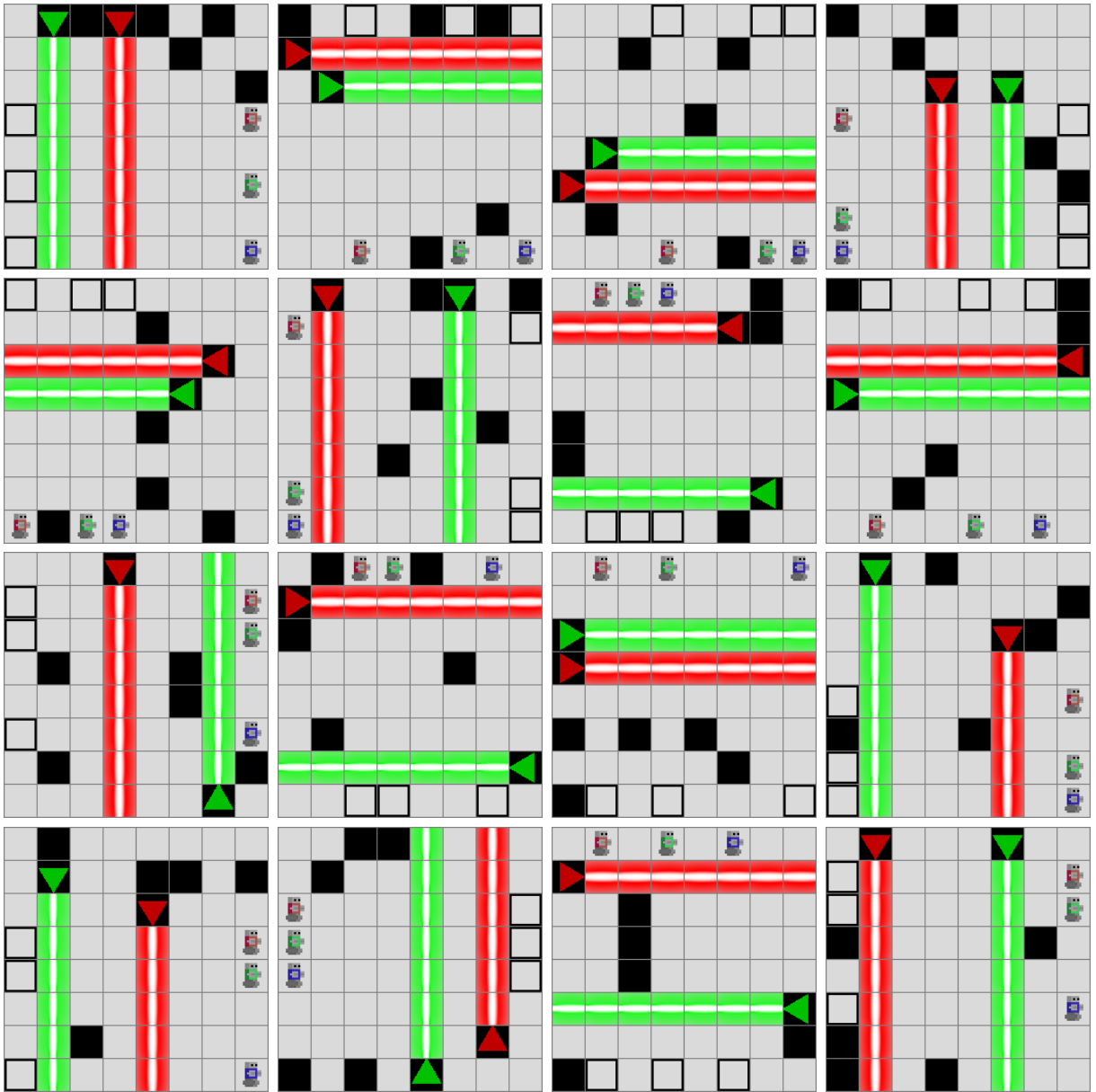


Figure 48: 16 profile-filtered (mutual) 8×8 (3 agents, 2 lasers) levels.

A.18.2 Constructive Cooperative — 10×10, 4 agents, 3 lasers, profile = distributed

Grid	10 × 10
Agents	4
Lasers	3
Horizon $T_{\max}$	20
Wall budget	8
Seed	20260642
Profile filter	distributed
Samples in pool	16

Table 40: Parameters and seed for the profile-filtered (distributed) 10×10 gallery pool.

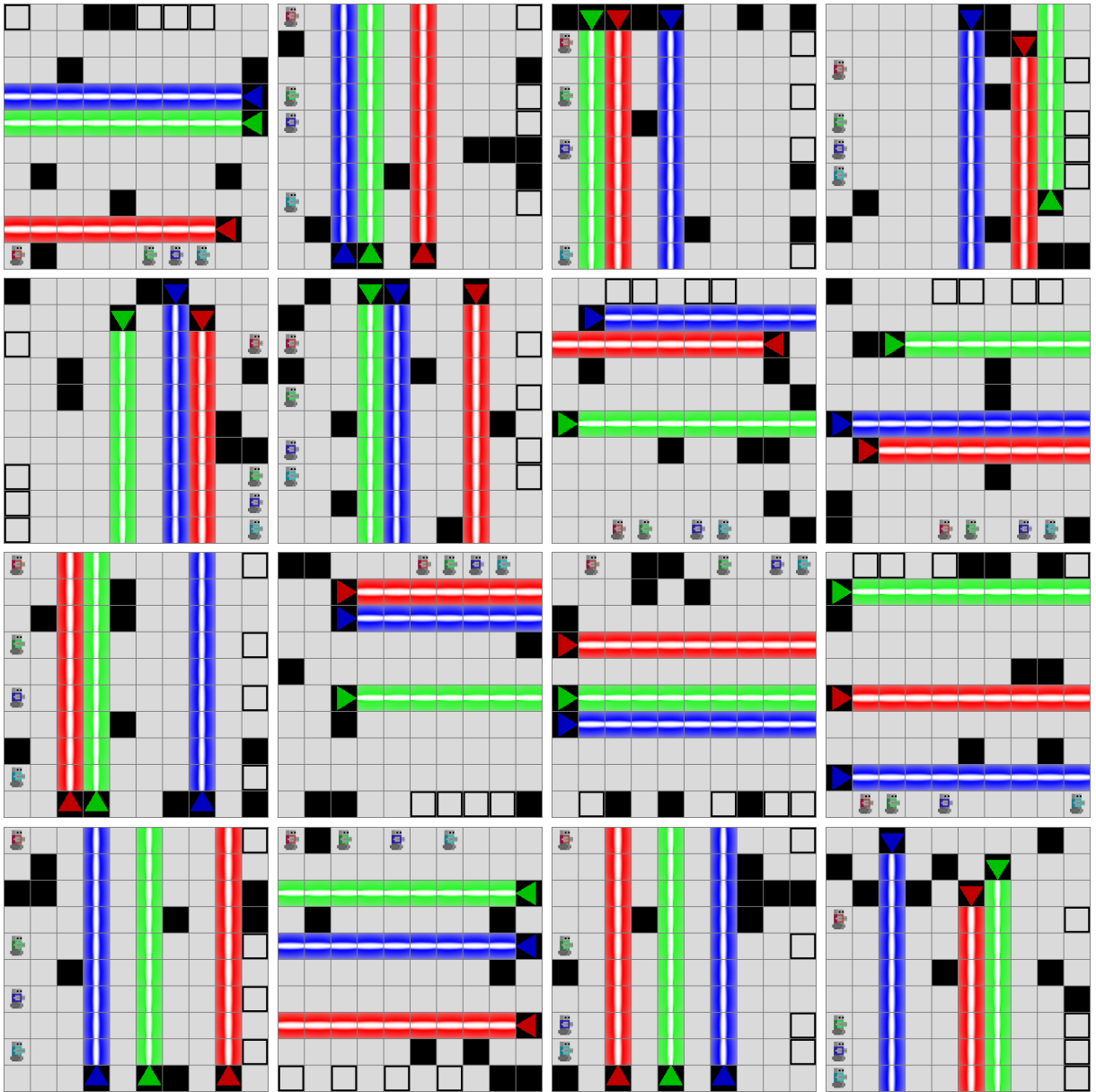


Figure 49: 16 profile-filtered (distributed) 10×10 (4 agents, 3 lasers) levels.

A.19 Generator Gallery — Level-6-Style Generator

Clustered starts and exits on opposing sides of the grid with a corridor of lasers in between, inspired by the hand-crafted LLE Level 6.

A.19.1 Level-6-Style — 8×8, 4 agents, 2 lasers

Grid	8 × 8
Agents	4
Lasers	2
Horizon $T_{\max}$	16
Wall budget	6
Seed	20260651
Samples in pool	16

Table 41: Parameters and seed for the Level-6-Style 8×8 gallery pool.

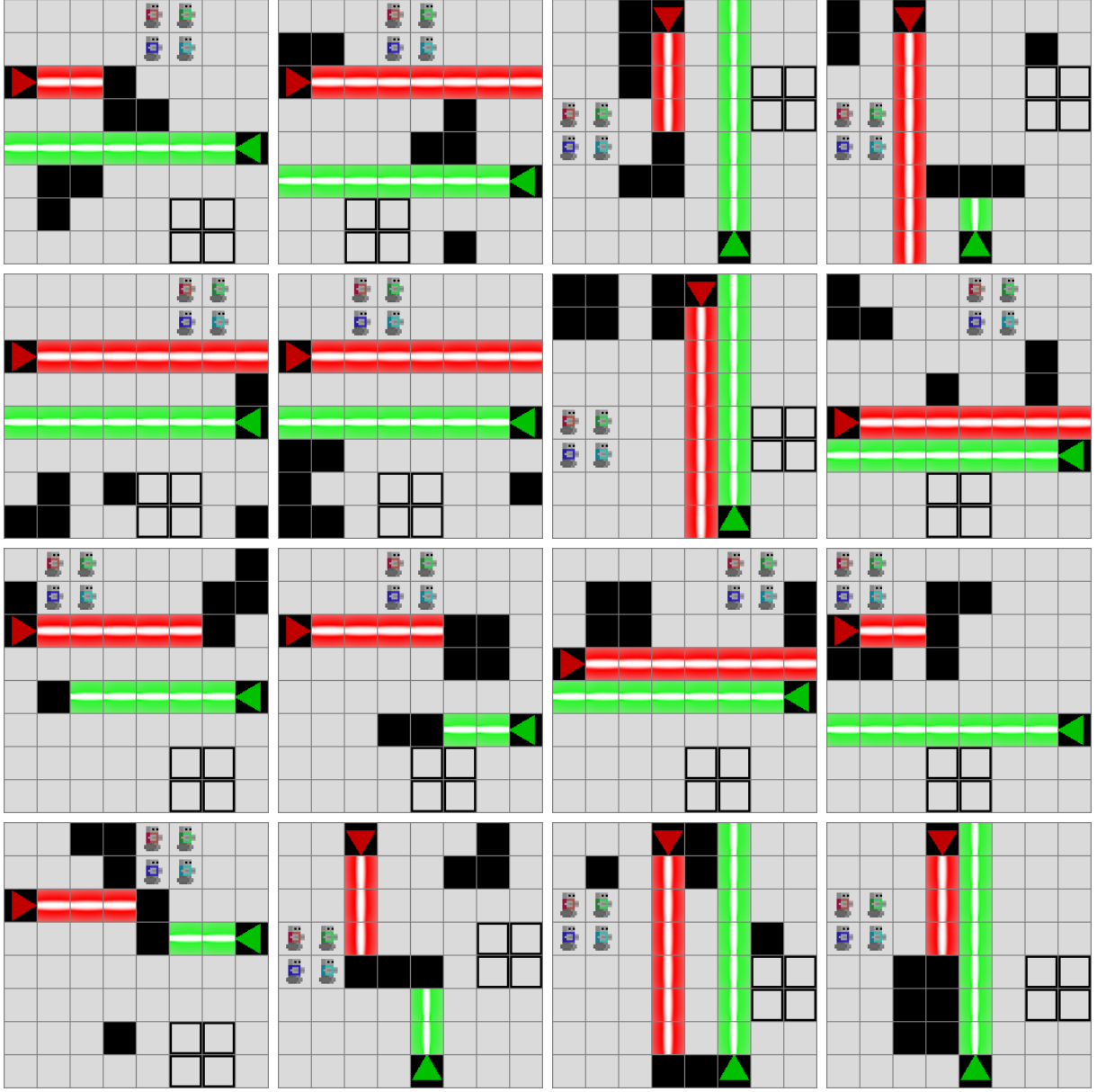


Figure 50: 16 Level-6-Style 8×8 (4 agents, 2 lasers) levels.

A.19.2 Level-6-Style — 10×10, 4 agents, 3 lasers

Grid	10 × 10
Agents	4
Lasers	3
Horizon $T_{\max}$	18
Wall budget	8
Seed	20260652
Samples in pool	16

Table 42: Parameters and seed for the Level-6-Style 10×10 gallery pool.

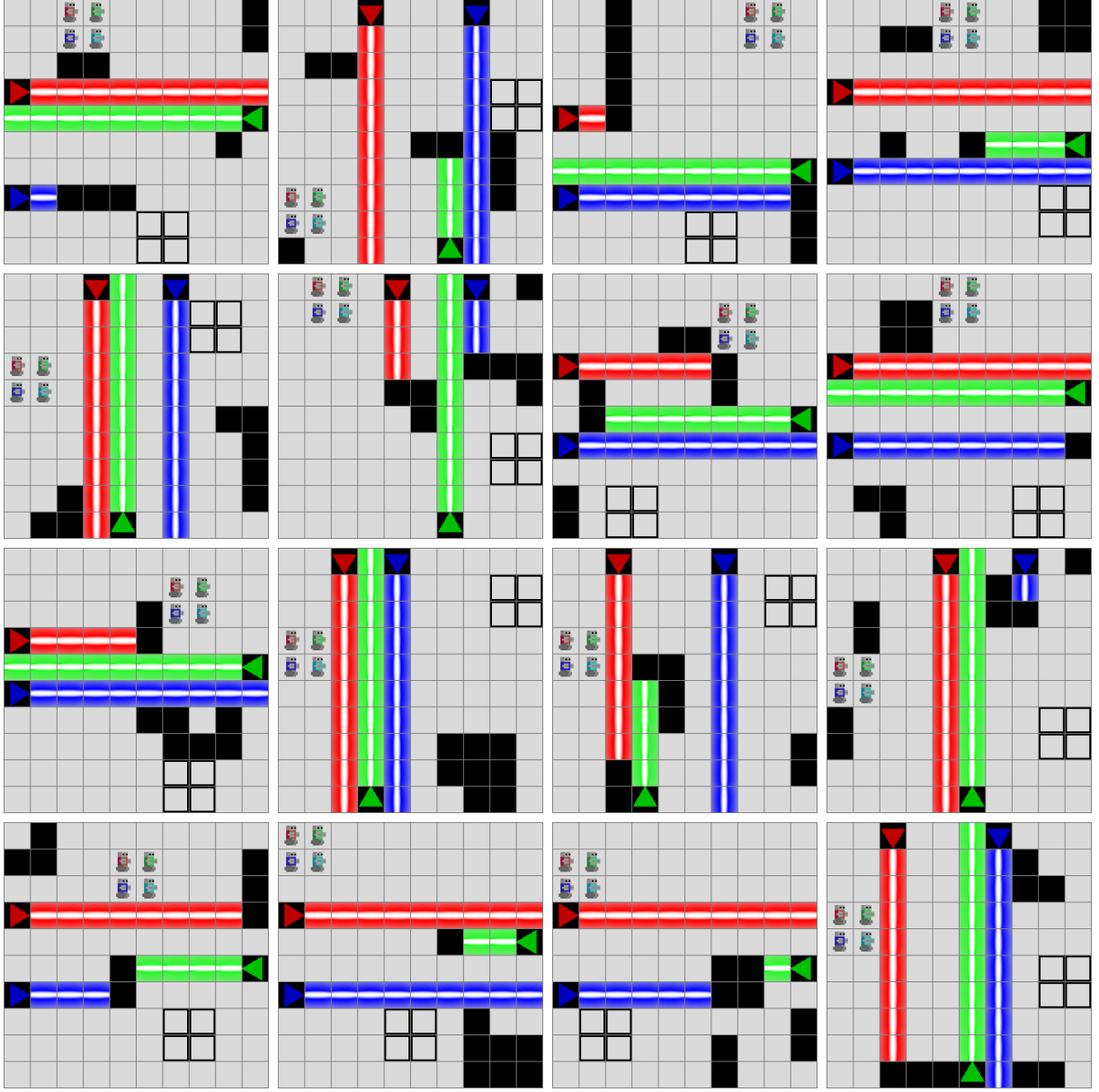


Figure 51: 16 Level-6-Style 10×10 (4 agents, 3 lasers) levels.

A.19.3 Level-6-Style — 12×13, 4 agents, 3 lasers (Level 6 footprint)

Grid	12 × 13
Agents	4
Lasers	3
Horizon $T_{\max}$	21
Wall budget	10
Seed	20260653
Samples in pool	16

Table 43: Parameters and seed for the Level-6-Style 12×13 gallery pool.

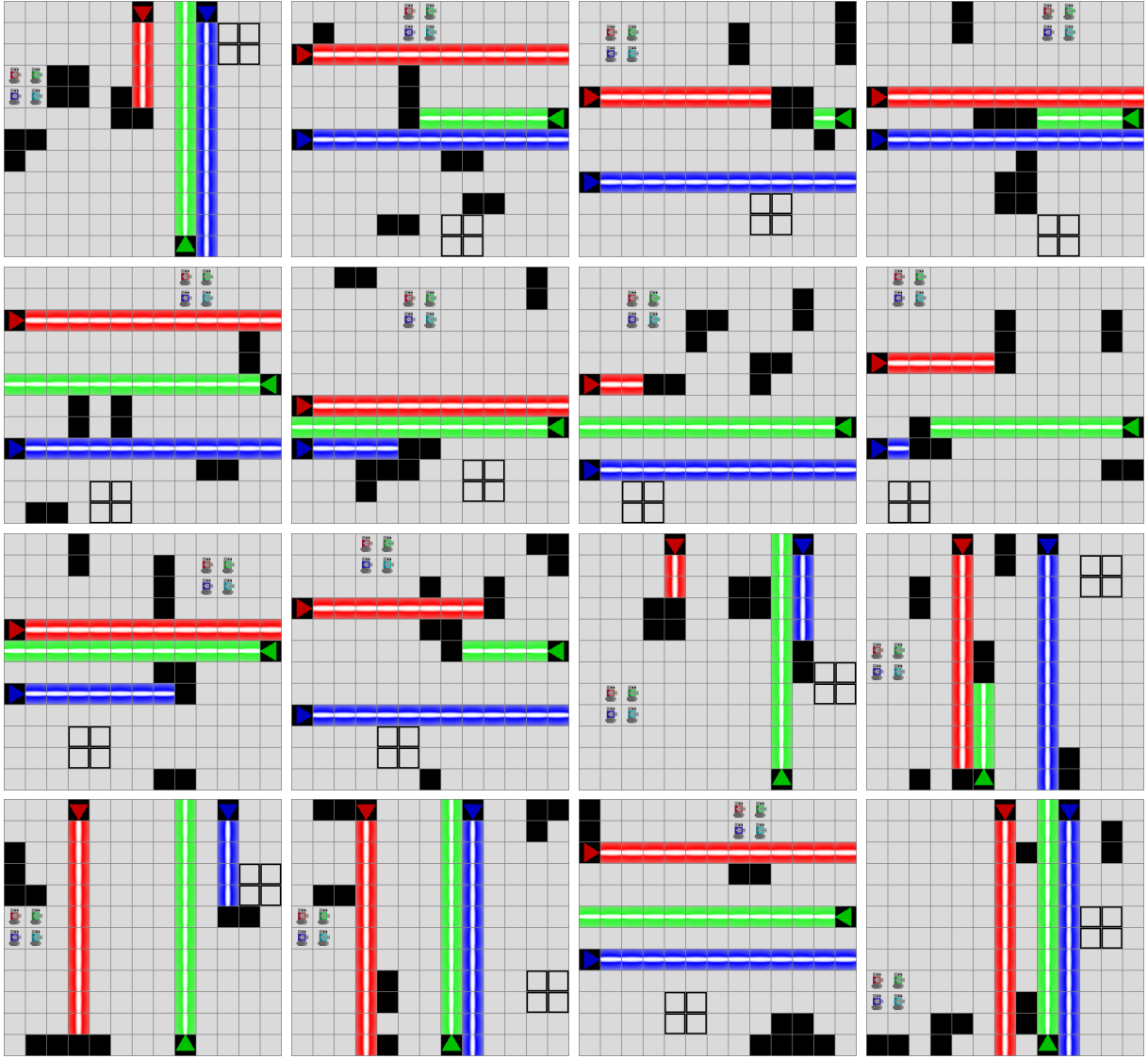


Figure 52: 16 Level-6-Style 12×13 (4 agents, 3 lasers) levels — same footprint as the hand-crafted LLE Level 6.

Bibliography

- [1] L. S. Shapley, “Stochastic Games,” *Proceedings of the National Academy of Sciences*, vol. 39, no. 10, pp. 1095–1100, 1953, doi: 10.1073/pnas.39.10.1095.
- [2] M. L. Littman, “Markov Games as a Framework for Multi-Agent Reinforcement Learning,” in *Machine Learning Proceedings 1994*, Morgan Kaufmann, 1994, pp. 157–163. doi: 10.1016/B978-1-55860-335-6.50027-1.
- [3] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA: MIT Press, 2018. [Online]. Available: <http://incompleteideas.net/book/the-book-2nd.html>
- [4] P. Sunehag *et al.*, “Value-Decomposition Networks For Cooperative Multi-Agent Learning Based On Team Reward,” in *Proceedings of the 17th International Conference on Autonomous Agents and MultiAgent Systems*, in AAMAS 2018. International Foundation for Autonomous Agents, Multi-agent Systems, 2018, pp. 2085–2087.
- [5] T. Rashid, M. Samvelyan, C. Schroeder de Witt, G. Farquhar, J. Foerster, and S. Whiteson, “QMIX: Monotonic Value Function Factorisation for Deep Multi-Agent Reinforcement Learning,” in *Proceedings of the 35th International Conference on Machine Learning*, in ICML 2018, vol. 80. PMLR, 2018, pp. 4295–4304.
- [6] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, “Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments,” in *Advances in Neural Information Processing Systems 30*, in NeurIPS 2017. 2017, pp. 6379–6390.
- [7] J. N. Foerster, G. Farquhar, T. Afouras, N. Nardelli, and S. Whiteson, “Counterfactual Multi-Agent Policy Gradients,” in *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*, in AAAI-18. AAAI Press, 2018, pp. 2974–2982.
- [8] A. Mahajan, T. Rashid, M. Samvelyan, and S. Whiteson, “MAVEN: Multi-Agent Variational Exploration,” in *Advances in Neural Information Processing Systems 32*, in NeurIPS 2019. 2019, pp. 7613–7624.
- [9] Y. Molinghen, R. Avalos, M. Van Achter, A. Nowé, and T. Lenaerts, “Laser Learning Environment: A New Environment for Coordination-Critical Multi-Agent Tasks,” in *Artificial Intelligence and Machine Learning*, F. A. Oliehoek, M. Kok, and S. Verwer, Eds., Cham: Springer Nature Switzerland, 2025, pp. 135–154.
- [10] M. Samvelyan *et al.*, “The StarCraft Multi-Agent Challenge,” in *Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS)*, 2019, pp. 2186–2188.
- [11] M. Carroll *et al.*, “On the Utility of Learning about Humans for Human-AI Coordination,” in *Advances in Neural Information Processing Systems*, 2019.
- [12] C. Guestrin, M. G. Lagoudakis, and R. Parr, “Coordinated Reinforcement Learning,” in *Proceedings of the Nineteenth International Conference on Machine Learning (ICML)*, 2002, pp. 227–234.
- [13] N. Shaker, J. Togelius, and M. J. Nelson, *Procedural Content Generation in Games*. Springer, 2016. doi: 10.1007/978-3-319-42716-4.
- [14] J. Togelius, G. N. Yannakakis, K. O. Stanley, and C. Browne, “Search-Based Procedural Content Generation: A Taxonomy and Survey,” *IEEE Transactions on Computational Intelligence and AI in Games*, vol. 3, no. 3, pp. 172–186, 2011, doi: 10.1109/TCIAIG.2011.2148116.
- [15] A. M. Smith and M. Mateas, “Answer Set Programming for Procedural Content Generation: A Design Space Approach,” *IEEE Transactions on Computational Intelligence and AI in Games*, vol. 3, no. 3, pp. 187–200, 2011, doi: 10.1109/TCIAIG.2011.2158545.

- [16] A. Summerville *et al.*, “Procedural Content Generation via Machine Learning (PCGML),” *IEEE Transactions on Games*, vol. 10, no. 3, pp. 257–270, 2018, doi: 10.1109/TG.2018.2846639.
- [17] S. Risi and J. Togelius, “Increasing Generality in Machine Learning through Procedural Content Generation,” *Nature Machine Intelligence*, vol. 2, no. 8, pp. 428–436, 2020, doi: 10.1038/s42256-020-0208-z.
- [18] A. Khalifa, P. Bontrager, S. Earle, and J. Togelius, “PCGRL: Procedural Content Generation via Reinforcement Learning,” in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, 2020, pp. 95–101. doi: 10.1609/aiide.v16i1.7416.
- [19] Y. Bengio, J. Louradour, R. Collobert, and J. Weston, “Curriculum Learning,” in *Proceedings of the 26th Annual International Conference on Machine Learning*, in ICML 2009. ACM, 2009, pp. 41–48. doi: 10.1145/1553374.1553380.
- [20] S. Narvekar, B. Peng, M. Leonetti, J. Sinapov, M. E. Taylor, and P. Stone, “Curriculum Learning for Reinforcement Learning Domains: A Framework and Survey,” *Journal of Machine Learning Research*, vol. 21, no. 181, pp. 1–50, 2020.
- [21] R. Wang, J. Lehman, J. Clune, and K. O. Stanley, “Paired Open-Ended Trailblazer (POET): Endlessly Generating Increasingly Complex and Diverse Learning Environments and Their Solutions,” in *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '19)*, ACM, 2019, pp. 142–151. doi: 10.1145/3321707.3321799.
- [22] M. Dennis *et al.*, “Emergent Complexity and Zero-shot Transfer via Unsupervised Environment Design,” in *Advances in Neural Information Processing Systems*, 2020, pp. 13049–13061.
- [23] H. Kautz and B. Selman, “Planning as Satisfiability,” in *Proceedings of the 10th European Conference on Artificial Intelligence*, in ECAI '92. John Wiley & Sons, 1992, pp. 359–363.
- [24] H. Kautz and B. Selman, “Pushing the Envelope: Planning, Propositional Logic, and Stochastic Search,” in *Proceedings of the Thirteenth National Conference on Artificial Intelligence*, in AAAI-96. AAAI Press, 1996, pp. 1194–1201.
- [25] J. Rintanen, K. Heljanko, and I. Niemelä, “Planning as satisfiability: parallel plans and algorithms for plan search,” *Artificial Intelligence*, vol. 170, no. 12–13, pp. 1031–1080, 2006, doi: 10.1016/j.artint.2006.08.002.
- [26] P. Surynek, “Problem Compilation for Multi-Agent Path Finding: A Survey,” in *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence*, in IJCAI-22. 2022, pp. 5615–5620.
- [27] G. Sharon, R. Stern, A. Felner, and N. R. Sturtevant, “Conflict-based search for optimal multi-agent pathfinding,” *Artificial Intelligence*, vol. 219, pp. 40–66, 2015, doi: 10.1016/j.artint.2014.11.006.
- [28] M. Frommknecht and P. Surynek, “Solving Multi-Agent Pathfinding with Stochastic Local Search SAT Algorithms,” in *Proceedings of the 21st International Conference on Informatics in Control, Automation and Robotics*, in ICINCO 2024, vol. 1. 2024, pp. 67–78. doi: 10.5220/0012944800003822.
- [29] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu, “Safe Reinforcement Learning via Shielding,” in *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI)*, 2018, pp. 2669–2678. doi: 10.1609/aaai.v32i1.11797.
- [30] S. A. Cook, “The Complexity of Theorem-Proving Procedures,” in *Proceedings of the Third Annual ACM Symposium on Theory of Computing*, 1971, pp. 151–158.

- [31] M. Davis, G. Logemann, and D. Loveland, “A Machine Program for Theorem-Proving,” *Communications of the ACM*, vol. 5, no. 7, pp. 394–397, 1962, doi: 10.1145/368273.368557.
- [32] N. Eén and N. Sörensson, “An Extensible SAT-solver,” in *Theory and Applications of Satisfiability Testing – SAT 2003*, in Lecture Notes in Computer Science, vol. 2919. Springer, 2004, pp. 502–518. doi: 10.1007/978-3-540-24605-3_37.
- [33] A. Ignatiev, A. Morgado, and J. Marques-Silva, “PySAT: A Python Toolkit for Prototyping with SAT Oracles,” in *Theory and Applications of Satisfiability Testing – SAT 2018*, in Lecture Notes in Computer Science, vol. 10929. Springer, 2018, pp. 428–437. doi: 10.1007/978-3-319-94144-8_26.